

ТАМЈЕНЈЕНИ КАБЕЛ ОД МЛУЈНАКА
ПРЕМА ПОБУДИ 1 ЗАШТИТА, ПОВЕЋАНИ ПРЕМА
ЗАТЋЕЊЕНОМ ТАЈУ

= 02 - 2 НААНН + ААНН - 3202 (НУЧУ 7x4, ПОБУДА)

ПОБУДА	ЖИЛА	СТРУЈНИЈАК
X0:1	1	X1:1
X0:2	2	X1:3
X0:3	3	X1:5
X0:4	4	X1:6

= 02 - 2 НААНН + ААНН - 3202 (НУЧУ 0 7x4, ЗАШТИТА)

ЗАШТИТА	ЖИЛА	СТРУЈНИЈАК
X101:1	1	X1:7
X101:2	2	X1:9
X101:3	3	X1:11
X101:4	4	X1:12

REVOLVING

Place for digital signature of Contractor

Place for digital signature of Revisor

*НАПОМЕНА: Ми кабели који су испоручени
у мучу 8x2,5 ти
мучу 16x2,5 имају радни
нтен гуње

Rev.	Revision Description	Drawn by	Checked by	Approved by Designer	Approved by Contractor	Date
02	After FAT, for installation	Tamara Trogrlić	Antonio Aleksić	Grga Kolić		27.7.2026.
01	For manufacturing	Tamara Trogrlić	Antonio Aleksić	Grga Kolić		7.5.2026.
00	First issue	Tamara Trogrlić	Antonio Aleksić	Grga Kolić		27.2.2026.

Document No. (CSD) PER-E-8-KAI&KA2-SCH-002-02-VHHR

Employer:  **epcg** Elektroprivreda Crne Gore NO

Consultant: **ME ENERGOPROJEKT** HIDROINŽENJERING a.d.

Contractor: **VOITH** VOITH HYDRO GmbH & Co KG St. Pölten

Project: Reconstruction and Modernisation of HPP Perucica - Phase II

Title: Unit KA2 control cubicle circuit diagrams

Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100
TITLE PAGE	Designer No.:	SALES No.:	PAGE
			PAGE
			1 / 158

Format ISO/DIN A3

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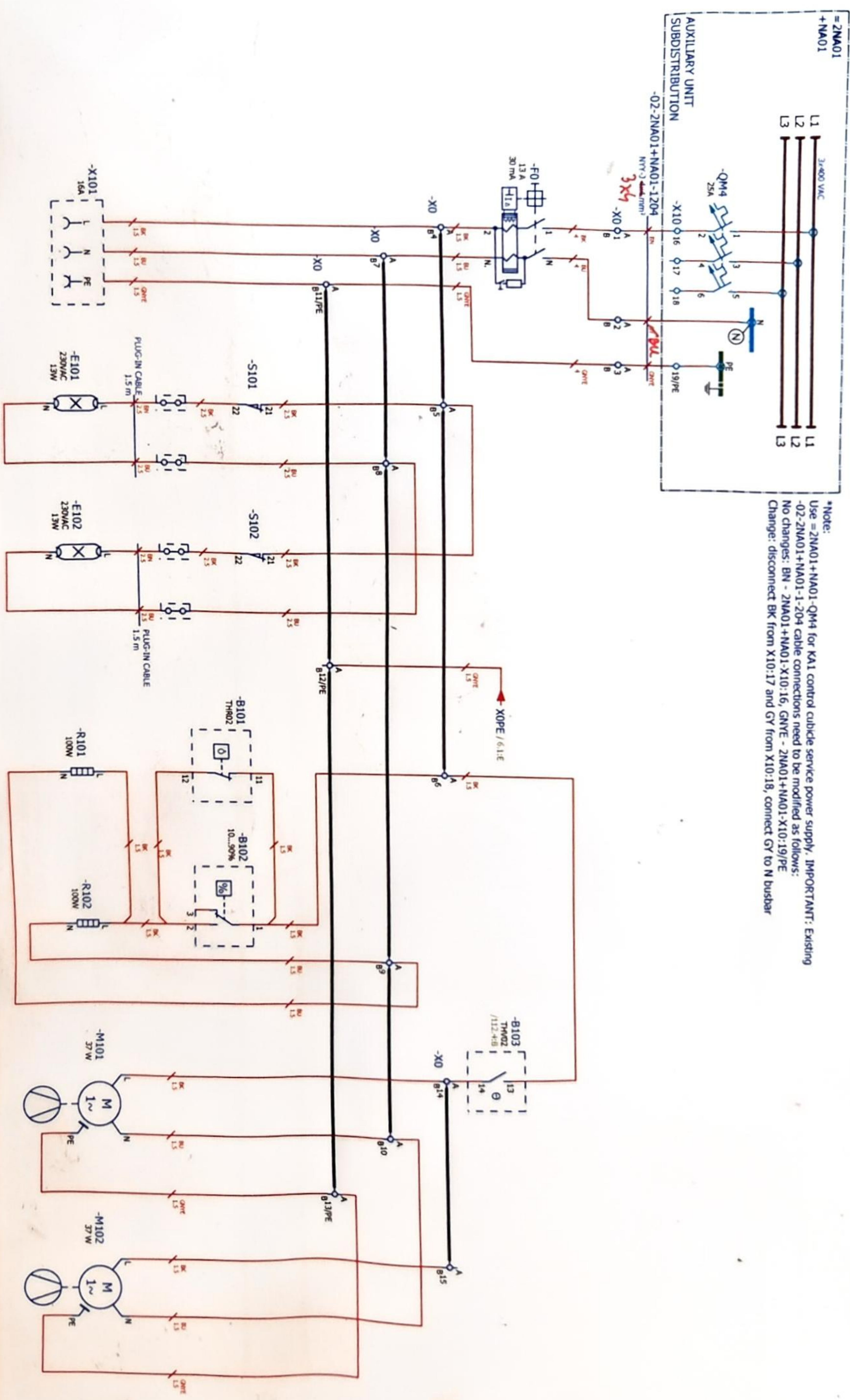
Approved by Contractor	Grga Kolić	27.7.2026.	Reconstruction and Modernisation of HPP Perucica - Phase II
Approved by Designer	Antonio Aleksić	27.7.2026.	
Checked by	Tamara Trogrlić	27.7.2026.	
Drawn by			

SERVICE
POWER SUPPLY

CUBICLE LIGHTING

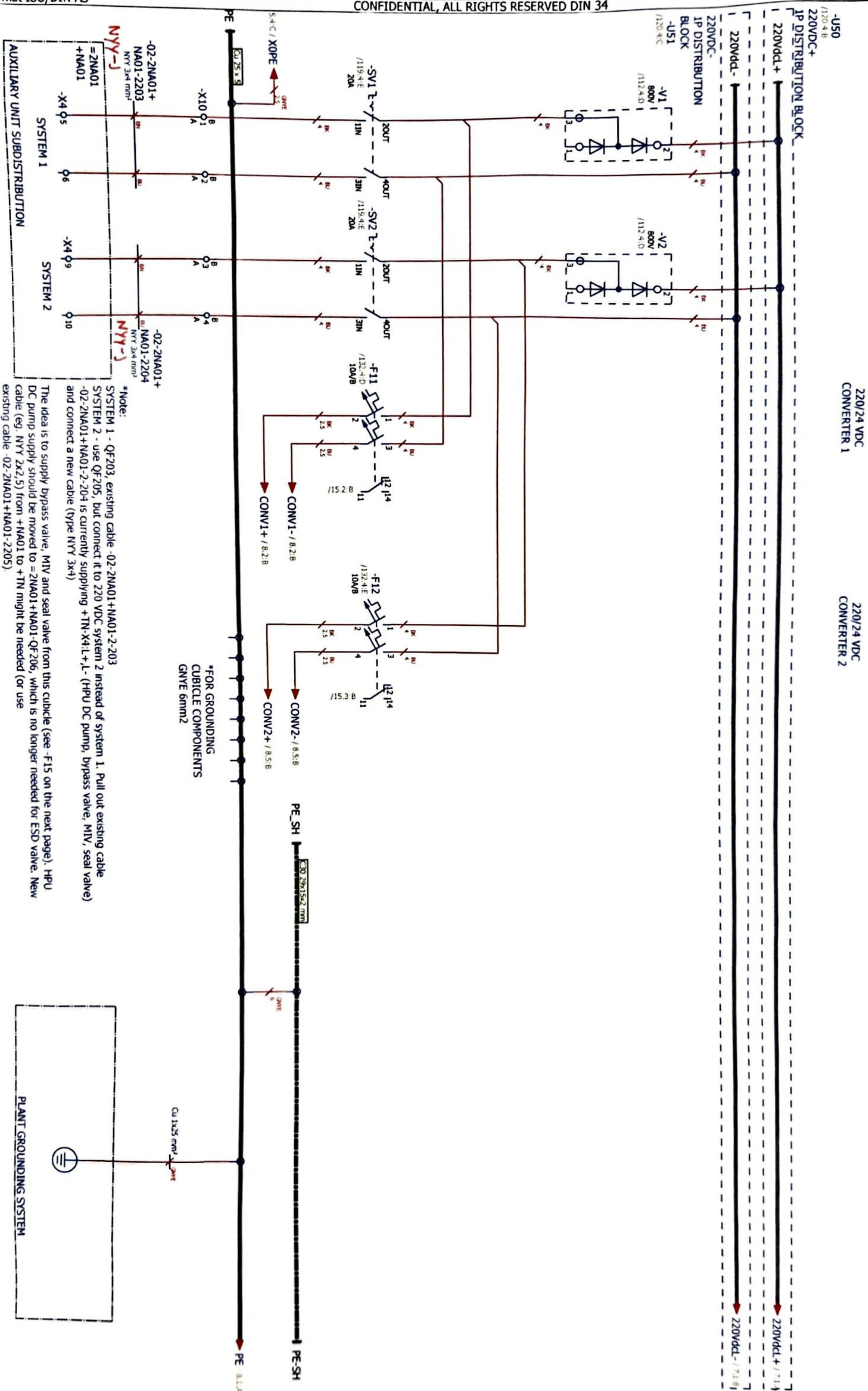
CUBICLE HEATER

CUBICLE FAN



*Note:
Use = 2NA01+NA01-QM4 for KA1 control cubicle service power supply. IMPORTANT: Existing
-02-2NA01+NA01-1-204 cable connections need to be modified as follows:
No changes: BN - 2NA01+NA01-X10:16, GNYE - 2NA01+NA01-X10:19/PE
Change: disconnect BK from X10:17 and GY from X10:18, connect GY to N busbar

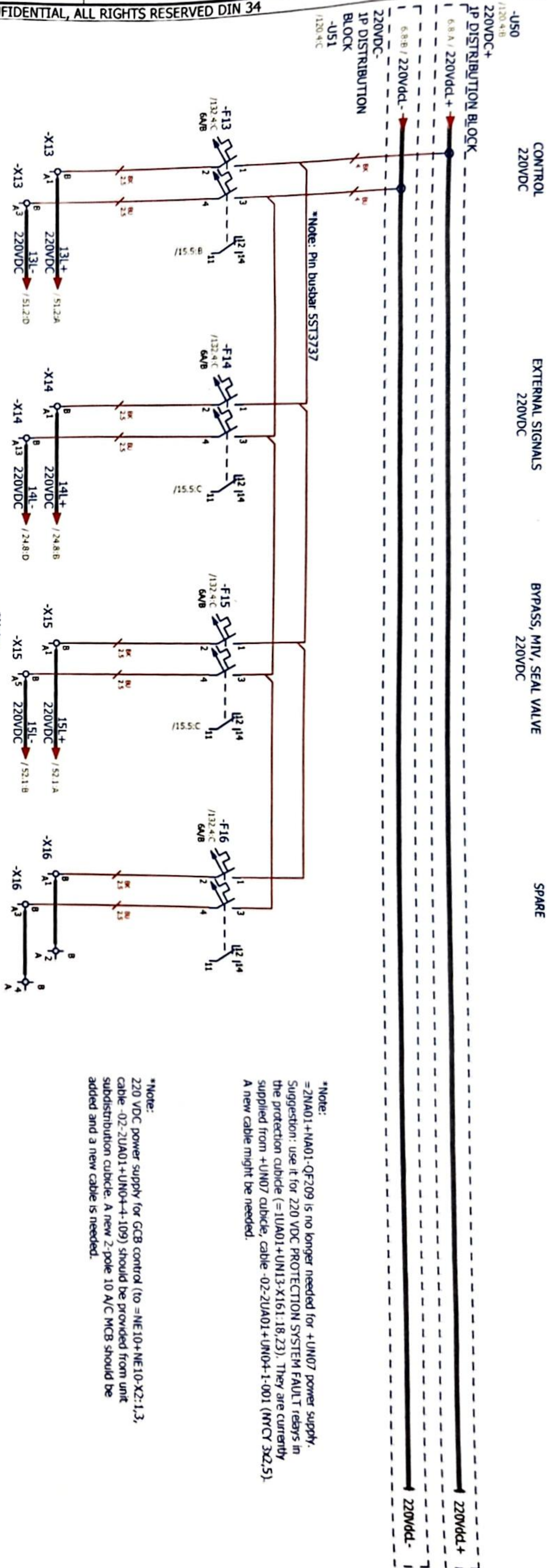
Approved by Contractor		Reconstruction and Modernisation		epcg		SINTAKSA		Unit KA2 control cubicle circuit diagrams		DRAWING No.: PERU-01-19-170-31-002		REV No.: 02		= UBCHA100H100	
Checked by Designer		Antonio Aleksic		27.7.2026.		DESIGNER		AC DISTRIBUTION		Designer No.:		SALES No.:		PAGE	
Drawn by		Tamara Trogrlic		27.7.2026.		PROJECT NAME		EMPLOYER		5		6		7	
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***FOR GROUNDING
CUBICLE COMPONENTS
GNYE 6mm²**

PLANT GROUNDING SYSTEM

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*Note:
=2UA01+NA01-QFZ09 is no longer needed for +UM07 power supply.
Suggestion: use it for 220 VDC PROTECTION SYSTEM FAULT relays in the protection cubicle (=1UA01+UM13-X161;18,23). They are currently supplied from +UM07 cubicle, cable -02-2UA01+UM04-1-001 (HYCY 3x2,5). A new cable might be needed.

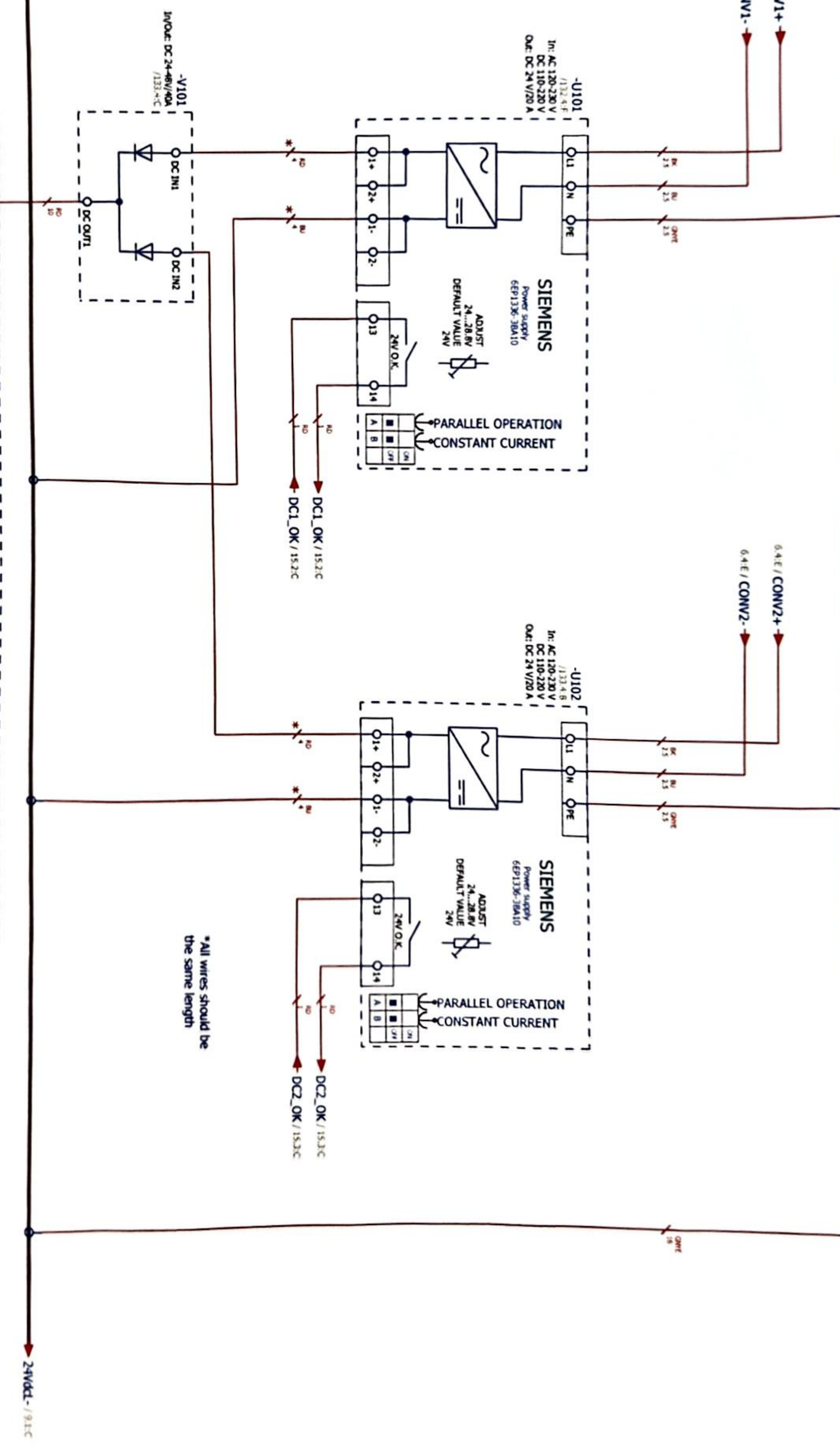
*Note: 220 VDC power supply for GCB control (to =NE10+NE10-XZ:1,3 cable -02-2U/A01+UN04+109) should be provided from unit subdistribution cubicle. A new 2-pole 10 A/C MCB should be added and a new cable is needed.

*Note:
This is currently supplied from +TN via +UN07. Since we are planning to supply it from here, -2WK1006 cable cores 1 and 2 are no longer used and also need to be disconnected in the +TN cubicle.

Approved by Contractor		  A world of connectivity	Unit K62 control cubicle circuit diagrams	DRAWING No.:	REV No.:
Approved by Designer	Griga Kolik			27.7.2026.	= UBCA100GH100
Checked by	Antonio Aleksic			27.7.2026.	+ UBCA10
Drawn by	Tamara Trojilic			27.7.2026.	PAGE
PROJECT NAME			DESIGNER	DC DISTRIBUTION 220VDC	SALES No.:
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Format ISO/DIN A3
24Vdcl-
24Vdcl+
24VDC+
1P DISTRIBUTION BLOCK
-U52
/120-4°C

Approved by Contractor	27.7.2026.	Reconstruction and Modernisation of HPP Perućka - Phase II	EMPLOYER	SINTAKSA	Unit KAZ control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	+ UBCA10
Checked by Designer	27.7.2026.	PROJECT NAME	DESIGNER	Unit KAZ control cubicle circuit diagrams	DC DISTRIBUTION 24VDC	Designer No.:	SALES No.:	PAGE
Drawn by	27.7.2026.							8 / 158



*All wires should be the same length

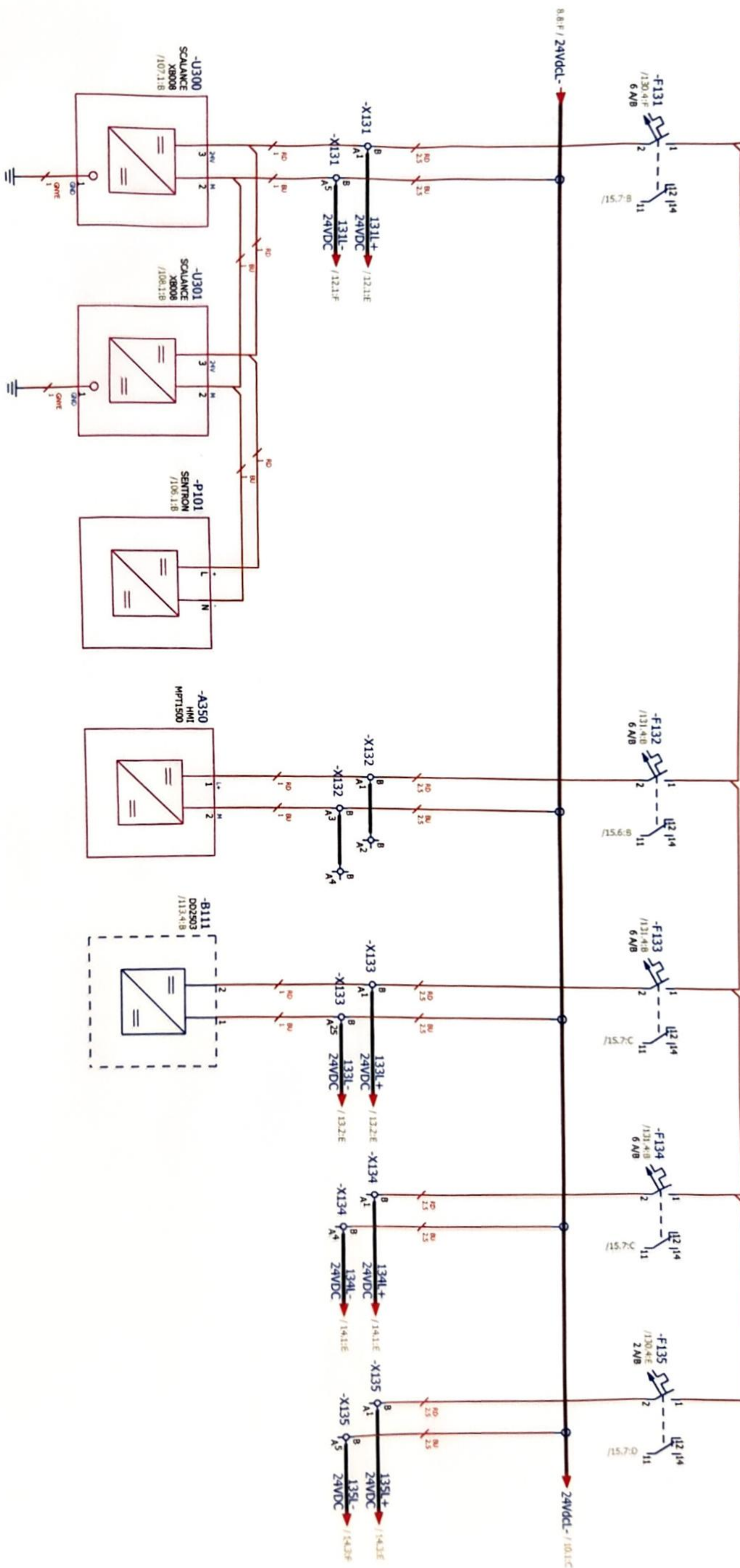
/120/4°C

24VDC+

IP DISTRIBUTION BLOCK

8.8.F / 24Vdcl1 + 

*Note: Pin busbar SST3733





SYNCHRONIZATION
CONTROLAUTOMATIC
SYNCHRONIZER

ALARM ANNUNCIATOR

SPARE

SPARE

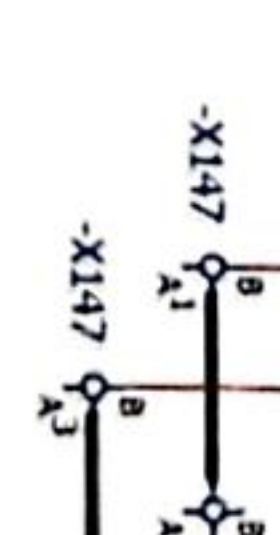
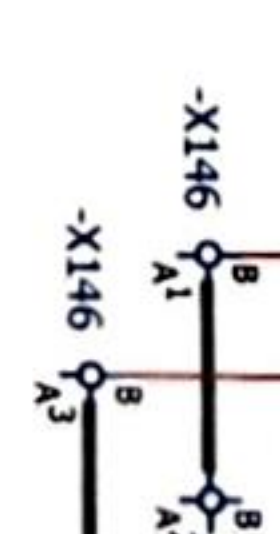
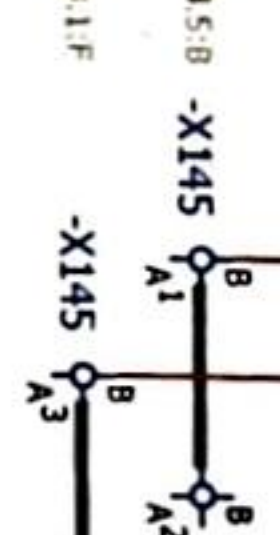
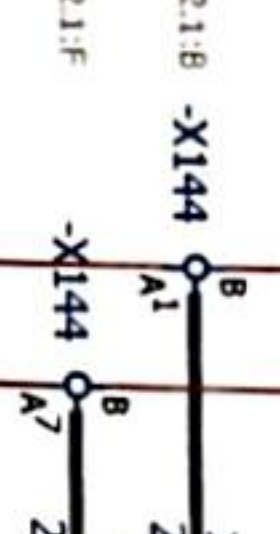
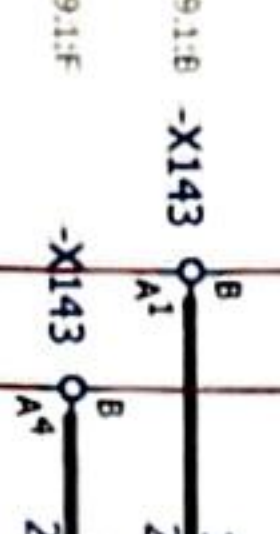
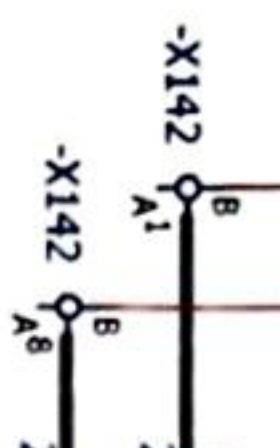
SPARE

-U52
/120+/-C
24VDC+

1P DISTRIBUTION BLOCK

10.8 A / 24Vdcl+ 24Vdcl+ 1

*Note: Pin busbar SST3733



GND 2.5

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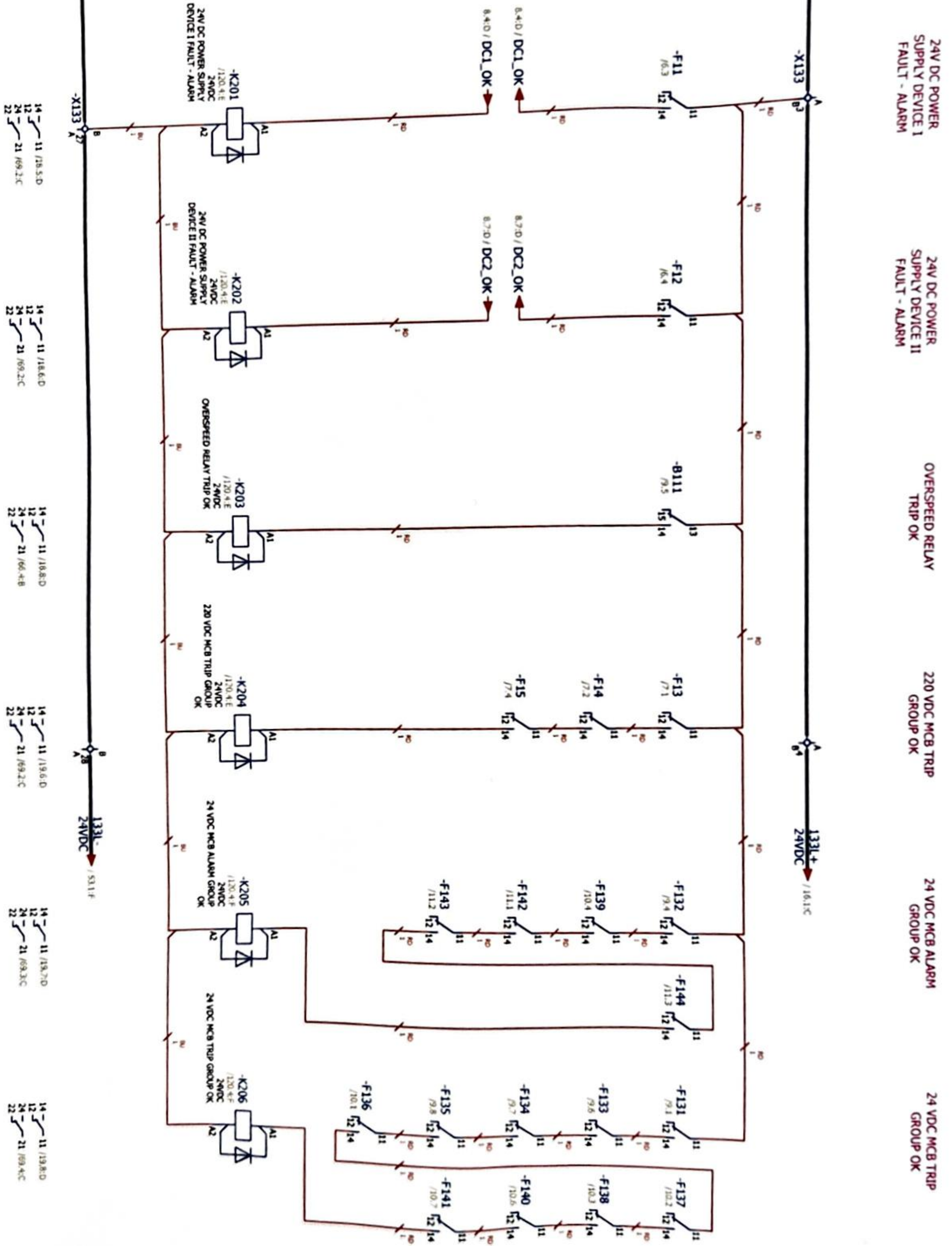
Approved by Contractor
Approved by Designer
Checked by
Drawn byGrga Kolić
Antonio Aleksić
Tamara Trojčić27.7.2026.
27.7.2026.
27.7.2026.Reconstruction and Modernisation
of HPP Perućica - Phase II
PROJECT NAME

EMPLOYER

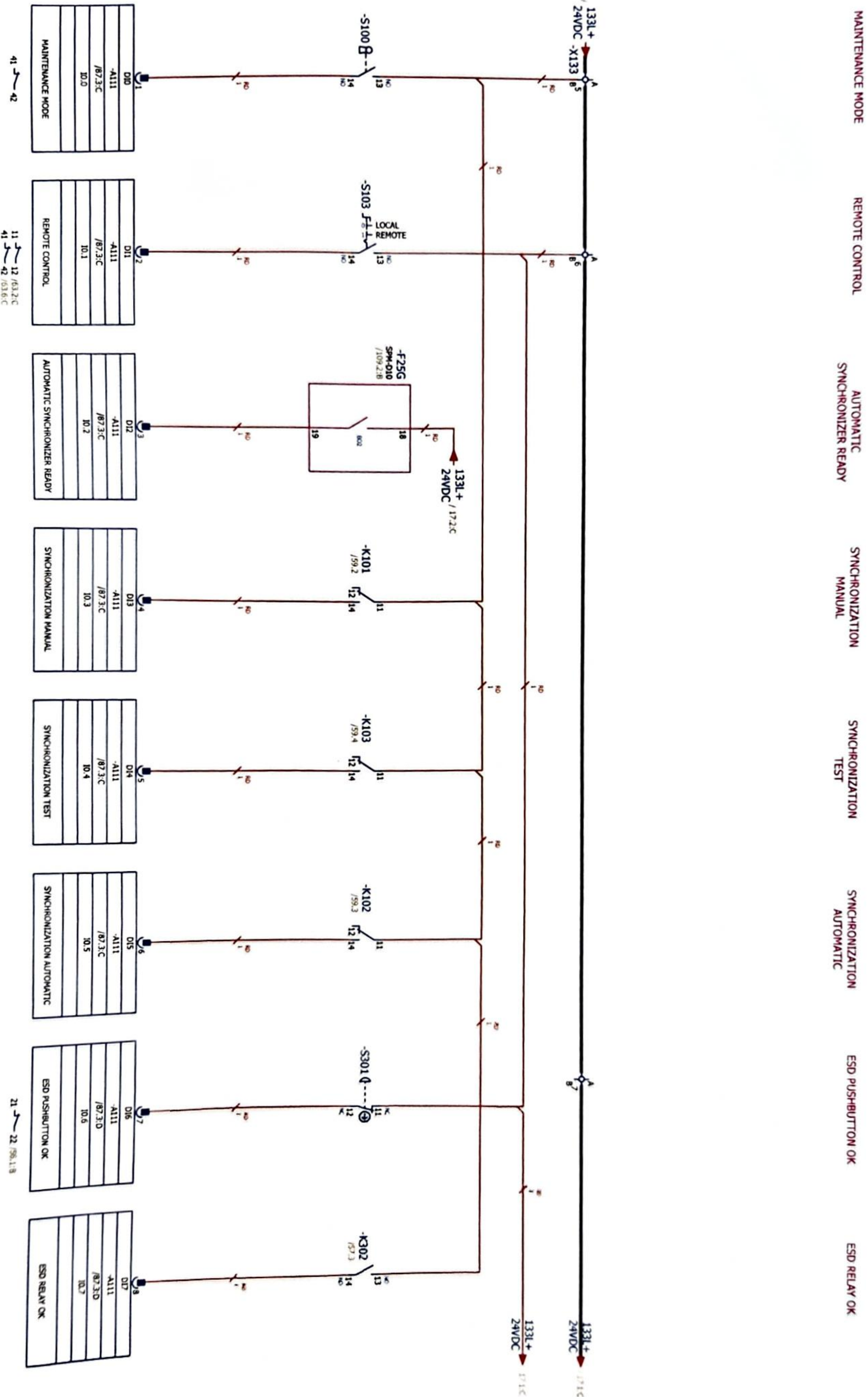
DESIGNER

Unit K02 control cabinet circuit diagrams
DC DISTRIBUTION 24VDCDRAWING No.:
PERU-01-19-170-31-002
Designer No.:REV No.:
02
SALES No.:= UBCIA10GH100
+ UBCIA10
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PAGE

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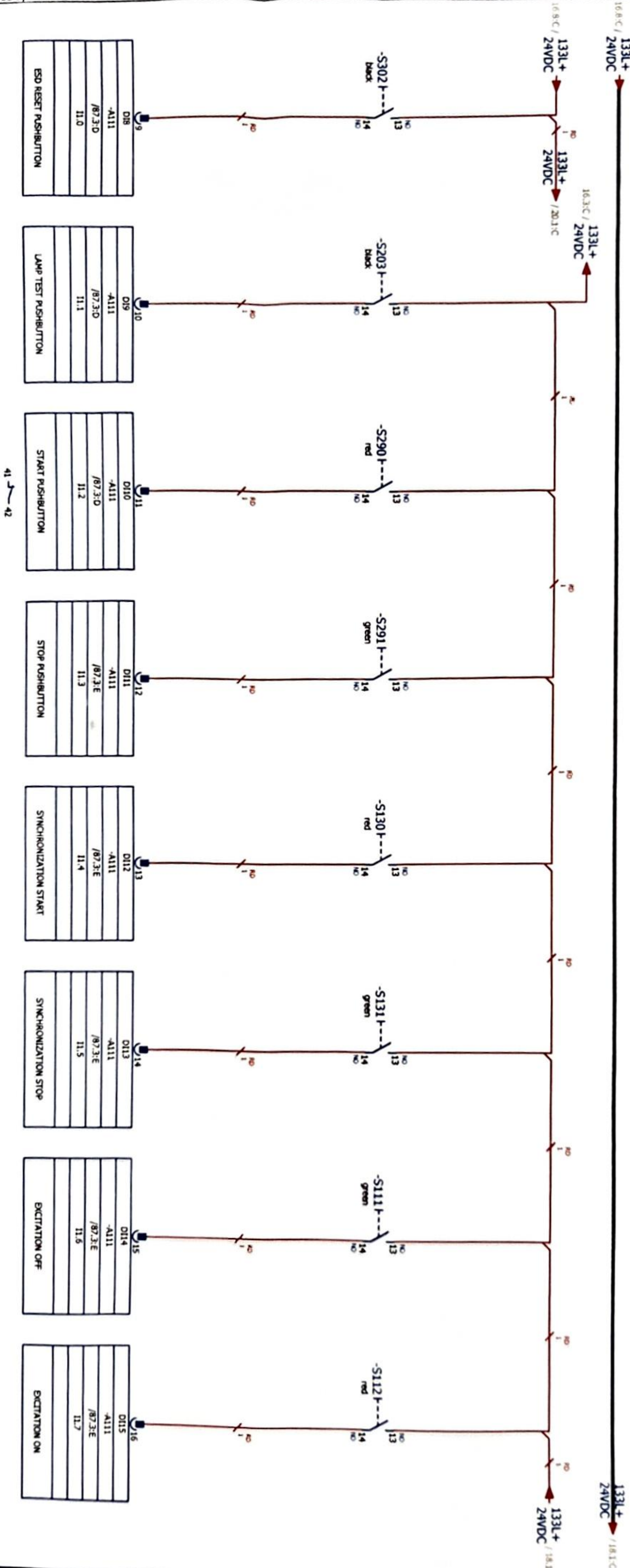


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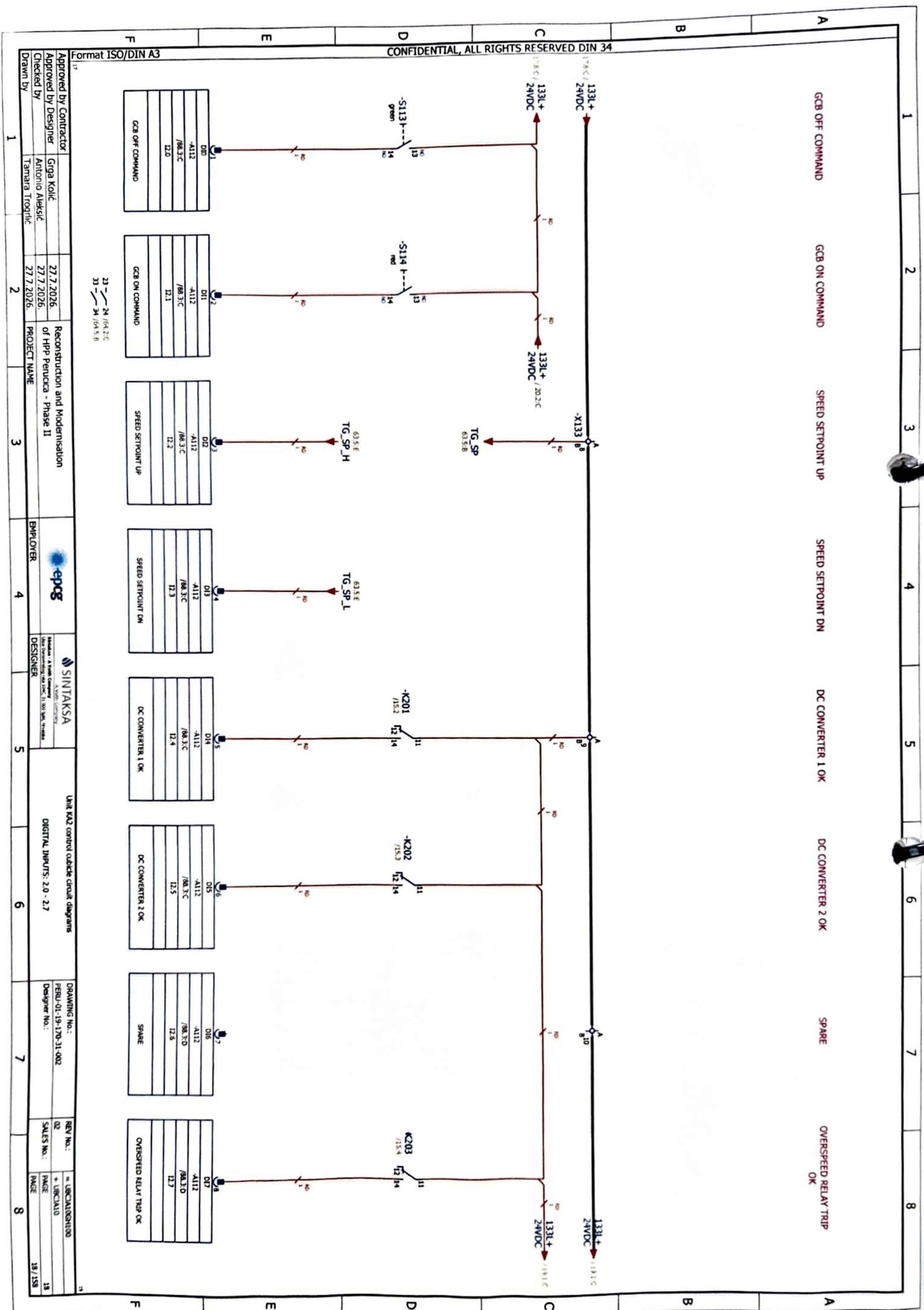


Approved by Contractor	Grga Kolic	27.7.2026.	Reconstruction and Modernisation	EMPLOYER	epcg	Unit KA2 control outside circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCIA10GHT100
Checked by Designer	Antonio Aleksić	27.7.2026.	of HPP Perucka - Phase II	DESIGNER	SINTAKSA	DIGITAL INPUTS: 0.0 - 0.7	Designer No.:	SALES No.:	+ UBCIA10
Drawn by	Tamara Trogljić	27.7.2026.	PROJECT NAME					PAGE	16
1	2	3	4	5	6	7	8		16 / 158

ESD RESET PUSHBUTTON LAMP TEST PUSHBUTTON START PUSHBUTTON STOP PUSHBUTTON SYNCHRONIZATION START SYNCHRONIZATION STOP EXCITATION OFF EXCITATION ON



Approved by Contractor	Grga Kolic	27.7.2026.	Reconstruction and Modernisation of HPP Peruca - Phase II	EMPLOYER	epcg	SINTAKSA	Unit KAZ control cabinet circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	SALES No.: + UBCA10
Checked by	Antonio Aleksic	27.7.2026.		DESIGNER			DIGITAL INPUTS: 1.0 - 1.7	Designer No.:		PAGE
Drawn by	Tamara Trogrlic	27.7.2026.	PROJECT NAME							PAGE



GENERATOR PROTECTION
ESD RELAY TRIP

GCB SYNCHRONIZATION
NOT SELECTED FEEDBACK

TRANSFORMER CB
SYNCHRONIZATION NOT
SELECTED FEEDBACK

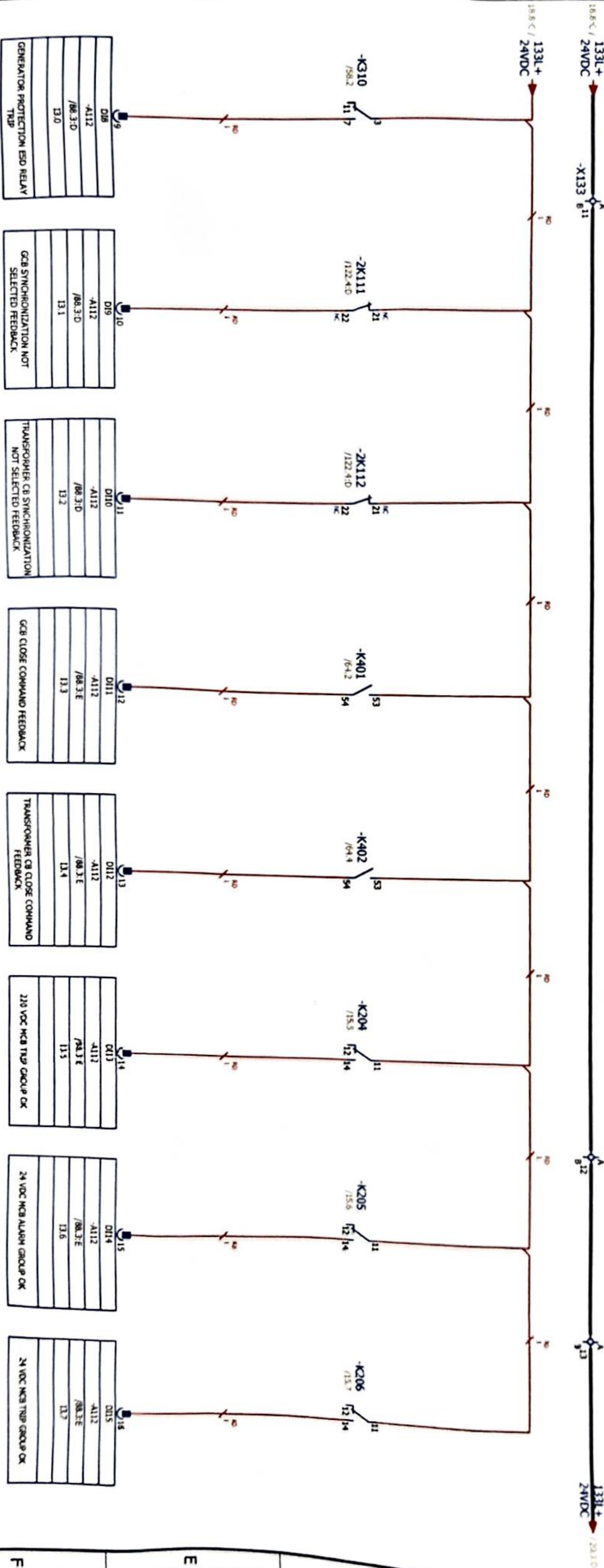
GCB CLOSE COMMAND
FEEDBACK

TRANSFORMER CB CLOSE
COMMAND FEEDBACK

220 VDC MCB TRIP GROUP
OK

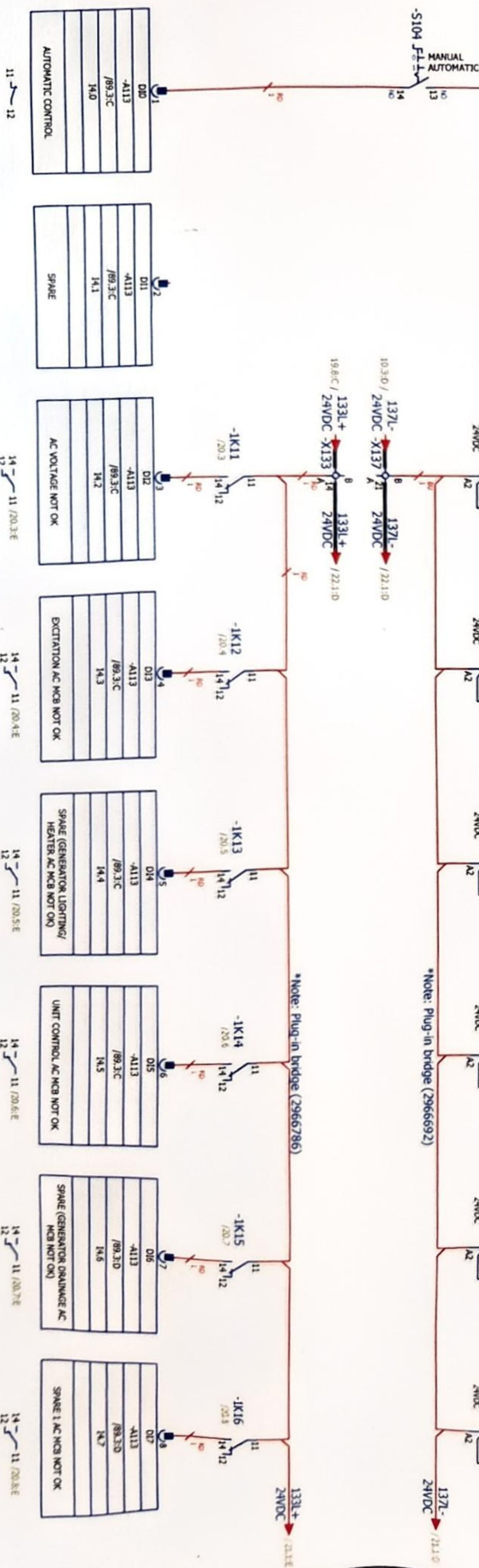
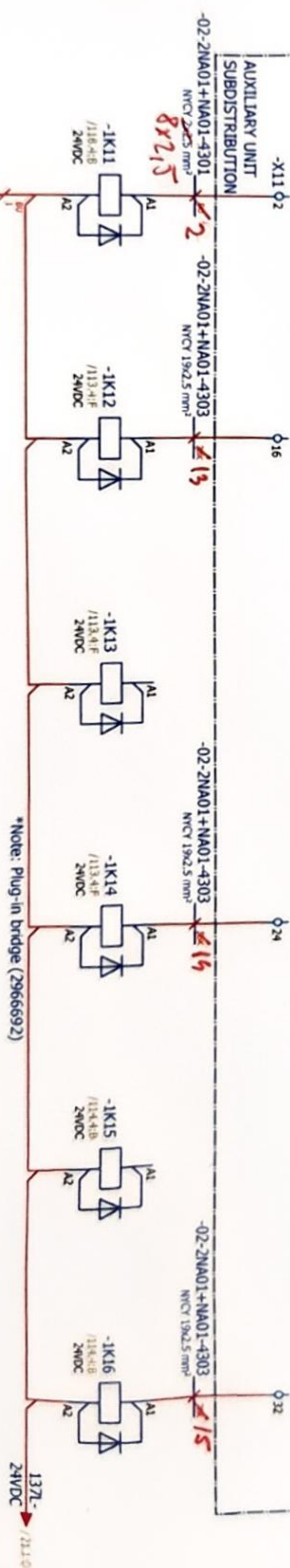
24 VDC MCB ALARM
GROUP OK

24 VDC MCB TRIP GROUP
OK



Format ISO/DIN A3	1		2		3		4		5		6		7		8	
Approved by Contractor	Grga Kolit		27.7.2026.		Reconstruction and Modernisation of HPP Peruca - Phase II		epog		SINTAKSA		Unit K42 control outside circuit diagrams		DRAWING No.: PERU-01-19-170-31-002		REV No.: 02	
Approved by Designer	Antonio Aleksic		27.7.2026.												PAGE + UBCA10	
Checked by	Tamara Trojic		27.7.2026.		PROJECT NAME		EMPLOYER		DESIGNER						PAGE 19 / 159	
Drawn by	1		2		3		4		5		6		7		8	

SPARE 1 AC MCB NOT OK

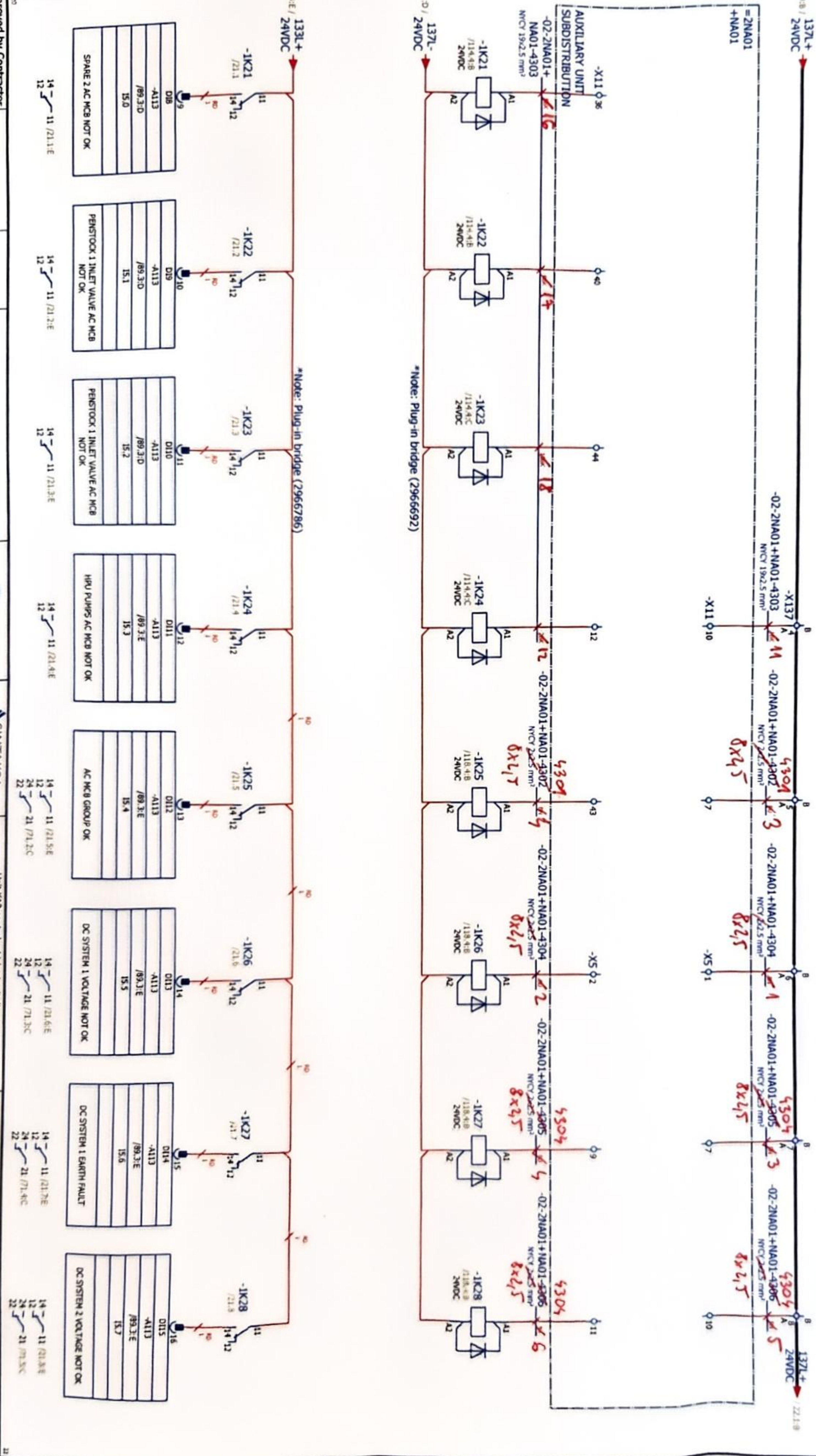


Approved by Contractor	Grga Kolić	27.7.2026.	Reconstruction and Modernisation of HPP Perućica - Phase II		NOVI CITY PERUĆICA A WHOLLY OWNED COMPANY	DIGITAL INPUTS: 4.0 - 4.7	PERU-01-19-170-31-002	02	+ URBIA10				
Approved by Designer													
Checked by	Antonio Aleksić	27.7.2026.											
Drawn by	Tamara Trogrlić	27.7.2026.											
PROJECT NAME			3	EMPLOYER	4	DESIGNER	5	6	7	8	9		
									Designer No.:	SALES NO.:	PAGE	PAGE	20 / 158

SPARE 2 AC MCB NOT OK PENSTOCK 1 INLET VALVE AC MCB NOT OK PENSTOCK 2 INLET VALVE AC MCB NOT OK HPU PUMPS AC MCB NOT OK AC MCB GROUP OK DC SYSTEM 1 VOLTAGE NOT OK DC SYSTEM 1 EARTH FAULT DC SYSTEM 2 VOLTAGE NOT OK

8x25 s 644E

Format ISO/DIN A3



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Approved by Contractor	Grga Kolić	27.7.2026.	Reconstruction and Modernisation of HPP Perućica - Phase II	EMPLOYER	SINTAKSA	Unit K02 control cabinet circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	SALES No.: 21
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	DESIGNER	Usta (Corporate) Ltd. (JSC) 21 000 Split, Croatia	DIGITAL INPUTS: 5.0 - 5.7	Designer No.:	+ URCALIO	PAGE
Drawn by	Tamara Troščić	27.7.2026.							PAGE

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DC SYSTEM 2 EARTH
FAULTEXCITATION 1 DC MCB
NOT OKEXCITATION 2 DC MCB
NOT OKUNIT CONTROL 1 DC MCB
NOT OKEXCITATION EXTERNAL
INVERTER DC MCB NOT
OKUNIT CONTROL 2 DC MCB
NOT OKHPU DC PUMP MCB NOT
OKEXCITATION FIELD
FLASHING DC MCB NOT
OKAUXILIARY UNIT
SUBDISTRIBUTION-02-2NA01+
NA01-4307
NVCY 2x25 mm²
118.40
24VDC-02-2NA01+NA01-4309
NVCY 2x25 mm²
16x2.5
118.40
24VDC-02-2NA01+NA01-4309
NVCY 2x25 mm²
16x2.5
118.40
24VDC-02-2NA01+NA01-4309
NVCY 2x25 mm²
16x2.5
118.40
24VDC-02-2NA01+NA01-4309
NVCY 2x25 mm²
16x2.5
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24VDC-02-2NA01+NA01-4309
NVCY 2x25 mm²
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24VDC-02-2NA01+NA01-4309
NVCY 2x25 mm²
16x2.5
118.40
24VDC-02-2NA01+NA01-4309
NVCY 2x25 mm²
16x2.5
118.40
24VDC

*Note: Plug-in bridge (2966692)

*Note: Plug-in bridge (2966786)

DC SYSTEM 2 EARTH FAULT
11
12
14
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EXCITATION 1 DC MCB NOT OK
11
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EXCITATION 2 DC MCB NOT OK
11
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UNIT CONTROL 1 DC MCB NOT OK
11
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EXCITATION EXTERNAL INVERTER DC MCB NOT OK
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UNIT CONTROL 2 DC MCB NOT OK
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HPU DC PUMP MCB NOT OK
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EXCITATION FIELD FLASHING DC MCB NOT OK
11
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Format ISO/DIN A3

14 11 /22.1.E
12 24 21 /71.6.C
2214 11 /22.1.E
12 24 21 /71.6.C
2214 11 /22.1.E
12 24 21 /71.6.C
2214 11 /22.1.E
12 24 21 /71.6.C
2214 11 /22.1.E
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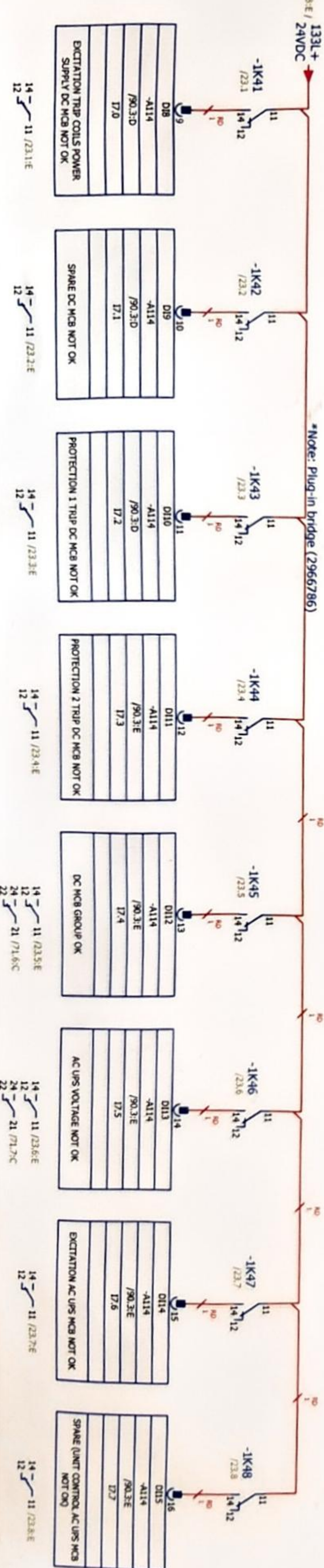
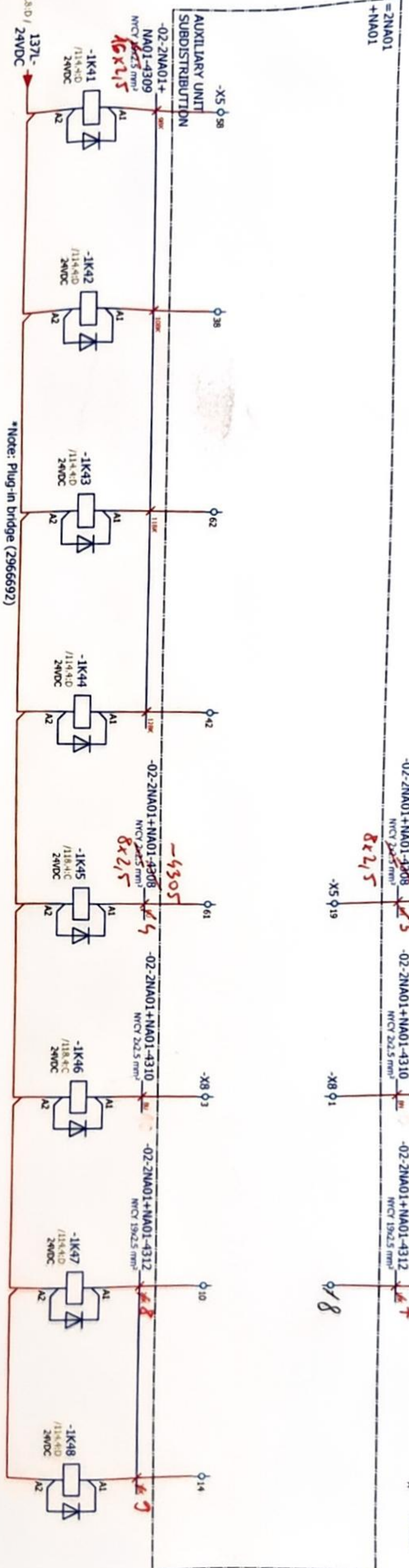
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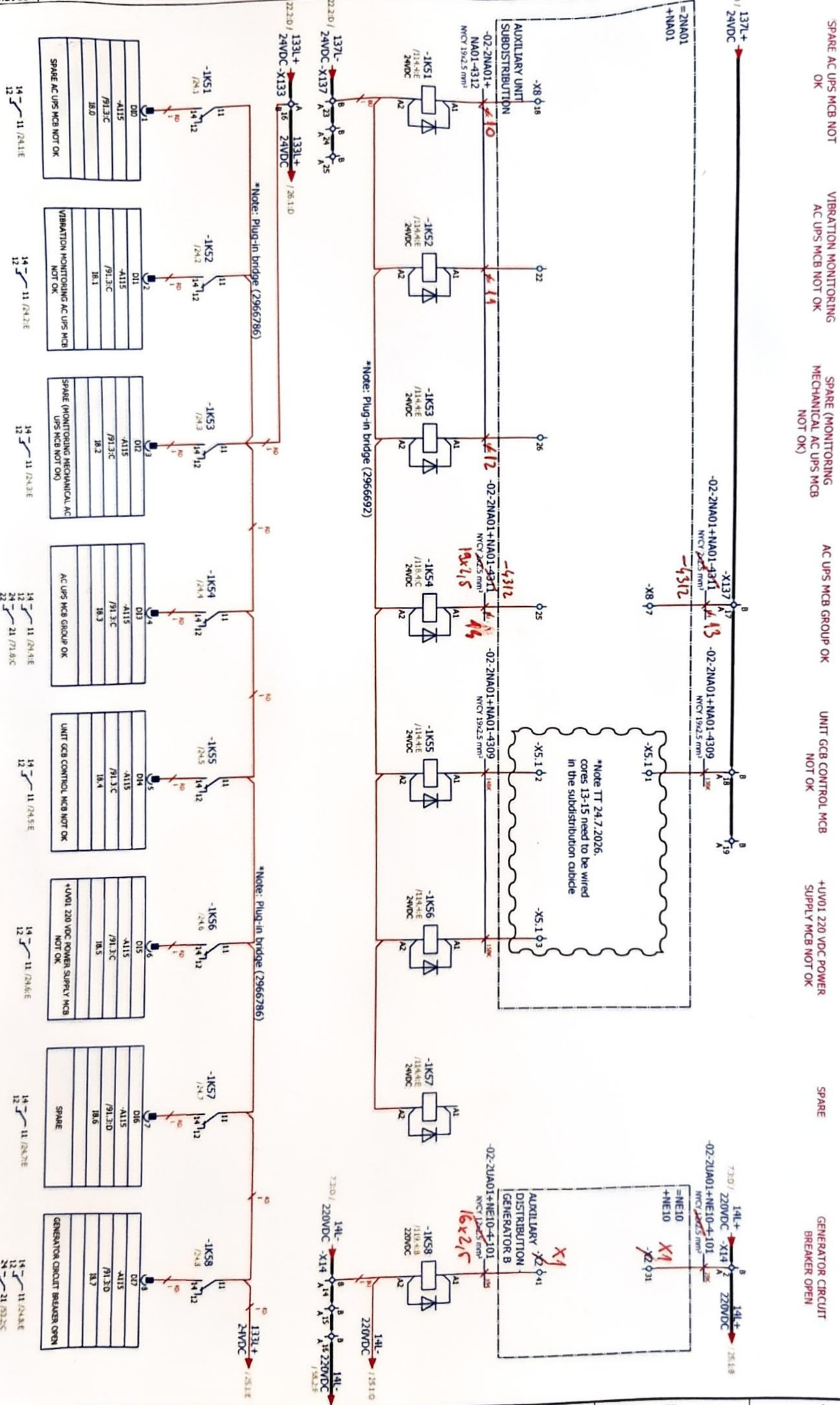
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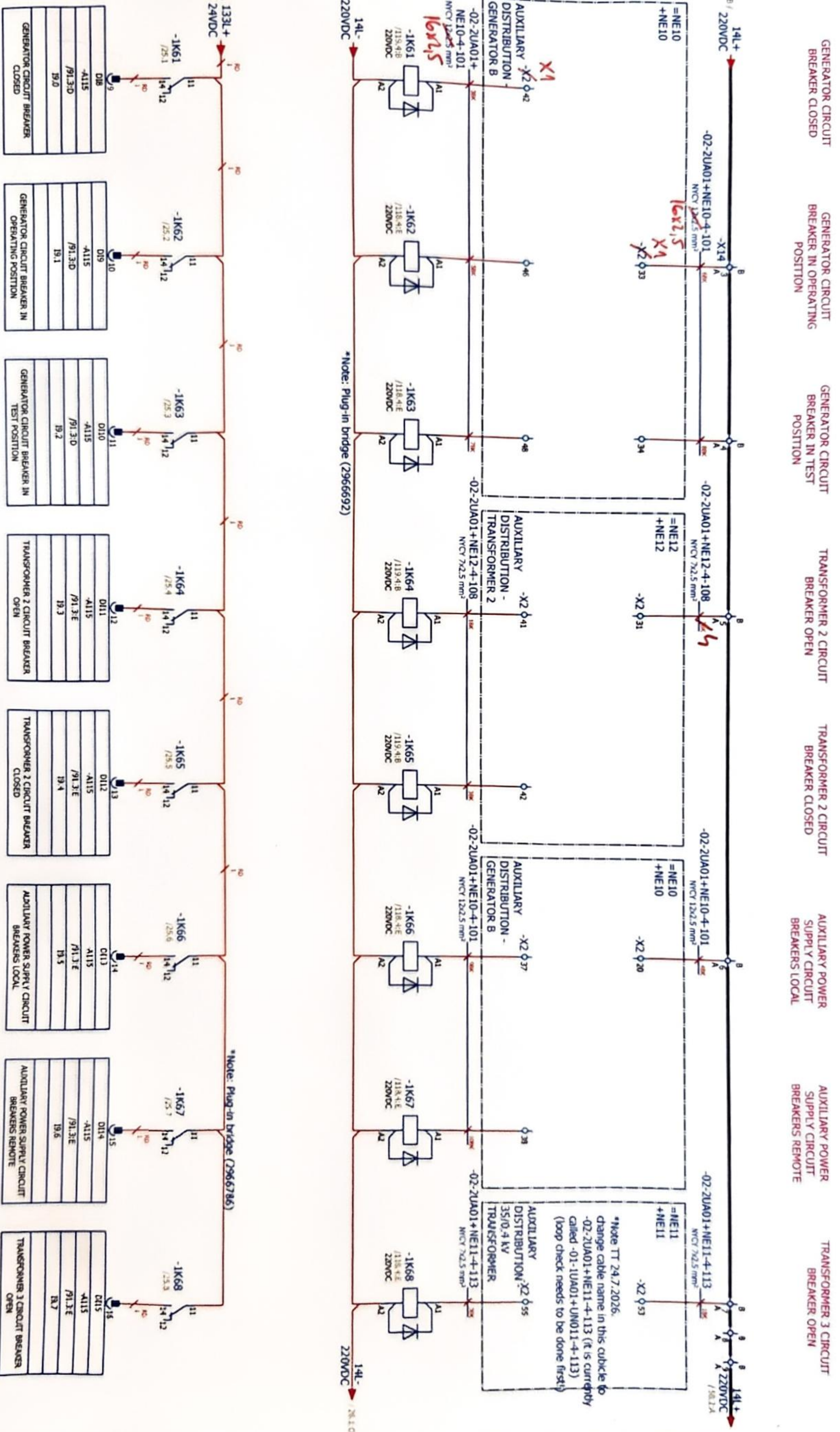
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Approved by Contractor		Reconstruction and Modernisation of HPP Perutka - Phase II		 A VITA COMPANY Marketa - A VITA Company Ulica Podzemlje 100, 1000 Ljubljana, Slovenia	Unit K02 control cabinet circuit diagrams	DIGITAL INPUTS: 7.0 - 7.7	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCAL06H100 + UBCAL10	
Approved by Designer	Grga Kolik									27.7.2026.
Checked by	Antonio Aleksić									27.7.2026.
Drawn by	Tamara Trogrlić									27.7.2026.
1	2	3	4	5	6	7	8			

[illegible]



14- 11 / 25.1 E
12- 21 / 53.3 C
22

14- 11 / 25.2 E
12- 21 / 53.3 C
22

14- 11 / 25.3 E
12- 21 / 53.3 C
22

14- 11 / 25.4 E
12- 21 / 53.4 C
22

14- 11 / 25.5 E
12- 21 / 53.5 C
22

14- 11 / 25.6 E
12- 21 / 53.6 C
22

14- 11 / 25.7 E
12- 21 / 53.7 C
22

14- 11 / 25.8 E
12- 21 / 53.8 C
22

GENERATOR CIRCUIT BREAKER CLOSED
DB8
-A115
/91.3.0
19.0
GENERATOR CIRCUIT BREAKER CLOSED

GENERATOR CIRCUIT BREAKER IN OPERATING POSITION
DB9
-A115
/91.3.0
19.1
GENERATOR CIRCUIT BREAKER IN OPERATING POSITION

GENERATOR CIRCUIT BREAKER IN TEST POSITION
DB10
-A115
/91.3.0
19.2
GENERATOR CIRCUIT BREAKER IN TEST POSITION

TRANSFORMER 2 CIRCUIT BREAKER OPEN
DB11
-A115
/91.3.E
19.3
TRANSFORMER 2 CIRCUIT BREAKER OPEN

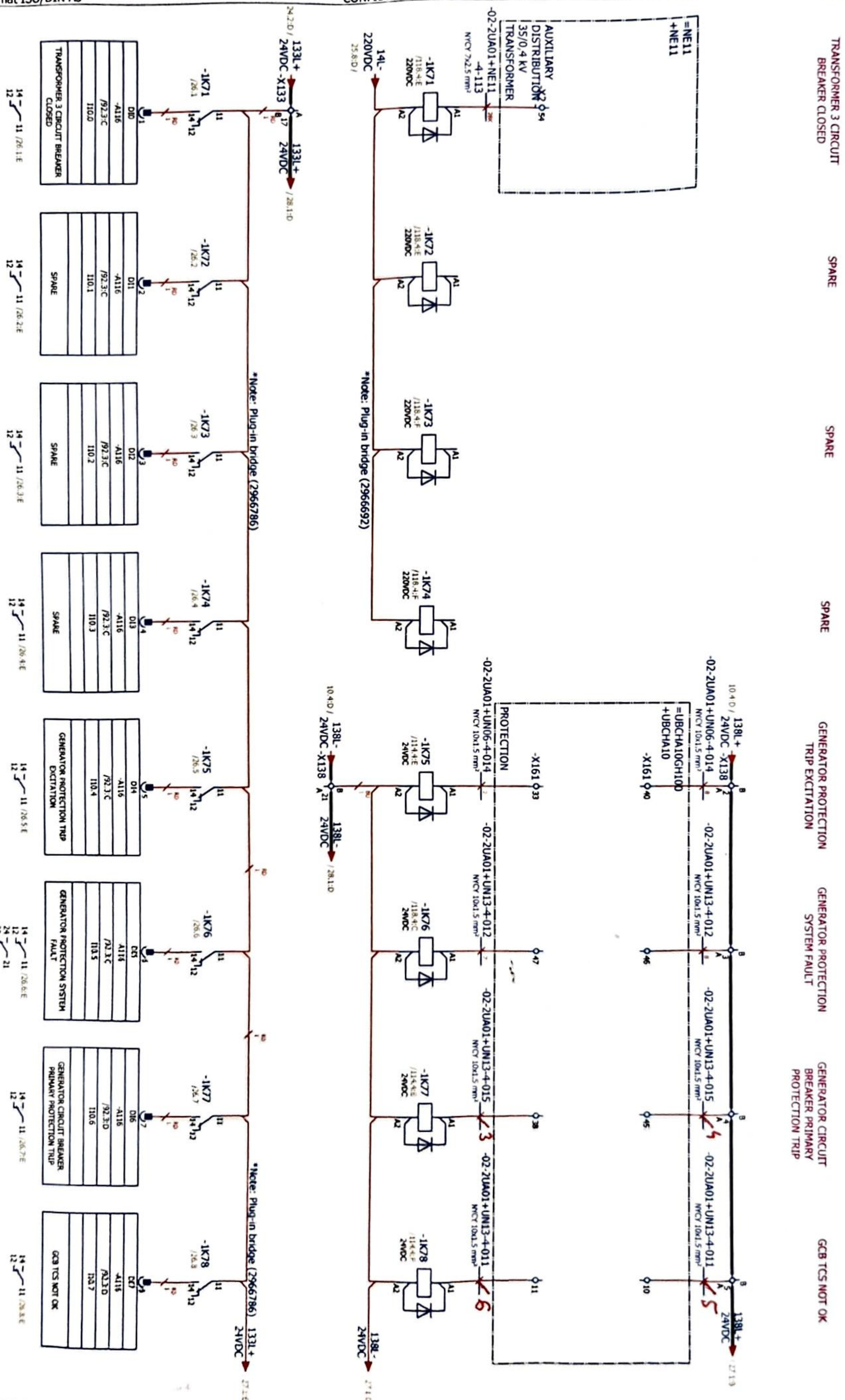
TRANSFORMER 2 CIRCUIT BREAKER CLOSED
DB12
-A115
/91.3.E
19.4
TRANSFORMER 2 CIRCUIT BREAKER CLOSED

AUXILIARY POWER SUPPLY CIRCUIT BREAKERS LOCAL
DB13
-A115
/91.3.E
19.5
AUXILIARY POWER SUPPLY CIRCUIT BREAKERS LOCAL

AUXILIARY POWER SUPPLY CIRCUIT BREAKERS REMOTE
DB14
-A115
/91.3.E
19.6
AUXILIARY POWER SUPPLY CIRCUIT BREAKERS REMOTE

TRANSFORMER 3 CIRCUIT BREAKER OPEN
DB15
-A115
/91.3.E
19.7
TRANSFORMER 3 CIRCUIT BREAKER OPEN

Approved by Contractor	27.7.2026.	Reconstruction and Modernisation	EMPLOYER	SINTAKSA	Unit KAZ control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02
Approved by Designer	27.7.2026.	of HPP Perucca - Phase II	DESIGNER	Unit KAZ control cubicle circuit diagrams	DIGITAL INPUTS: 9.0 - 9.7	Designer No.: 25	SALES No.: 25
Checked by	Tamara Trojic	PROJECT NAME	4	5	6	7	8
Drawn by	1	2	3	4	5	6	7



Approved by Contractor
Approved by Designer
Checked by
Drawn by

27.7.2026.
27.7.2026.
27.7.2026.

Reconstruction and Modernisation of HPP Perucca - Phase II
PROJECT NAME



EMPLOYER

SINTAKSA
A Voth Company
Skutumpah - A Voth Company
Utara, Corresponding Area 1000, 21 000 1000, 1000
DESIGNER

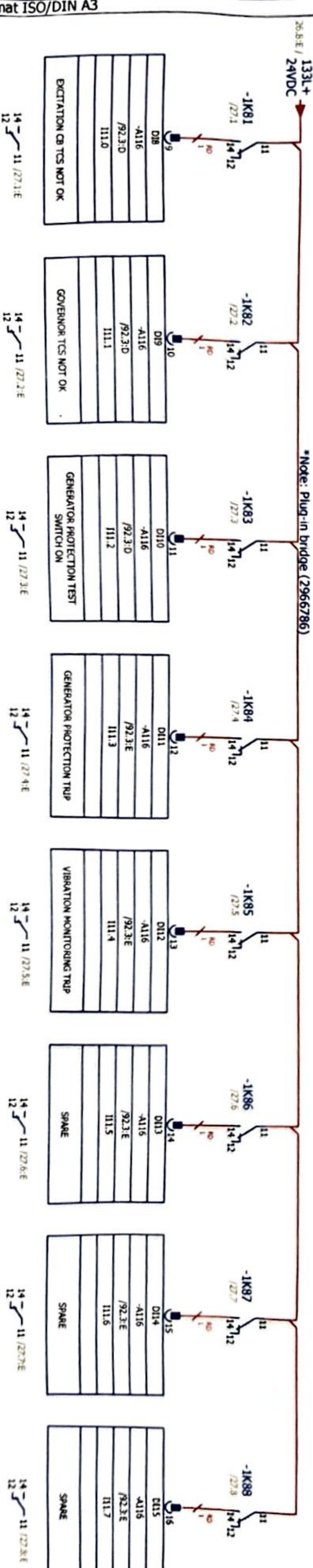
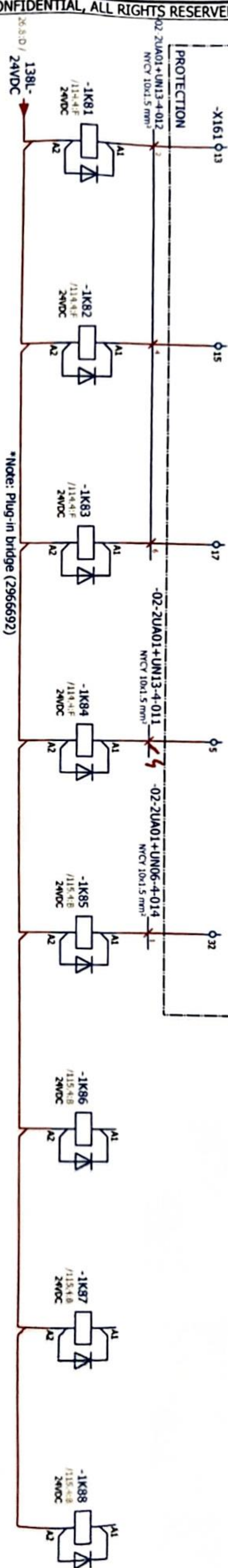
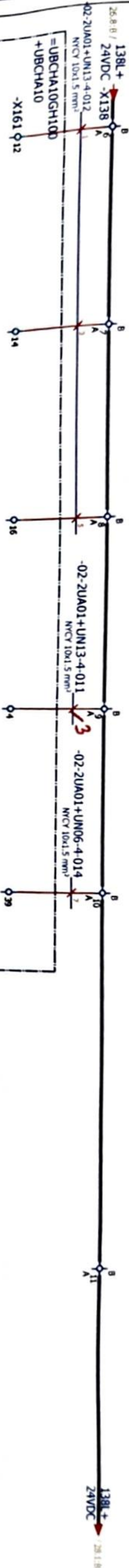
DIGITAL INPUTS: 10.0 - 10.7

DRAWING NO.:
 PERU-01-19-170-31-002
 Designer No.:
 7

SALES NO.	02
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CONCERNED PARTY	
+ UBCALD	
PAGE	26
PAGE	26 / 158

EXCITATION CB TCS NOT OK GOVERNOR TCS NOT OK GENERATOR PROTECTION TEST SWITCH ON GENERATOR PROTECTION TRIP VIBRATION MONITORING TRIP SPARE SPARE SPARE



EXCITATION CB TCS NOT OK	GOVERNOR TCS NOT OK	GENERATOR PROTECTION TEST SWITCH ON	GENERATOR PROTECTION TRIP	VIBRATION MONITORING TRIP	SPARE	SPARE	SPARE
DI8 -A116 /92.3.0 111.0	DI9 -A116 /92.3.0 111.1	DI10 -A116 /92.3.0 111.2	DI11 -A116 /92.3.E 111.3	DI12 -A116 /92.3.E 111.4	DI13 -A116 /92.3.E 111.5	DI14 -A116 /92.3.E 111.6	DI15 -A116 /92.3.E 111.7

14 11 /27.1.E 14 11 /27.2.E 14 11 /27.3.E 14 11 /27.4.E 14 11 /27.5.E 14 11 /27.6.E 14 11 /27.7.E 14 11 /27.8.E

Approved by Contractor: Grđa Koljć 27.7.2026. Reconstruction and Modernisation of HPP Peručka - Phase II EMPLOYER: epog SINTAKSA Unit KAZ control cabinet circuit diagrams DRAWING No.: PERU-01-19-170-31-002 REV No.: 02 SALES No.: PAGE: 27

Checked by: Antonio Aleksić 27.7.2026. PROJECT NAME: DESIGNER: DIGITAL INPUTS: 11.0 - 11.7 Designer No.: SALES No.: PAGE: 27

Drawn by: Tamara Trogrlić 27.7.2026. PROJECT NAME: DESIGNER: DIGITAL INPUTS: 11.0 - 11.7 Designer No.: SALES No.: PAGE: 27

Format ISO/DIN A3

SPARE
SPARE
SPARE
SPARE
SPARE
SPARE

SPARE
SPARE
SPARE
SPARE
SPARE
SPARE

EXCITATION CIRCUIT BREAKER OPEN
EXCITATION CIRCUIT BREAKER OPEN
EXCITATION CIRCUIT BREAKER OPEN
EXCITATION CIRCUIT BREAKER OPEN
EXCITATION CIRCUIT BREAKER OPEN
EXCITATION CIRCUIT BREAKER OPEN

EXCITATION CIRCUIT BREAKER CLOSED
EXCITATION CIRCUIT BREAKER CLOSED
EXCITATION CIRCUIT BREAKER CLOSED
EXCITATION CIRCUIT BREAKER CLOSED
EXCITATION CIRCUIT BREAKER CLOSED
EXCITATION CIRCUIT BREAKER CLOSED

EXCITATION ESD PUSHBUTTON PRESSED
EXCITATION ESD PUSHBUTTON PRESSED
EXCITATION ESD PUSHBUTTON PRESSED
EXCITATION ESD PUSHBUTTON PRESSED
EXCITATION ESD PUSHBUTTON PRESSED
EXCITATION ESD PUSHBUTTON PRESSED

EXCITATION THYRISTRON CRITICAL
EXCITATION THYRISTRON CRITICAL
EXCITATION THYRISTRON CRITICAL
EXCITATION THYRISTRON CRITICAL
EXCITATION THYRISTRON CRITICAL
EXCITATION THYRISTRON CRITICAL

EXCITATION 24 V SIGNAL VOLTAGE
EXCITATION 24 V SIGNAL VOLTAGE
EXCITATION 24 V SIGNAL VOLTAGE
EXCITATION 24 V SIGNAL VOLTAGE
EXCITATION 24 V SIGNAL VOLTAGE
EXCITATION 24 V SIGNAL VOLTAGE

EXCITATION ALARM OK
EXCITATION ALARM OK
EXCITATION ALARM OK
EXCITATION ALARM OK
EXCITATION ALARM OK
EXCITATION ALARM OK

14 11 / 28.1 E
12

14 11 / 28.2 E
12

14 11 / 28.3 E
12

14 11 / 28.4 E
12

14 11 / 28.5 E
12

14 11 / 28.6 E
12 24 21

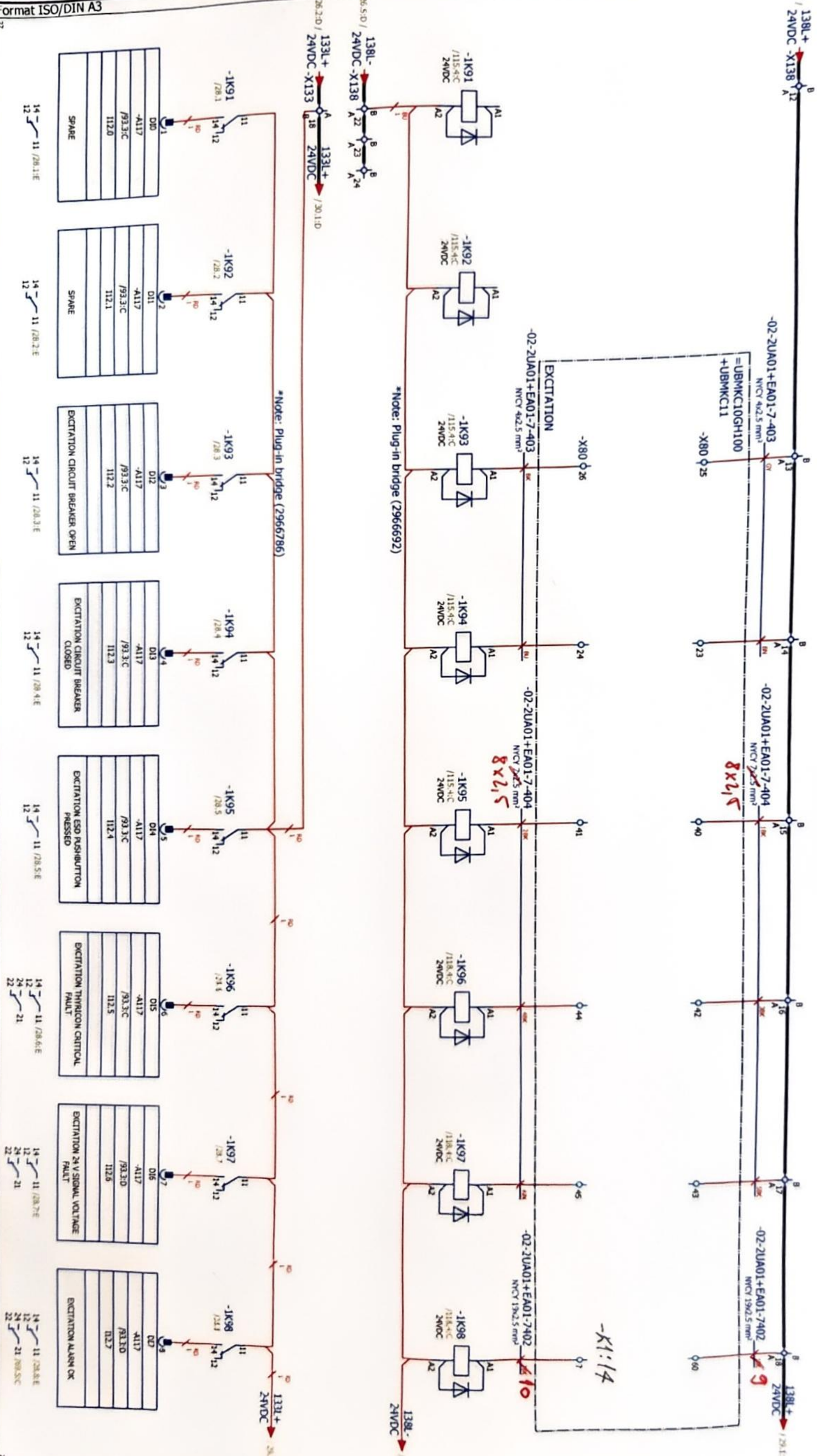
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12 24 21

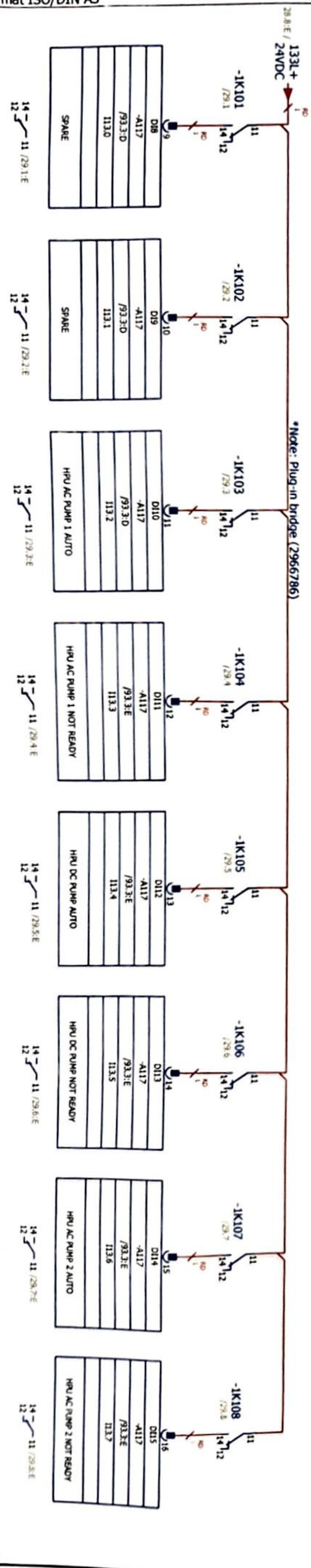
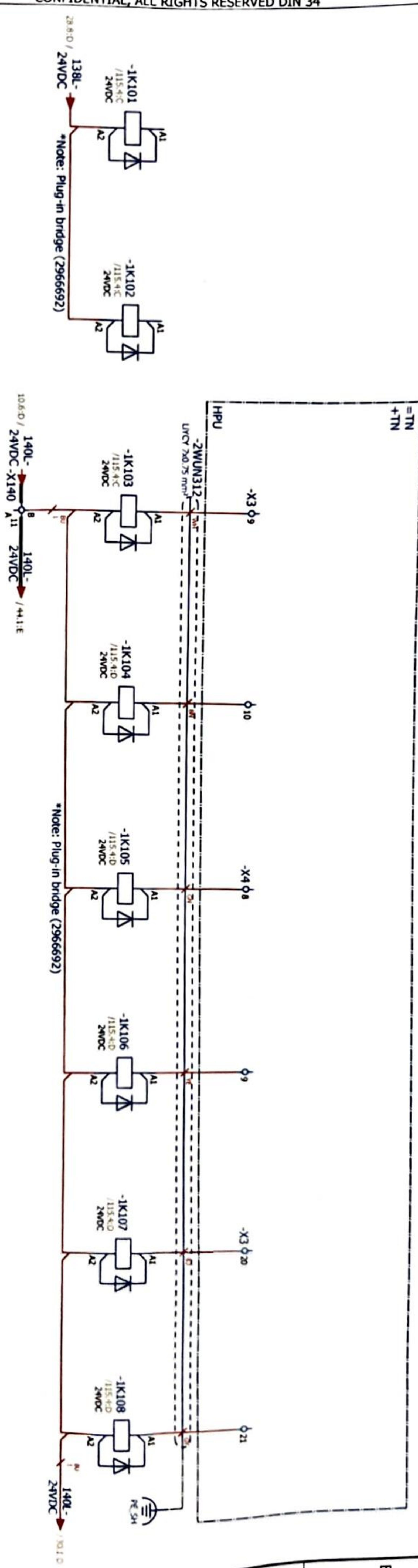
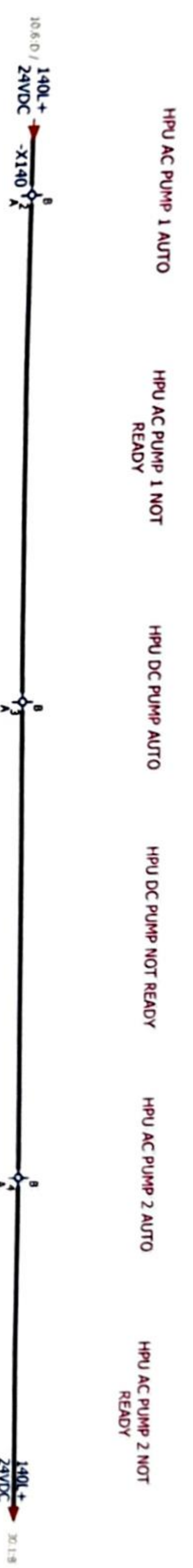
14 11 / 28.8 E
12 24 21 22 23 24 25 26 27 28 29

*Note: Plug-in bridge (2966786)

*Note: Plug-in bridge (2966692)

Approved by Contractor	Grga Kolic	27.7.2026.	Reconstruction and Modernisation	 	Unit K02 control cabinet circuit diagrams	DRAWING NO.: PERU-01-19-170-31-002	REV NO.: 02	SALES NO.: + UBMCAL100	
Checked by Designer	Antonio Aleksic	27.7.2026.	of HPP Penca - Phase II		DESIGNER	DIGITAL INPUTS: 12.0 - 12.7	Designer No.:		PAGE 28
Drawn by	Tamara Trojicic	27.7.2026.	PROJECT NAME		EMPLOYER				PAGE 28 / 158





Approved by Contractor	Giga Kolic	27.7.2026.	Reconstruction and Modernisation of HPP Perucica - Phase II	EMPLOYER	SINTAKSA	Unit K42 control cabinet circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	SALES No.: + UBCA10	PAGE 29
Checked by Designer	Antonio Aleksić	27.7.2026.		DESIGNER	epcg	DIGITAL INPUTS: 13.0 - 13.7	Designer No.:			
Drawn by	Tamara Trogrlić	27.7.2026.	PROJECT NAME							

Format ISO/DIN A3

~~MIN CLOSED~~
SPARE

~~SPARE~~
MIN CLOSED

MTV OPEN

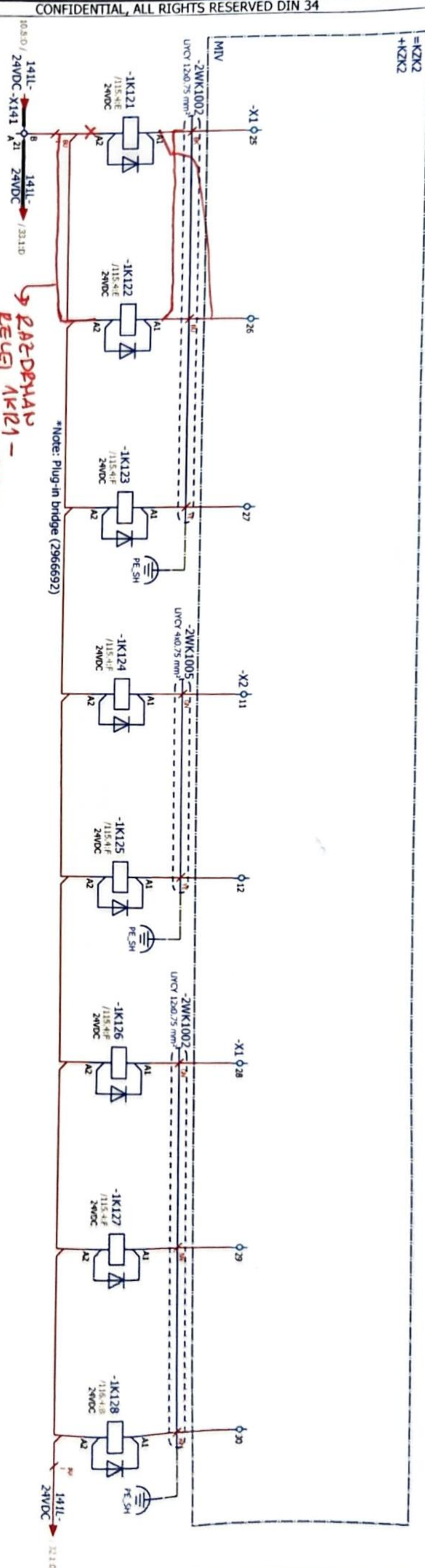
BYPASS VALVE OPEN

BYPASS VALVE CLOSED

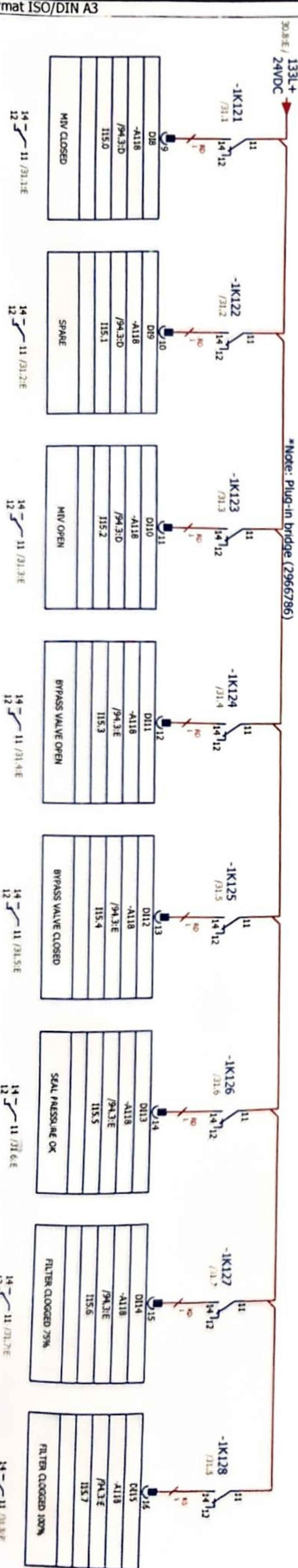
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FILTER CLOGGED 75%

FILTER CLOGGED 100%

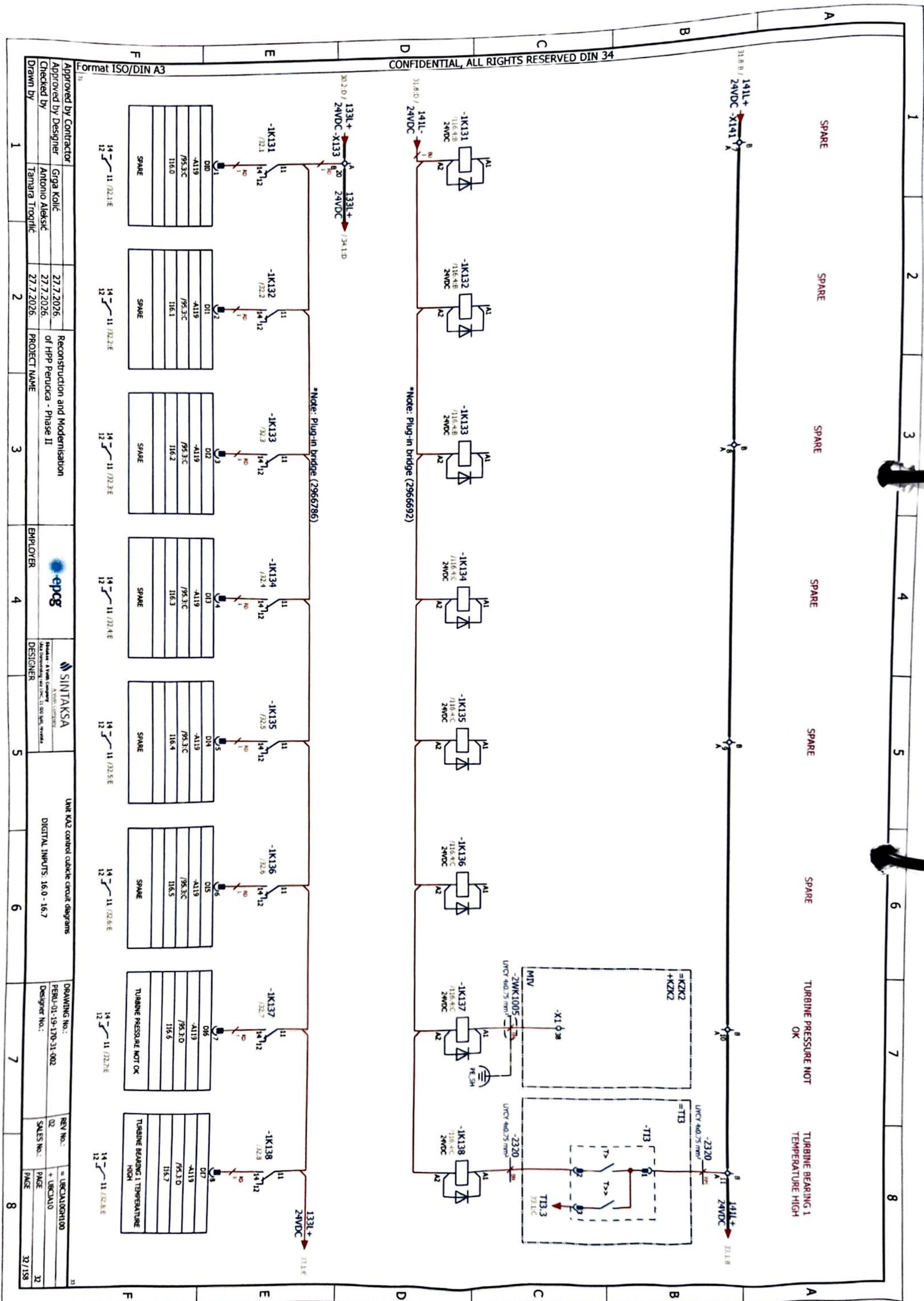


PAZDEMAN
LEUQ KR1-
SALO JE TIRALO



5

Approved by Contractor		Reconstruction and Modernisation of HPP Perucica - Phase II		 SINTAKSA A VOB Contractor Strada 7 - 4 Vahid Company 1600 Chondroghilly Road, PMB 11, 900 Sgo., Nevada	Unit K22 control cable circuit diagrams		DRAWING No.:	REV No.:	= UBCAL0GH1.00 + UBCAL10
Approved by Designer	Griga Koilk				27.7.2026.	DIGITAL INPUTS: 15.0 - 15.7		PERU-01-19-170-31-002	
Checked by	Antonio Aleksić				27.7.2026.	DESIGNER	DESIGNER No.:	SALES No.:	PAGE
Drawn by	Tamara Trogrlić				27.7.2026.	EMPLOYER			31



TURBINE BEARING 1
TEMPERATURE HIGH
HIGH

TURBINE BEARING 2
TEMPERATURE HIGH
HIGH

TURBINE BEARING 2
TEMPERATURE HIGH
HIGH

GENERATOR BEARING 1
TEMPERATURE HIGH
HIGH

GENERATOR BEARING 1
TEMPERATURE HIGH
HIGH

GENERATOR BEARING 2
TEMPERATURE HIGH
HIGH

GENERATOR BEARING 2
TEMPERATURE HIGH
HIGH

SPARE

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Format ISO/DIN A3

14 11 / 33.1E
12

14 11 / 33.2E
12

14 11 / 33.3E
12

14 11 / 33.4E
12

14 11 / 33.5E
12

14 11 / 33.6E
12

14 11 / 33.7E
12

14 11 / 33.8E
12

DIB	-A119	/95.3:0	117.0
TURBINE BEARING 1 TEMPERATURE HIGH HIGH			

DIB	-A119	/95.3:0	117.1
TURBINE BEARING 2 TEMPERATURE HIGH HIGH			

DIB	-A119	/95.3:0	117.2
TURBINE BEARING 2 TEMPERATURE HIGH HIGH			

DIB	-A119	/95.3:0	117.3
GENERATOR BEARING 1 TEMPERATURE HIGH HIGH			

DIB	-A119	/95.3:0	117.4
GENERATOR BEARING 1 TEMPERATURE HIGH HIGH			

DIB	-A119	/95.3:0	117.5
GENERATOR BEARING 2 TEMPERATURE HIGH HIGH			

DIB	-A119	/95.3:0	117.6
GENERATOR BEARING 2 TEMPERATURE HIGH HIGH			

DIB	-A119	/95.3:0	117.7
SPARE			

*Note: Plug-in bridge (2966786)

*Note: Plug-in bridge (2966692)

Approved by Contractor

Approved by Designer

Checked by

Drawn by

Grga Koic

Antonio Aleksić

Tamara Trojic

27.7.2026.

27.7.2026.

27.7.2026.

Reconstruction and Modernisation
of HPP Peruca - Phase II

PROJECT NAME

EMPLOYER

epcg

DESIGNER

SINTAKSA

Unit K42 control cabinet circuit diagrams

DIGITAL INPUTS: 17.0 - 17.7

DRAWING No.: PERU-01-19-170-31-002

Designer No.:

REV No.:

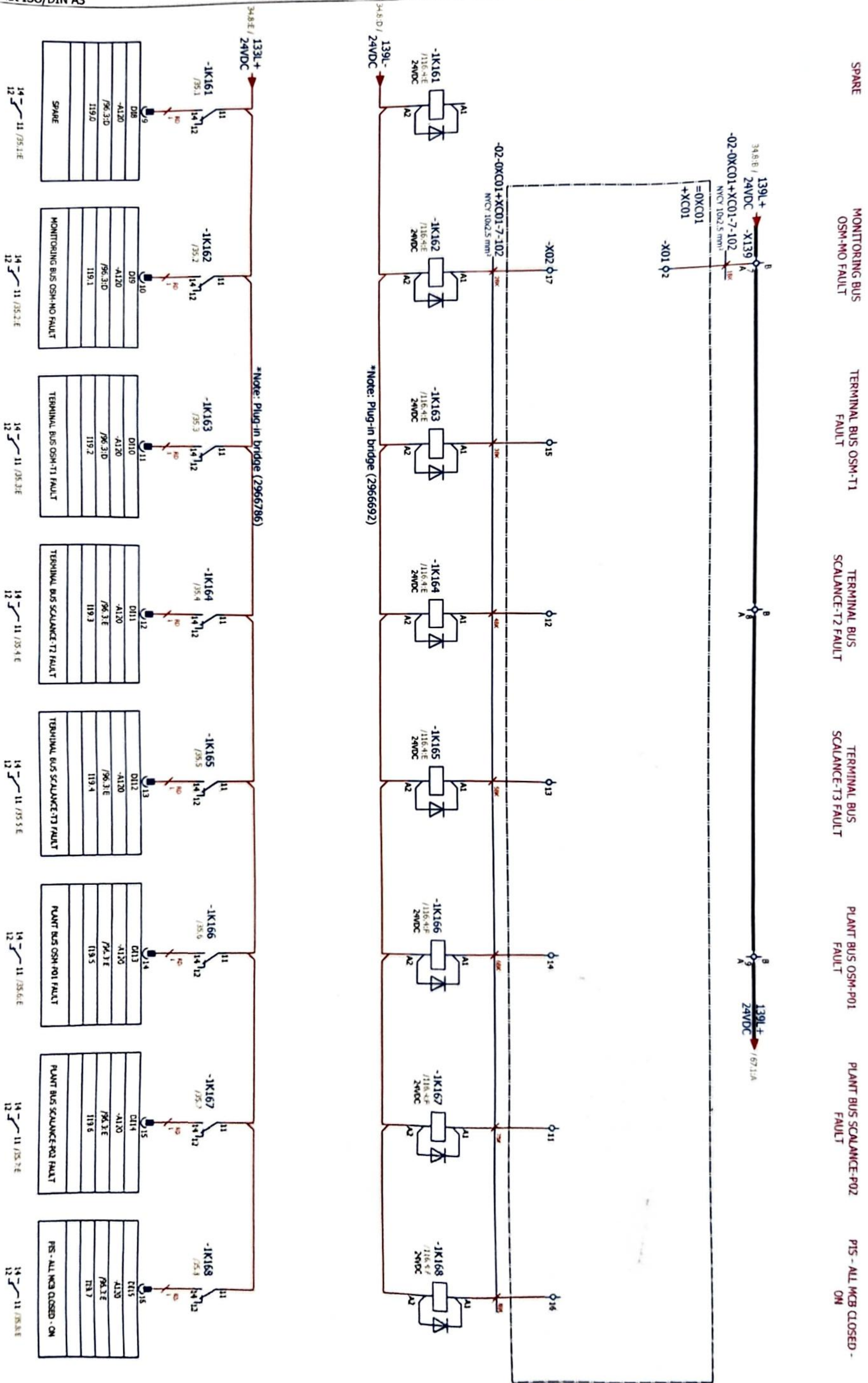
02

SALES No.:

PAGE

33

33 / 158



Approved by Contractor		Reconstruction and Modernisation of HPP Perucica - Phase II		 A 100% COMPANY SINTAKSA Standard : 8 Year Computer Udaa : Computerizing with Udaa : 21,000 Udaa, 17,000 Udaa	Unit K02 control cubicle circuit diagrams	DIGITAL INPUTS: 19.0 - 19.7	DRAWING No. :	REV No. :	= UBCA10CH100 + UBCA10		
Approved by Designer	Giga Kolić						27.7.2026.	PERU-01-19-170-31-002		02	
Checked by	Antonio Aleksić						27.7.2026.	Designer No. :		SALES No. :	35
Drawn by	Tamara Trojčić						27.7.2026.	PROJECT NAME		EMPLOYER	DESIGNER

HPU ISOLATION VALVE
CLOSE COMMAND

HPU LOAD VALVE ON
COMMAND

NOZZLE OPEN
COMMAND

NOZZLE CLOSE
COMMAND

ACTIVATE ESD VALVE

BYPASS VALVE OPEN
COMMAND

BYPASS VALVE CLOSE
COMMAND

MTV OPEN COMMAND

HPU ISOLATION VALVE CLOSE COMMAND
Q1.0
/97.3:D
-A131
DQ8

HPU LOAD VALVE ON COMMAND
Q1.1
/97.3:D
-A131
DQ9

NOZZLE OPEN COMMAND
Q1.2
/97.3:D
-A131
DQ10

NOZZLE CLOSE COMMAND
Q1.3
/97.3:E
-A131
DQ11

ACTIVATE ESD VALVE
Q1.4
/97.3:E
-A131
DQ12

BYPASS VALVE OPEN COMMAND
Q1.5
/97.3:E
-A131
DQ13

BYPASS VALVE CLOSE COMMAND
Q1.6
/97.3:E
-A131
DQ14

MTV OPEN COMMAND
Q1.7
/97.3:E
-A131
DQ15



13 - 14 /50.3:B
23 - 24 /50.3:C
33 - 34 /50.3:C
43 - 44

13 - 14 /50.4:B
23 - 24 /50.4:C
33 - 34 /50.4:C
43 - 44

13 - 14 /50.5:B
23 - 24 /50.5:C
33 - 34 /50.5:C
43 - 44

13 - 14 /50.6:B
23 - 24 /50.6:C
33 - 34 /50.6:C
43 - 44

13 - 14 /51.3:B
23 - 24 /51.3:C
33 - 34 /51.3:C
43 - 44

14 - 11 /52.7:B

14 - 11 /52.8:B

13 - 14 /52.4:B
23 - 24 /52.4:C
33 - 34 /52.4:C
43 - 44

SEAL VALVE OPEN
COMMAND

LUBRICATION PUMP ON
COMMAND

GENERATOR CIRCUIT
BREAKER OPEN
COMMAND

SPARE

SYNCHRONIZATION RUN

AUTOMATIC
SYNCHRONIZER ON
COMMAND

GCB SYNCHRONIZATION
SELECTED

TRANSFORMER CB
SYNCHRONIZATION
SELECTED

SEAL VALVE OPEN COMMAND
Q2.0
/98.3.C
-A132
DQ0

LUBRICATION PUMP ON COMMAND
Q2.1
/98.3.C
-A132
DQ1

GENERATOR CIRCUIT BREAKER OPEN COMMAND
Q2.2
/98.3.C
-A132
DQ2

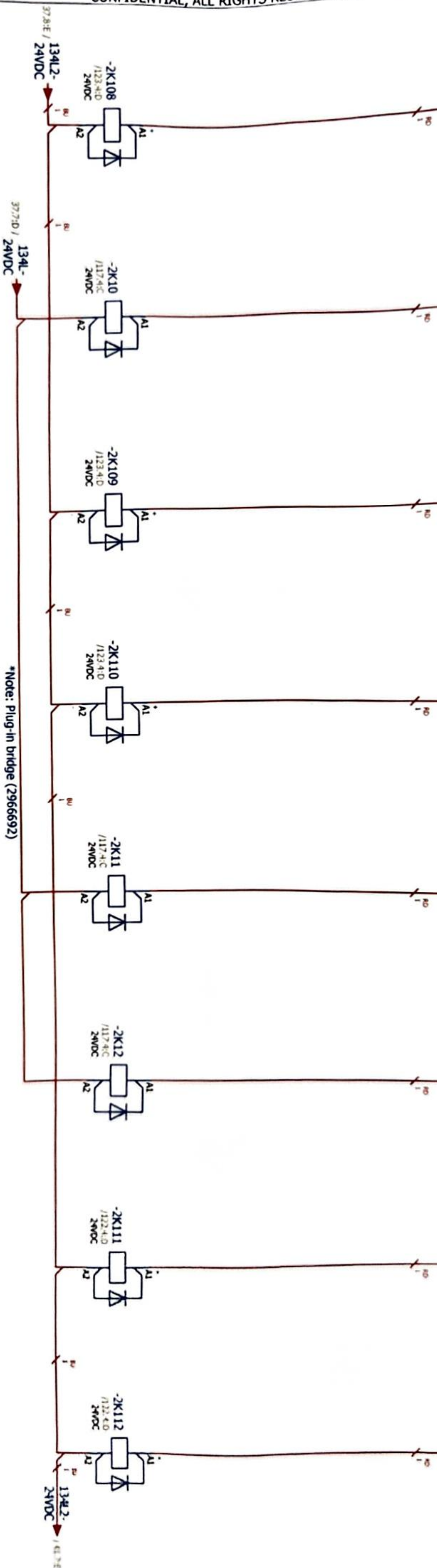
SPARE
Q2.3
/98.3.C
-A132
DQ3

SYNCHRONIZATION RUN
Q2.4
/98.3.C
-A132
DQ4

AUTOMATIC SYNCHRONIZER ON COMMAND
Q2.5
/98.3.C
-A132
DQ5

GCB SYNCHRONIZATION SELECTED
Q2.6
/98.3.D
-A132
DQ6

TRANSFORMER CB SYNCHRONIZATION SELECTED
Q2.7
/98.3.D
-A132
DQ7



*Note: Plug-in bridge (2966692)

13 - 14 /52.6.B
23 - 24 /52.6.C
33 - 34 /52.6.C
43 - 44

14 - 11 /49.6.B
12 - 11

13 - 14 /54.3.B
23 - 24 /54.3.B
33 - 34 /54.3.C
43 - 44

13 - 14
23 - 24
33 - 34
43 - 44

14 - 11 /59.7.B
12 - 11

14 - 11 /62.2.B
12 - 11

13 - 14 /62.3.B
21 - 22 /19.2.D
33 - 34 /64.2.D
43 - 44 /59.5.B

13 - 14 /62.4.B
21 - 22 /19.2.D
33 - 34 /64.2.D
43 - 44 /59.5.B

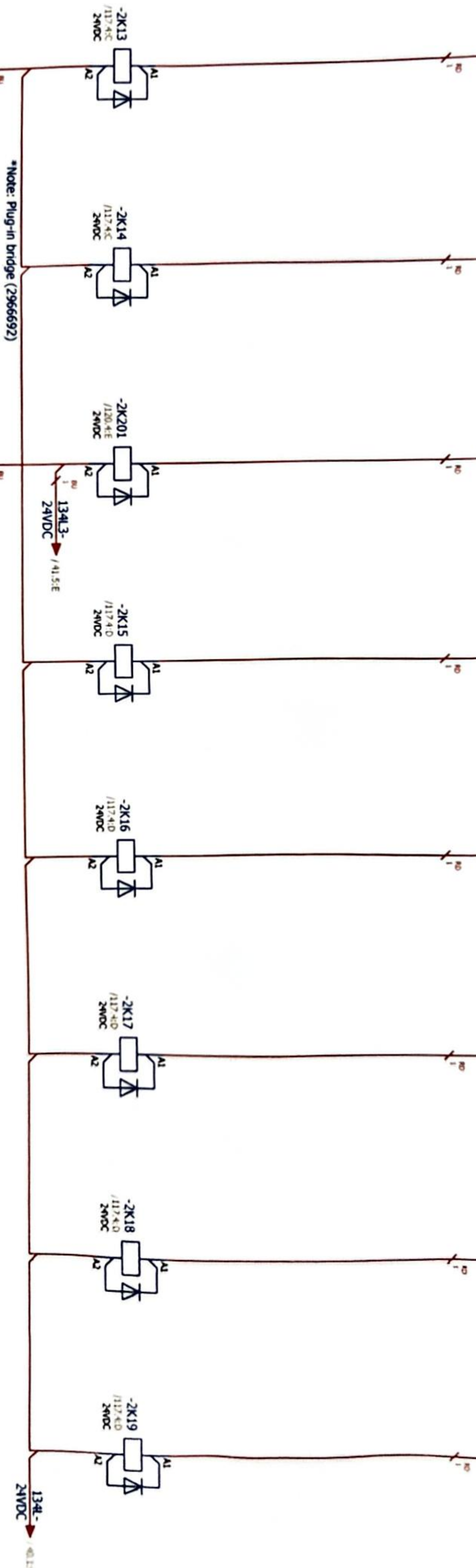
Format ISO/DIN A3

Approved by Contractor	Grga Kolic	27.7.2026.	Reconstruction and Modernisation of HPP Perucca - Phase II	EMPLOYER	SINTAKSA	Unit KA2 control cabinet circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCA10GH100
Checked by Designer	Antonio Aleksic	27.7.2026.			DESIGNER	DIGITAL OUTPUTS: 2.0 - 2.7	Designer No.:	SALES No.:	+ UBCA10
Drawn by	Tamara Tropicic	27.7.2026	PROJECT NAME						PAGE
1	2	3	4	5	6	7	8		38 / 158

1 2 3 4 5 6 7 8

GCB DIRECT CLOSE ENABLE
GCB AUTO DIRECT CLOSE COMMAND
AUTOMATIC/TEST SYNCHRONIZATION ON
EXCITATION ON
EXCITATION OFF
SPEED > EXCITATION START LIMIT
EXCITATION RESET OF ALARMS
START COMMAND ON

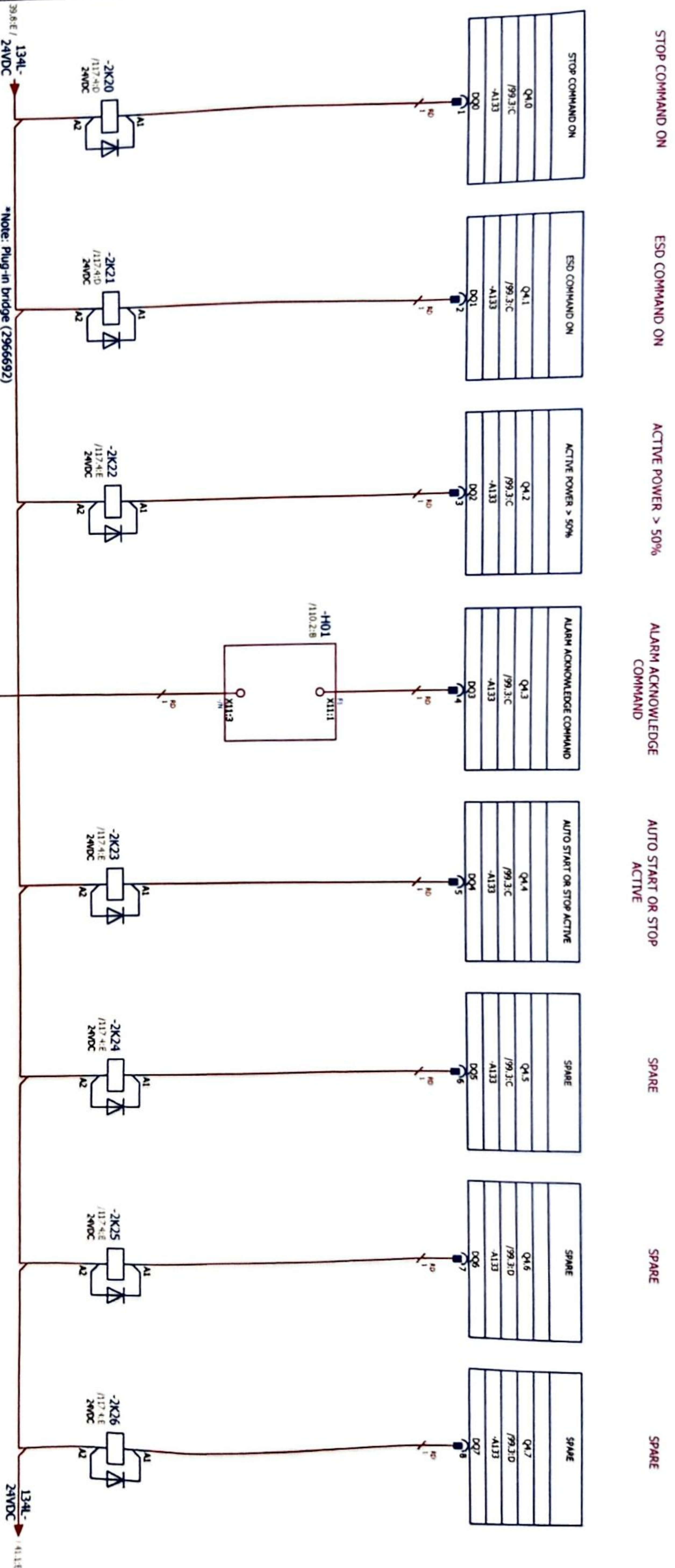
GCB DIRECT CLOSE ENABLE			
Q3.0	/98.3.0	-A132	DQ8
GCB AUTO DIRECT CLOSE COMMAND			
Q3.1	/98.3.0	-A132	DQ9
AUTOMATIC/TEST SYNCHRONIZATION ON			
Q3.2	/98.3.0	-A132	DQ10
EXCITATION ON			
Q3.3	/98.3.E	-A132	DQ11
EXCITATION OFF			
Q3.4	/98.3.E	-A132	DQ12
SPEED > EXCITATION START LIMIT			
Q3.5	/98.3.E	-A132	DQ13
EXCITATION RESET OF ALARMS			
Q3.6	/98.3.E	-A132	DQ14
START COMMAND ON			
Q3.7	/98.3.E	-A132	DQ15



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Format ISO/DIN A3

Approved by Contractor	Grga Kolic	27.7.2026.	Reconstruction and Modernisation	EPG	SINTAKSA	Unit K42 control cabinet circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	+ UBCA10
Approved by Designer	Antonio Aleksic	27.7.2026.	of HPP Peruca - Phase II			DIGITAL OUTPUTS: 3.0-3.7	Designer No.:	SALES No.:	PAGE
Checked by	Tamara Trojic	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE
Drawn by	1	2	3	4	5	6	7	8	39 / 158



14 11 /67.3:B

14 11 /67.4:B

14 11 /67.5:B

14 11 /68.6:C

14 11

14 11

14 11

Format ISO/DIN A3

Approved by Contractor	Grga Kolic	27.7.2026.	Reconstruction and Modernisation of HPP Perucica - Phase II	EMPLOYER	SINTAKSA	Unit KAZ control cubicle circuit diagrams	DRAWING NO.: PERU-01-19-170-31-002	REV No.: 02	SALES No.: = UBRCAL06H100
Approved by Designer	Antonio Aleksić	27.7.2026.		DESIGNER	Unit KAZ control cubicle circuit diagrams	DIGITAL OUTPUTS: 4.0 - 4.7	Designer No.: 40		
Checked by	Tamara Trojčić	27.7.2026.							
Drawn by									

1

2

3

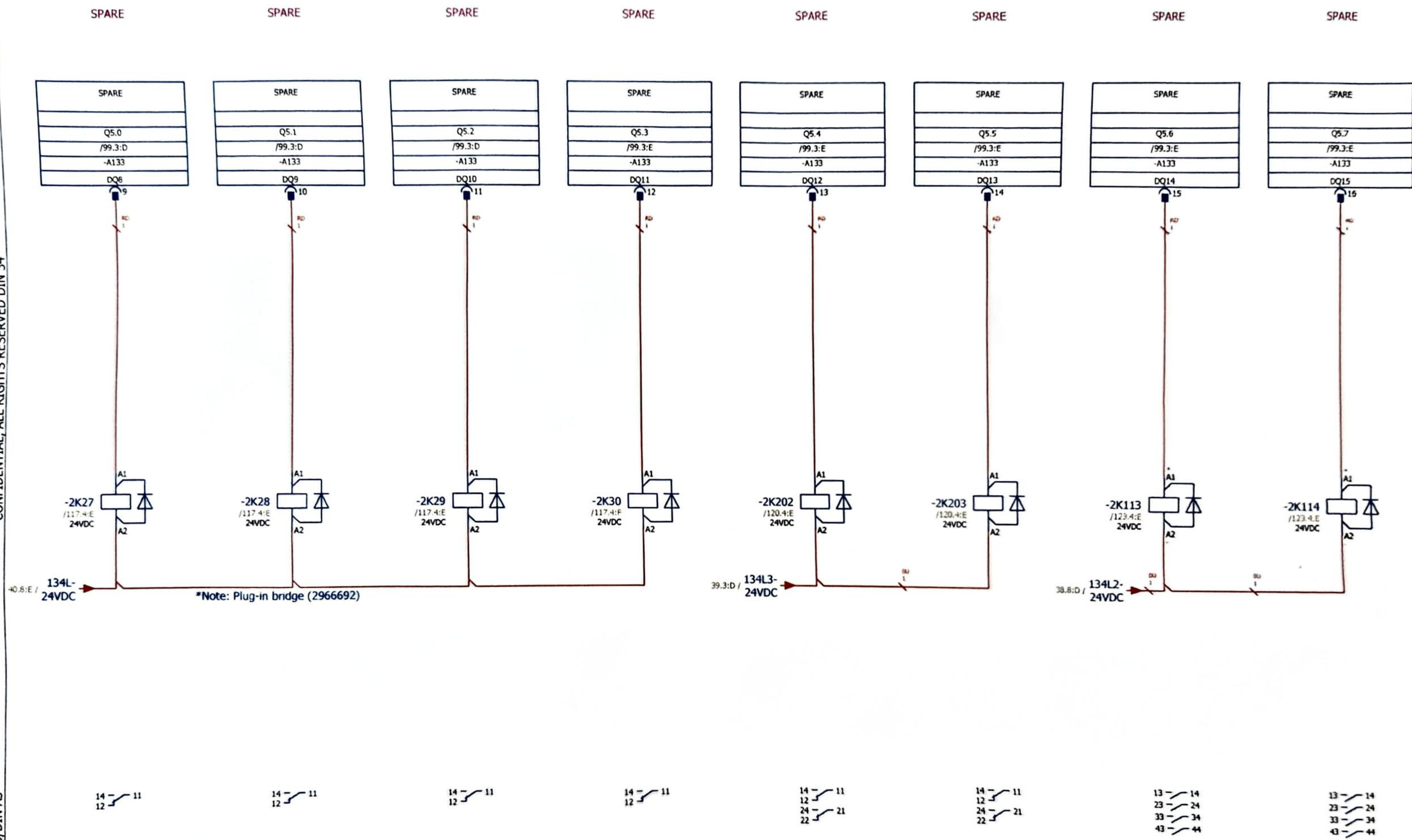
4

5

6

7

8



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Format ISO/DIN A3

SIGNALIZATION
EXCITATION CIRCUIT
BREAKER OFF

SIGNALIZATION
EXCITATION CIRCUIT
BREAKER ON

SIGNALIZATION
EXCITATION OFF

SIGNALIZATION
EXCITATION ON

SIGNALIZATION GCB OFF

SIGNALIZATION GCB ON

SPARE

SPARE

SIGNALIZATION EXCITATION CIRCUIT BREAKER OFF
Q6.0
/100.3:C
-A134
DQ0

SIGNALIZATION EXCITATION CIRCUIT BREAKER ON
Q6.1
/100.3:C
-A134
DQ1

SIGNALIZATION EXCITATION OFF
Q6.2
/100.3:C
-A134
DQ2

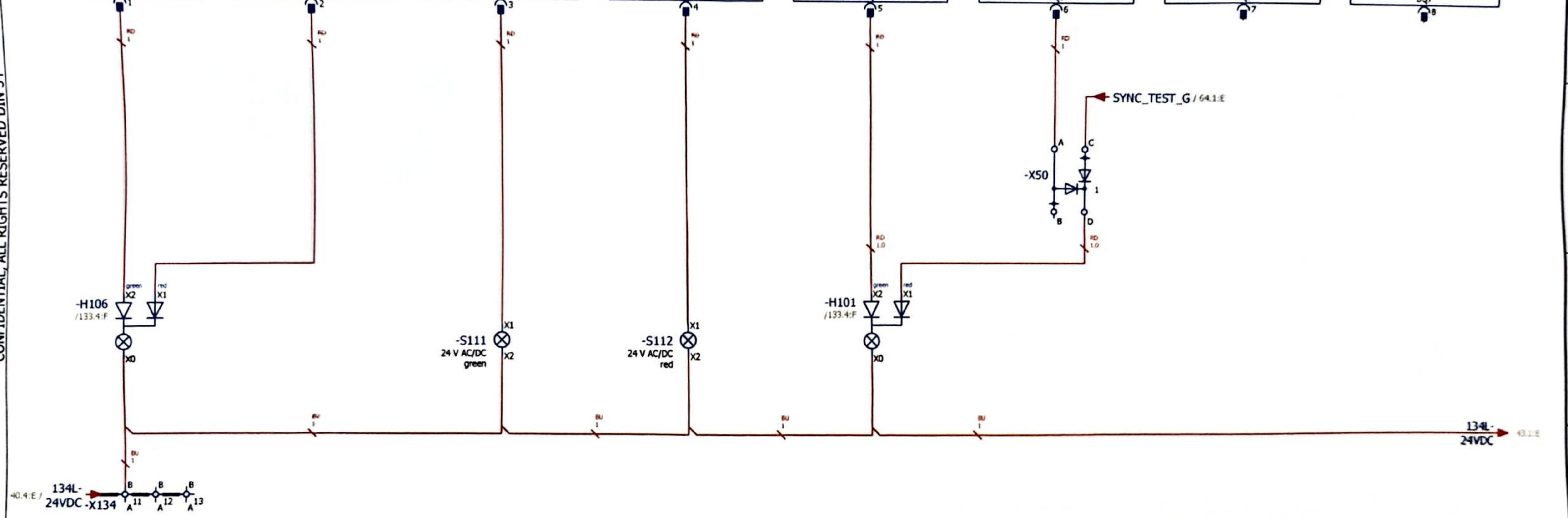
SIGNALIZATION EXCITATION ON
Q6.3
/100.3:C
-A134
DQ3

SIGNALIZATION GCB OFF
Q6.4
/100.3:C
-A134
DQ4

SIGNALIZATION GCB ON
Q6.5
/100.3:C
-A134
DQ5

SPARE
Q6.6
/100.3:D
-A134
DQ6

SPARE
Q6.7
/100.3:D
-A134
DQ7



Approved by Contractor					Unit KA2 control cubicle circuit diagrams	DRAWING No.:	REV No.:	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.				PERU-01-19-170-31-002	02	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.				Designer No.:	SALES No.:	PAGE 42
Drawn by	Tamara Trogrlić	27.7.2026.						PAGE 42 / 158
1	2	3	4	5	6	7	8	

SIGNALIZATION
SYNCHRONIZATION
START

SIGNALIZATION SYNCHRONIZATION START
Q7.0
/100.3:D
-A134
DQ8

SIGNALIZATION
SYNCHRONIZATION
STOP

SIGNALIZATION SYNCHRONIZATION STOP
Q7.1
/100.3:D
-A134
DQ9

SIGNALIZATION
GOVERNOR ALARM

SIGNALIZATION GOVERNOR ALARM
Q7.2
/100.3:D
-A134
DQ10

SIGNALIZATION
GOVERNOR TRIP

SIGNALIZATION GOVERNOR TRIP
Q7.3
/100.3:E
-A134
DQ11

SIGNALIZATION
GOVERNOR READY

SIGNALIZATION GOVERNOR READY
Q7.4
/100.3:E
-A134
DQ12

SIGNALIZATION START

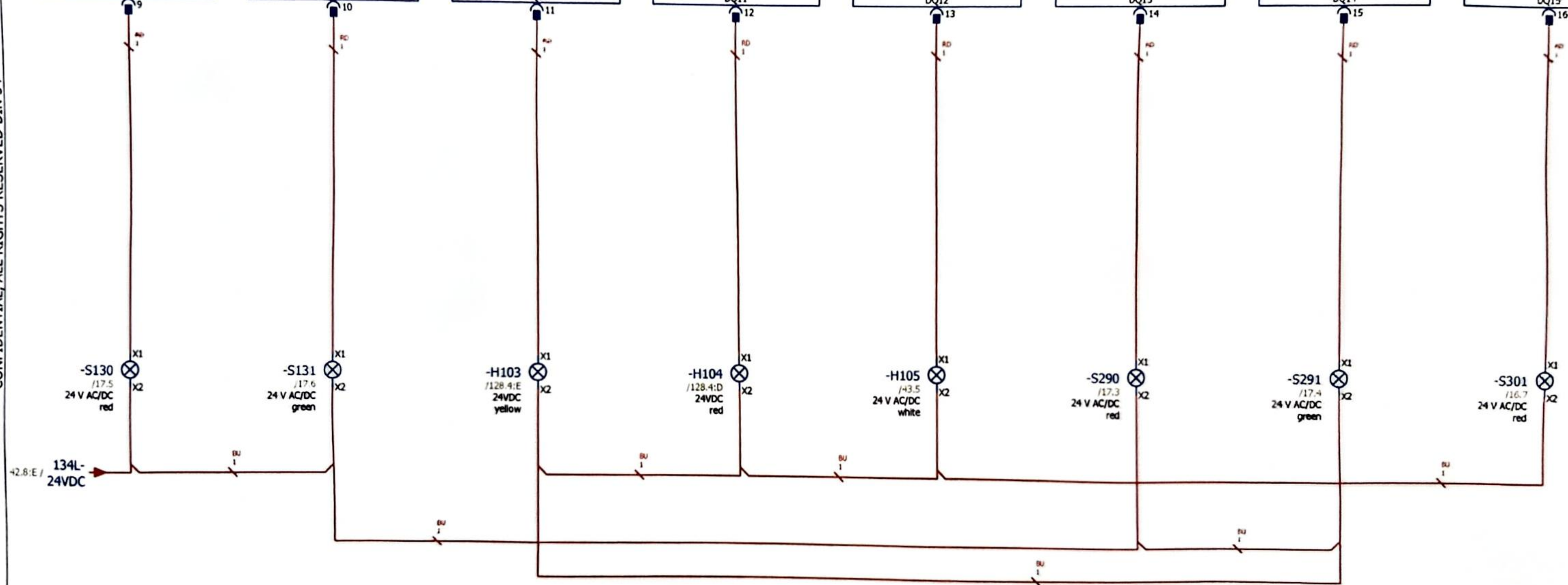
SIGNALIZATION START
Q7.5
/100.3:E
-A134
DQ13

SIGNALIZATION STOP

SIGNALIZATION STOP
Q7.6
/100.3:E
-A134
DQ14

SIGNALIZATION ESD

SIGNALIZATION ESD
Q7.7
/100.3:E
-A134
DQ15



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A Voth Company Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100 + UBCJA10
Approved by Designer	Grga Kolić	27.7.2026.				DIGITAL OUTPUTS: 7.0 - 7.7	Designer No.:	SALES No.:	PAGE 43
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER				PAGE 43 / 158
Drawn by	Tamara Trogrlić	27.7.2026.							

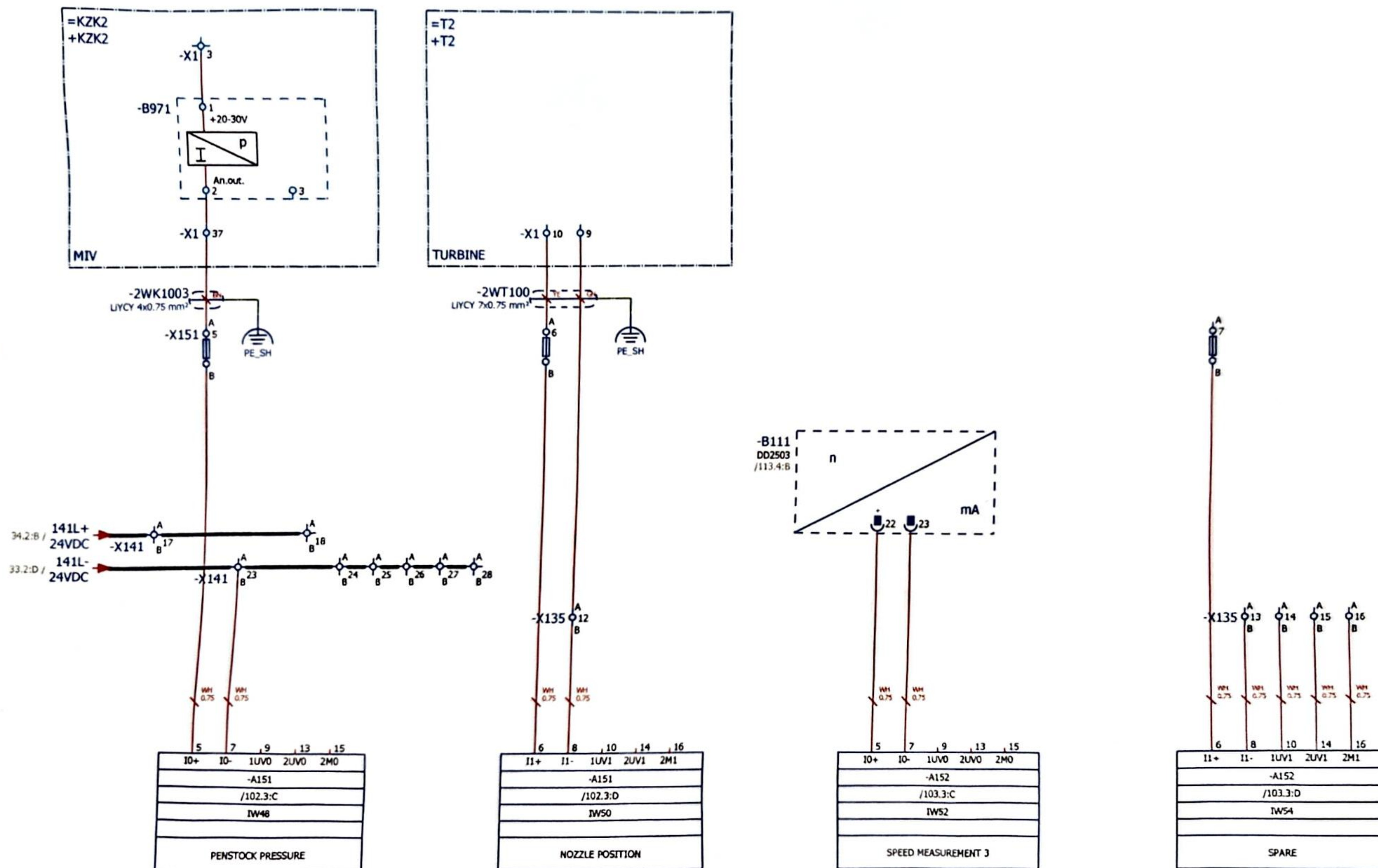
44 / 158

PENSTOCK PRESSURE

NOZZLE POSITION

SPEED MEASUREMENT 3

SPARE



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.					Designer No.:	SALES No.:	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER	ANALOG INPUTS: 5 - 8			PAGE 45
Drawn by	Tamara Trogrlić	27.7.2026.							PAGE 45 / 158

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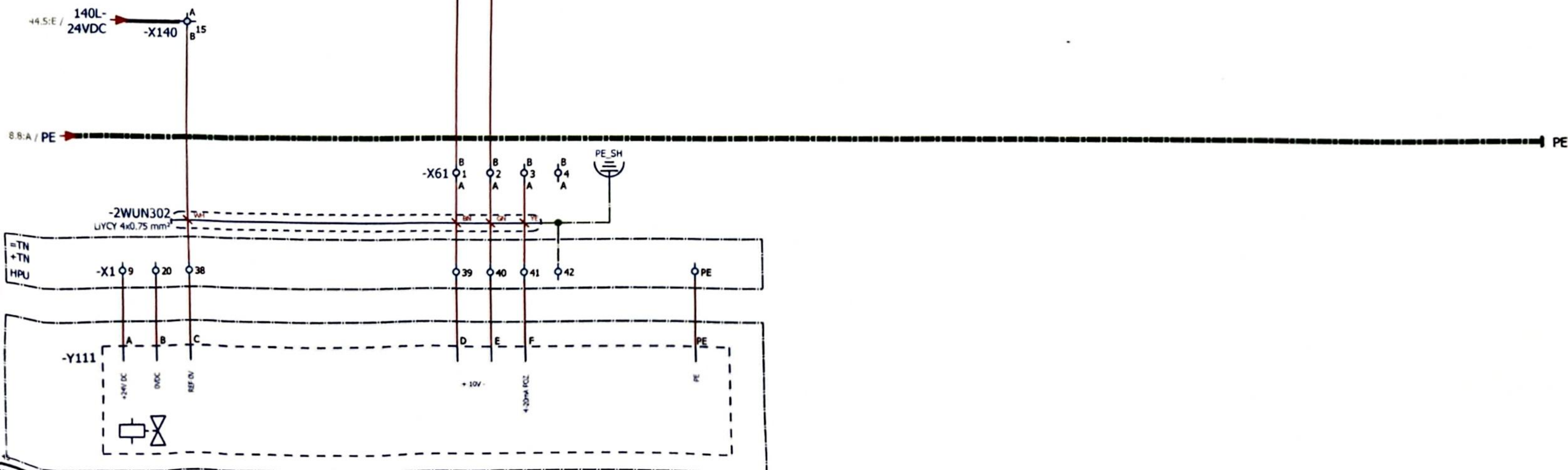
Format ISO/DIN A3

PROPORTIONAL VALVE COMMAND

TURBINE SPEED

PROPORTIONAL VALVE COMMAND			
QW40			
/104.3:C			
-A161			
Q0+	Q0-	S0+	S0-
1	5	9	13

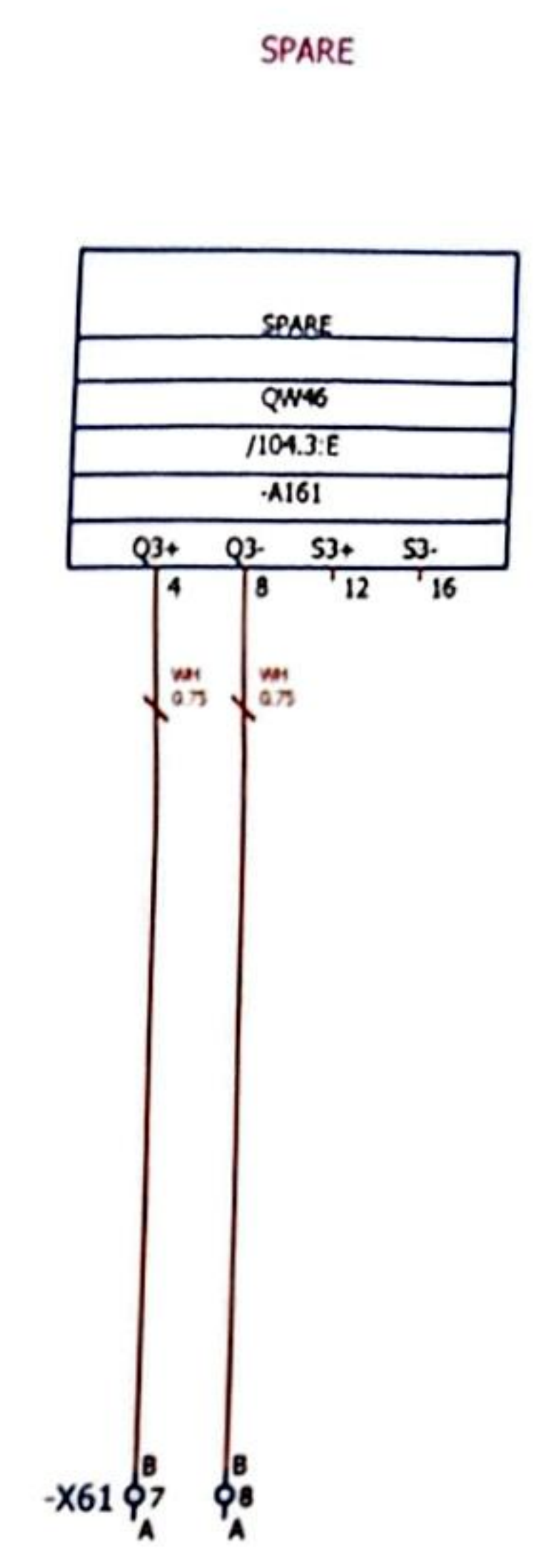
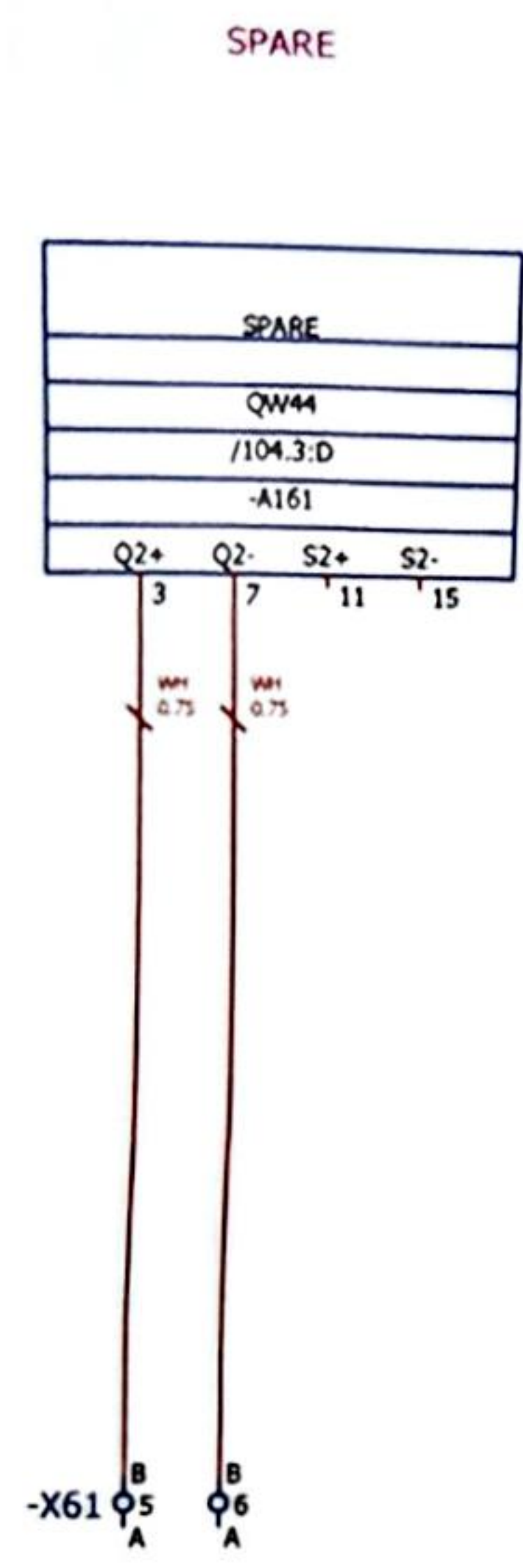
TURBINE SPEED			
QW42			
/104.3:C			
-A161			
Q1+	Q1-	S1+	S1-
2	6	10	14



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		 Sintaksa - A Voth Company Ulica Danovinskog rata 104C, 21 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams	DRAWING No.:	REV No.:	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.					PERU-01-19-170-31-002	02	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.					Designer No.:	SALES No.:	PAGE
Drawn by	Tamara Trogrlić	27.7.2026.					PROJECT NAME	EMPLOYER	DESIGNER

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Approved by Contractor
Approved by Designer
Checked by
Drawn by

Grga Kolić
Antonio Aleksić
27.7.2026.
27.7.2026.

Reconstruction and Modernisation
of HPP Perucica - Phase II



SINTAKSA
A Vard Group Company
Sintaksa - A Vard Group Company
Ulica Domovinskog rata 104C, 21 000 Split, HR

Unit KA2 control cubicle circuit diagrams

ANALOG OUTPUTS: 3 - 4

DRAWING No.:
PERU-01-19-170-31-002
Designer No.:

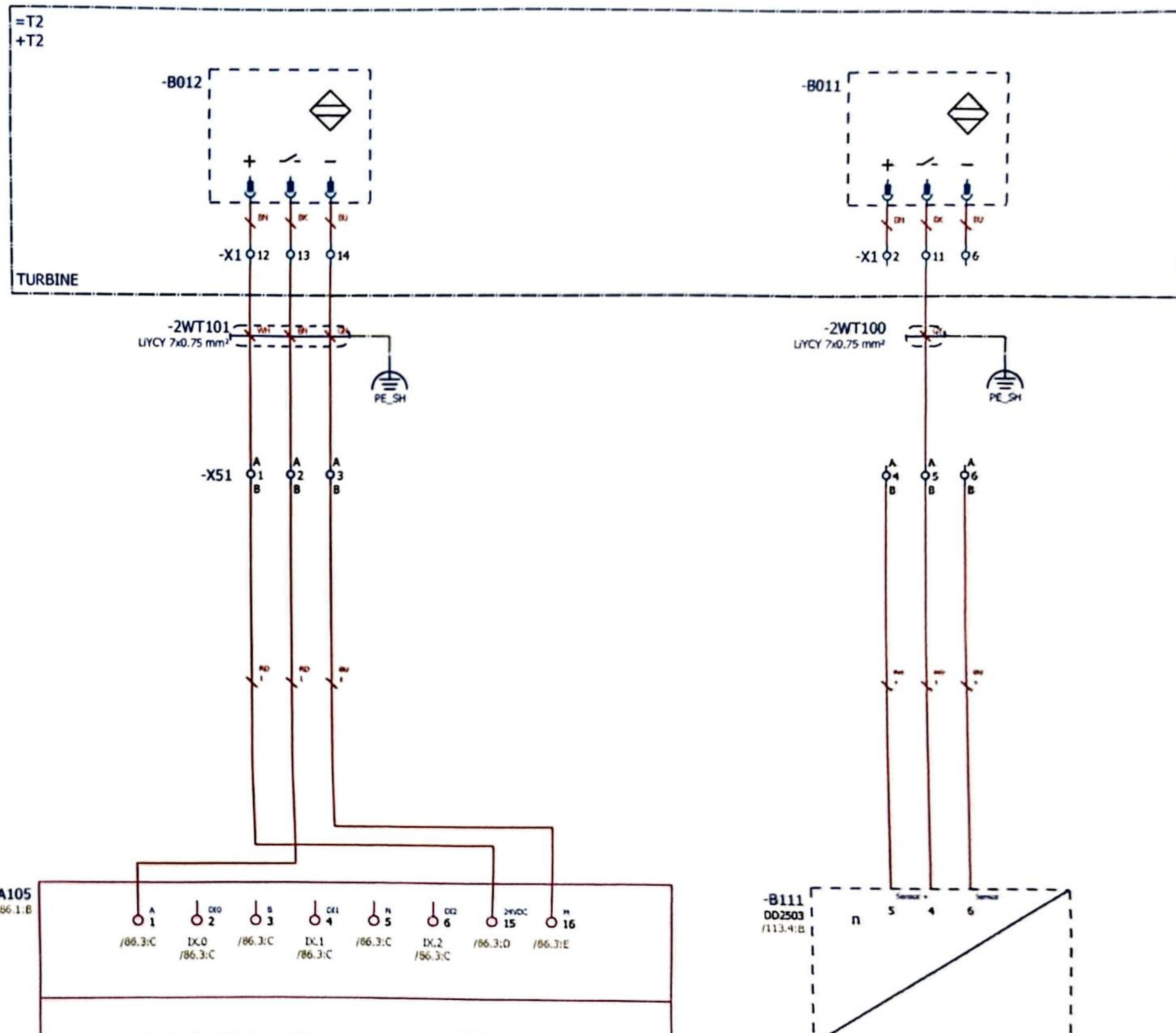
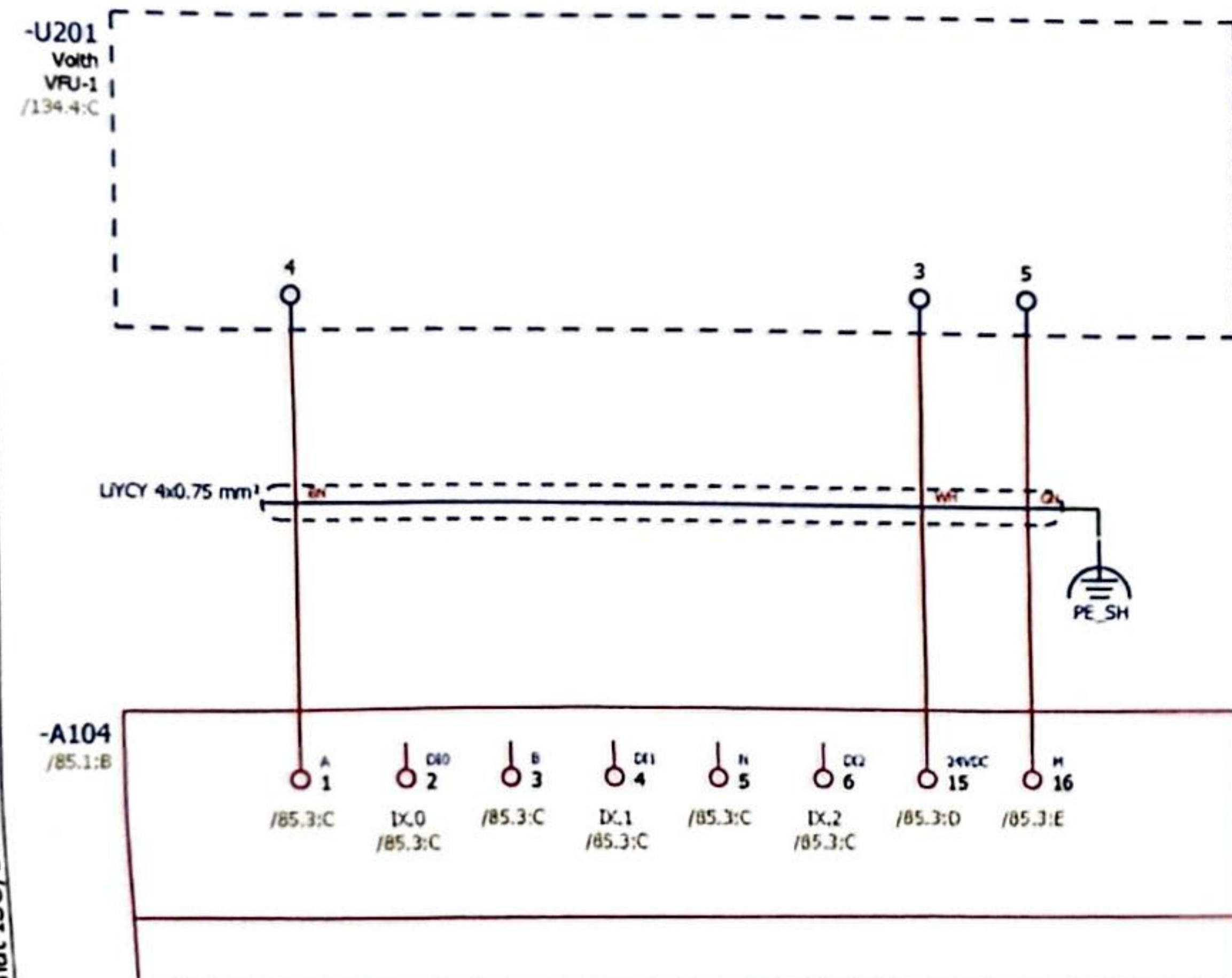
REV No.:
02
SALES No.:

= UBCJA10GH100
+ UBCJA10
PAGE
PAGE
47
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SPEED
MEASUREMENT 1

SPEED
MEASUREMENT 2

SPEED
MEASUREMENT 3



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E

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II			Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.					Designer No.:	SALES No.:	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER	SPEED MEASUREMENT			PAGE 48
Drawn by	Tamara Trogrlić	27.7.2026.							PAGE 48 / 158

1 2 3 4 5 6 7 8

A

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C

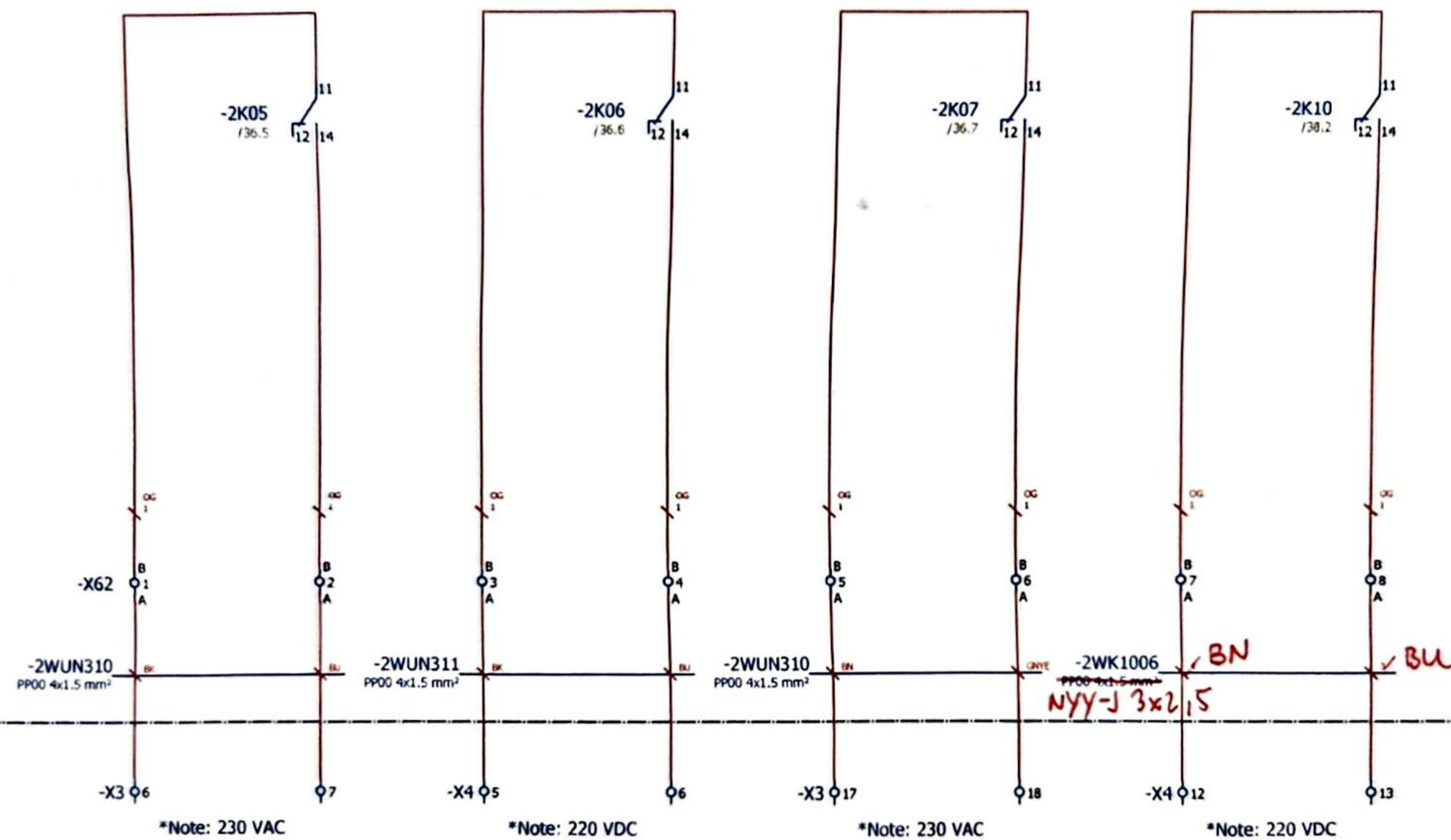
D

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Format ISO/DIN A3

HPU AC PUMP ON
COMMANDHPU DC PUMP ON
COMMANDHPU AC PUMP 3 ON
COMMANDLUBRICATION PUMP ON
COMMAND

A

B

C

D

E

F

Approved by Contractor	Grga Kolić	27.7.2026.	Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A Youth Company	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100 + UBCJA10
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER	HPU CONTROL	Designer No.:	SALES No.:	PAGE 49
Drawn by	Tamara Trogrlić	27.7.2026.							PAGE 49 / 158

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HPU ISOLATION VALVE
OPEN COMMAND

HPU ISOLATION VALVE
CLOSE COMMAND

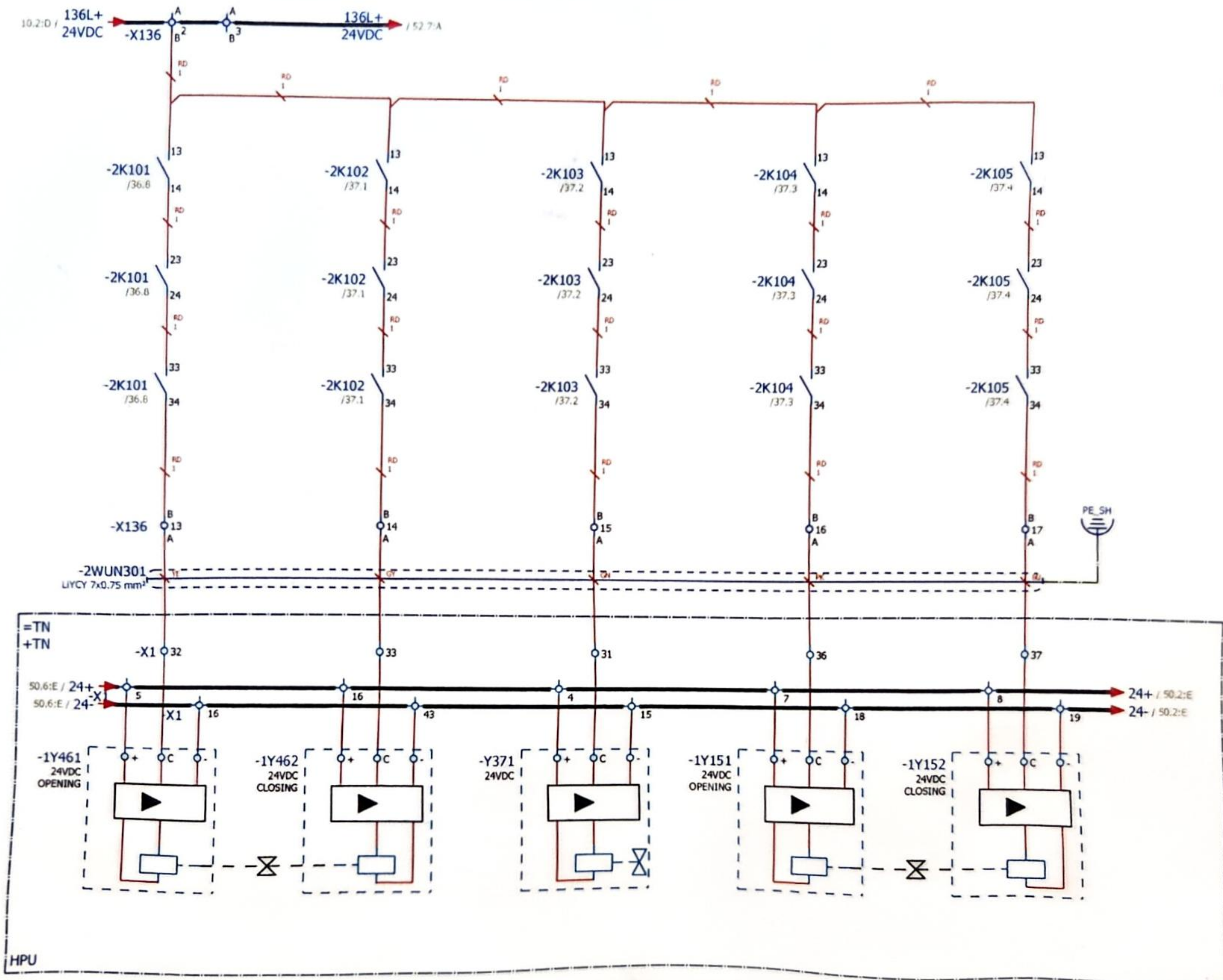
HPU LOAD VALVE ON
COMMAND

NOZZLE ~~OPEN~~
COMMAND

NOZZLE ~~CLOSE~~
COMMAND

* 28.8.2026.

Zamjenila u SW-u
komande za otvori/zatvori
jer su krivo označeni elem
u HPU shemi.



Approved by Contractor
Approved by Designer
Checked by
Drawn by

Grga Kolić
Antonio Aleksić
Tamara Trogrlić

27.7.2026.
27.7.2026.
27.7.2026.

Reconstruction and Modernisation
of HPP Perucica - Phase II
PROJECT NAME

EMPLOYER

SINTAKSA
A Veth Company
Sintaksa - A Veth Company
Ulica Domovinskog rata 104C, 21 000 Split, Hrvatska

DESIGNER

Unit KA2 control cubicle circuit diagrams
HPU CONTROL

DRAWING No.:
PERU-01-19-170-31-002
Designer No.:

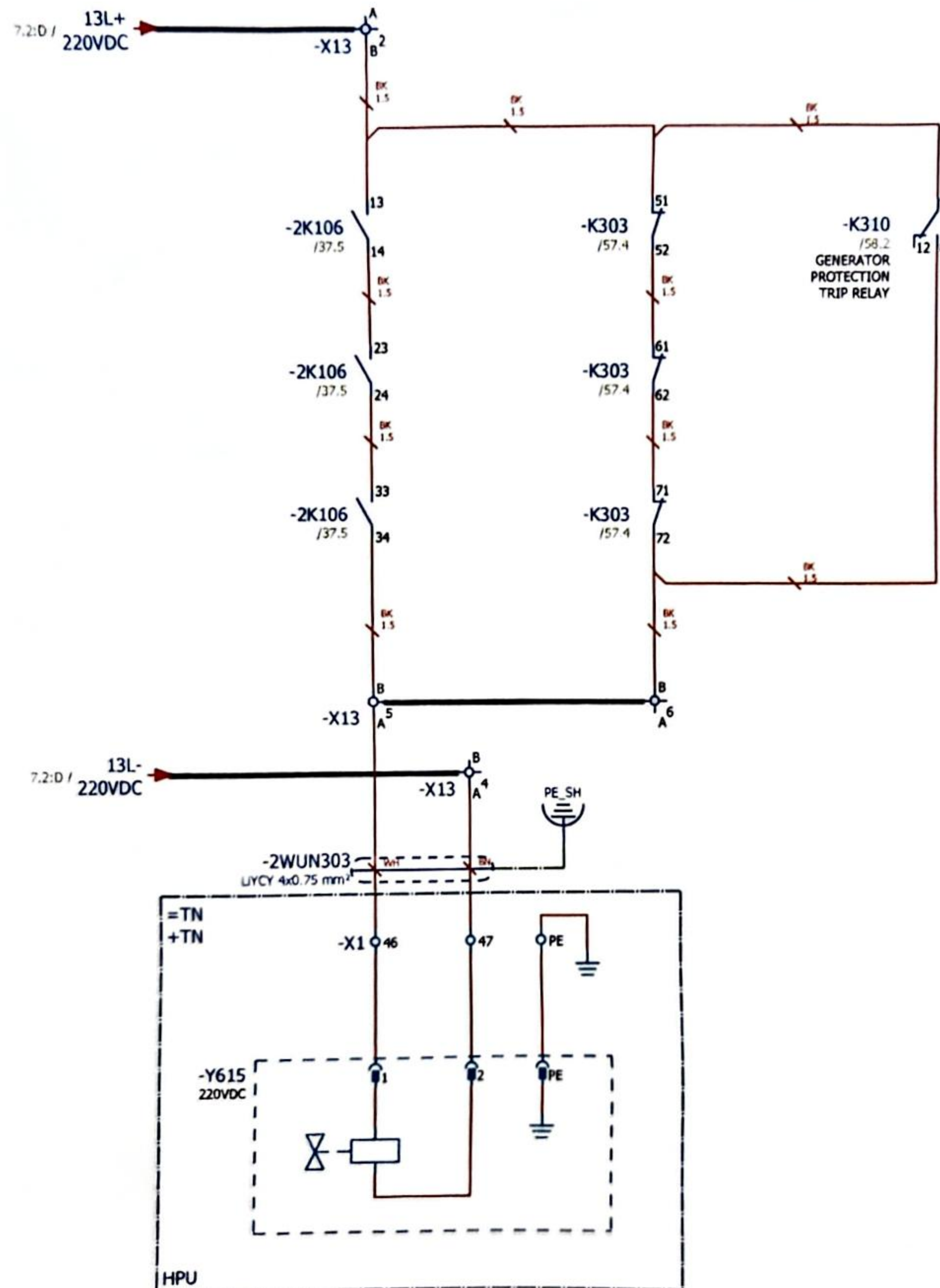
REV No.:
02
SALES No.:

= UBCJA10GH100
+ UBCJA10
PAGE 50
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ACTIVATE ESD VALVE

ESD RELAY

GENERATOR
PROTECTION ESD
ACTIVE



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E

Format ISO/DIN A3

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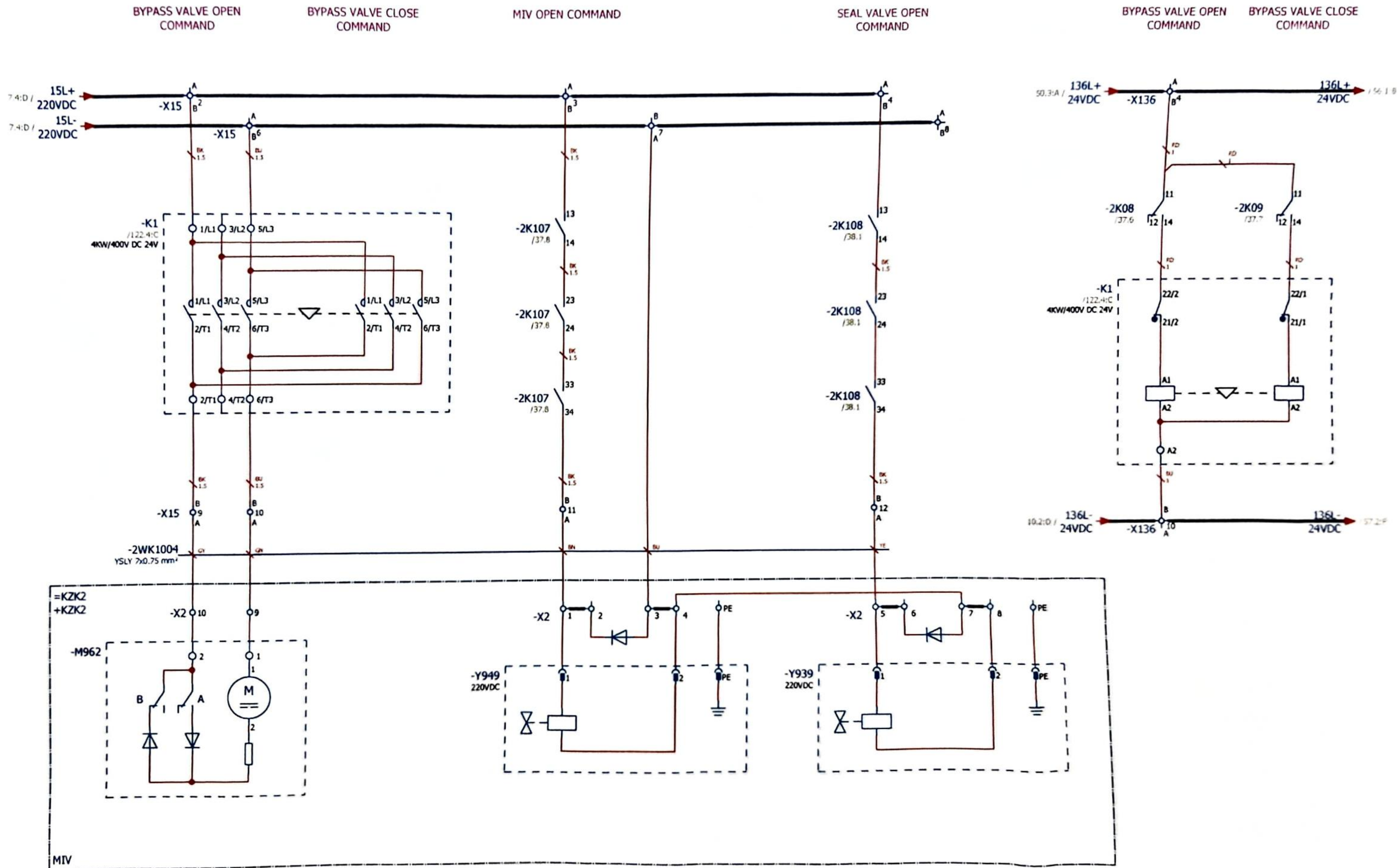
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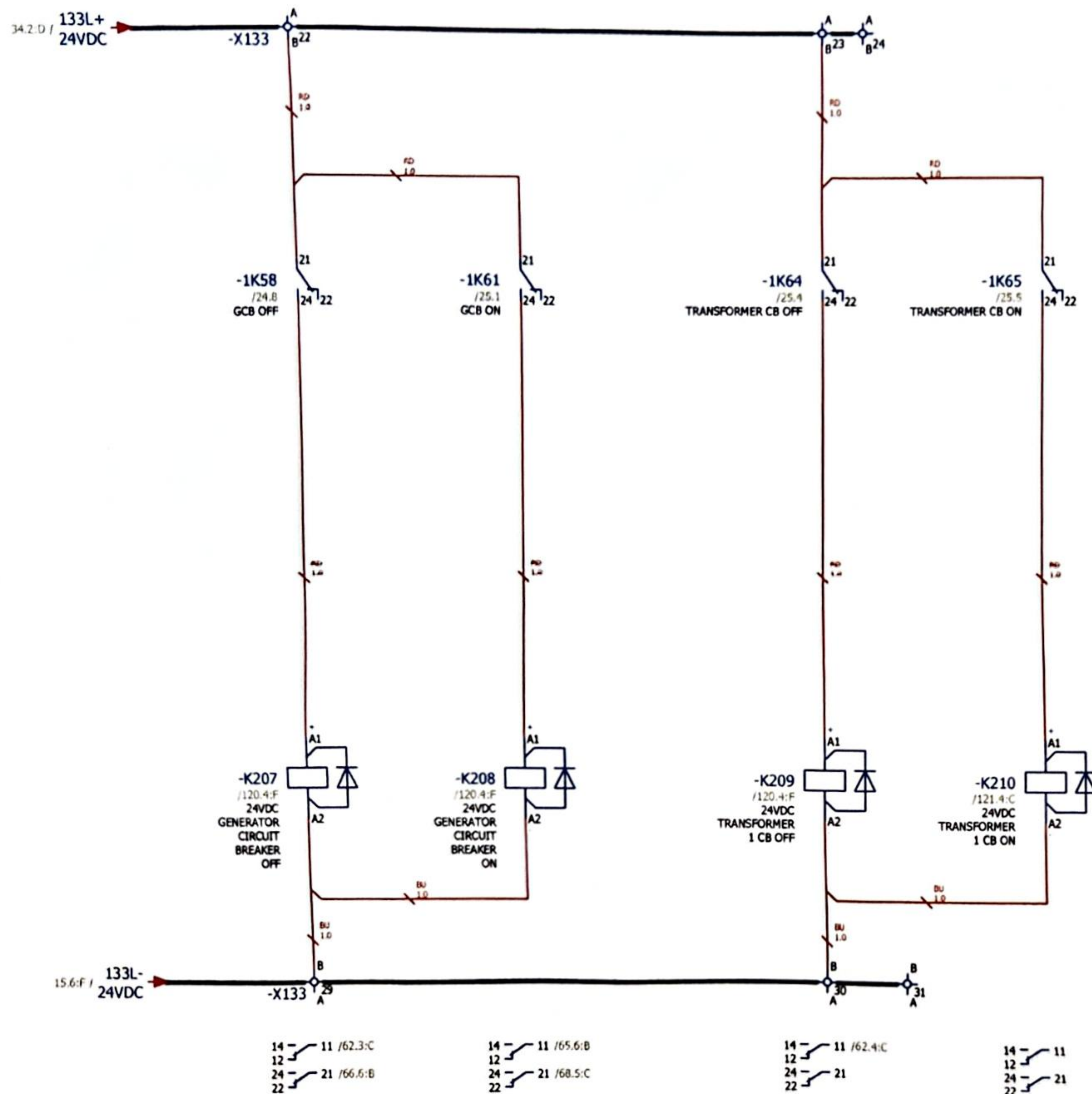


GCB OFF

GCB ON

TRANSFORMER CB OFF

TRANSFORMER CB ON



Approved by Contractor

Approved by Designer

Checked by

Drawn by

Grga Kolić

Antonio Aleksić

Tamara Trogrlić

27.7.2026.

27.7.2026.

27.7.2026.

Reconstruction and Modernisation
of HPP Perucica - Phase II

PROJECT NAME

epcg

SINTAKSA

Sintaksa - A Voth Company
Ulica Danovinskog rata 104C, 21 000 Split, Hrvatska

DESIGNER

Unit KA2 control cubicle circuit diagrams

LOCAL CONTROL

DRAWING No.:

PERU-01-19-170-31-002

Designer No.:

REV No.:

02

SALES No.:

= UBCJA10GH100

+ UBCJA10

PAGE

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E

F

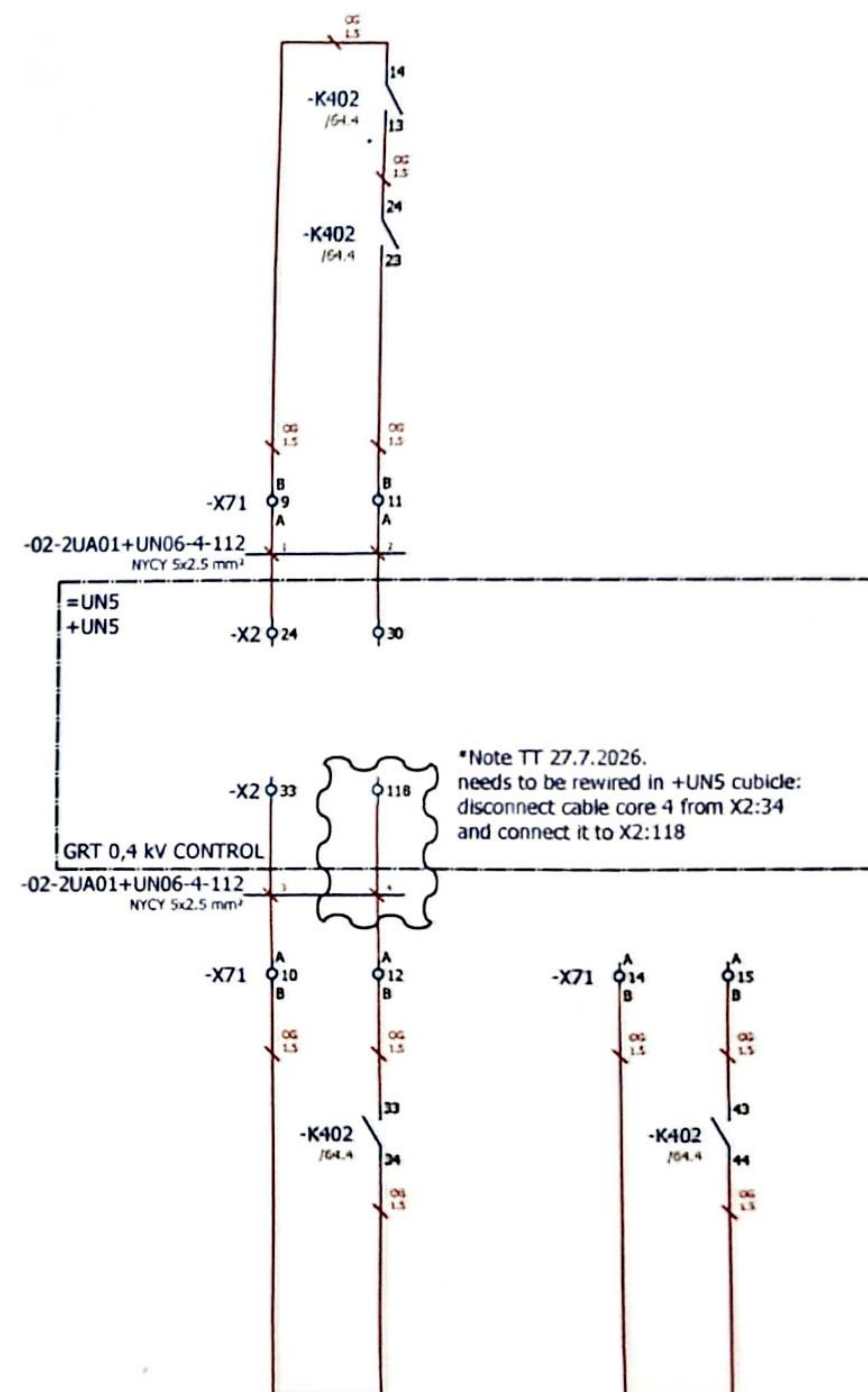
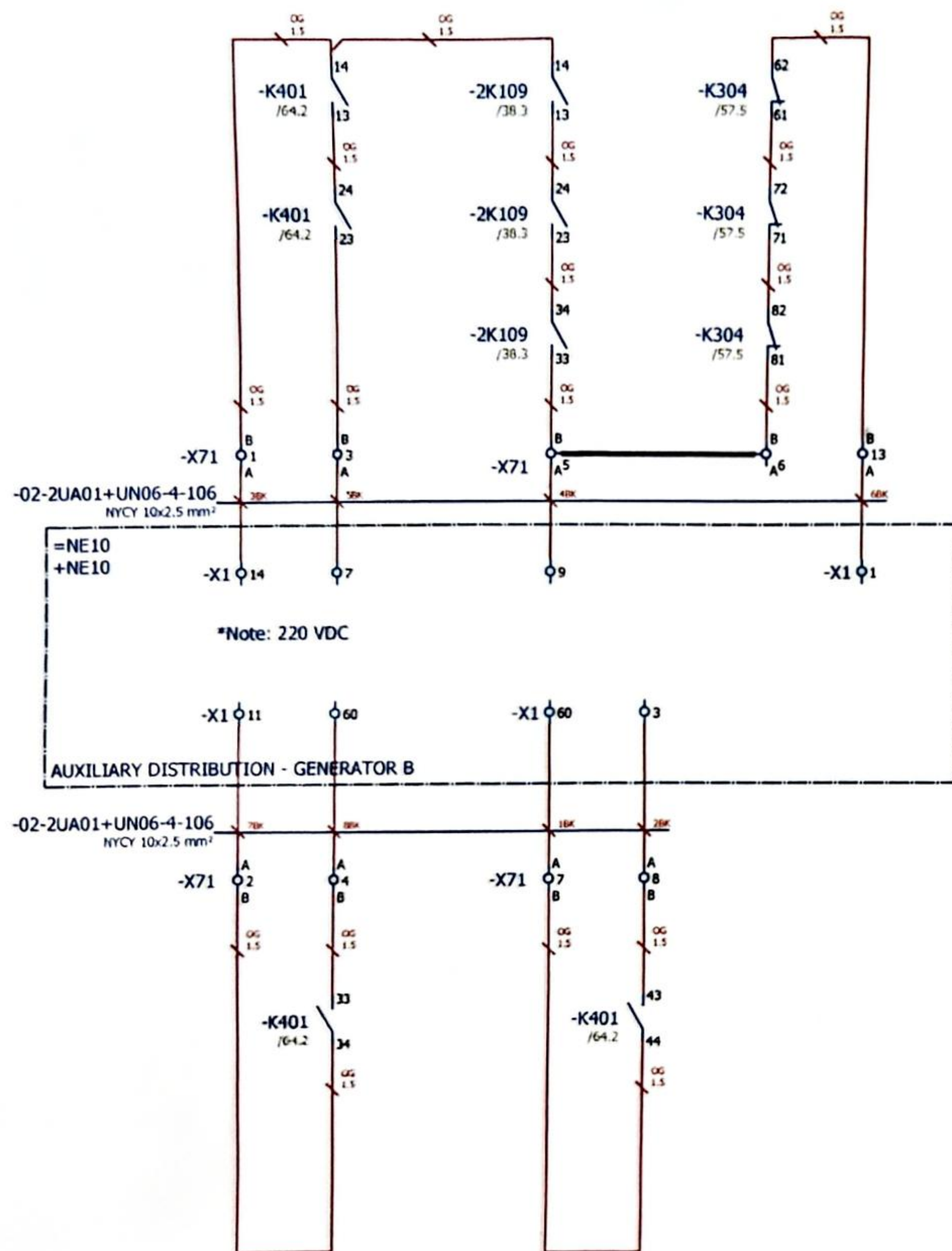
Format ISO/DIN A3

GENERATOR CB
CLOSING

GENERATOR CIRCUIT
BREAKER OPEN
COMMAND

GCB OFF
COMMAND - ESD

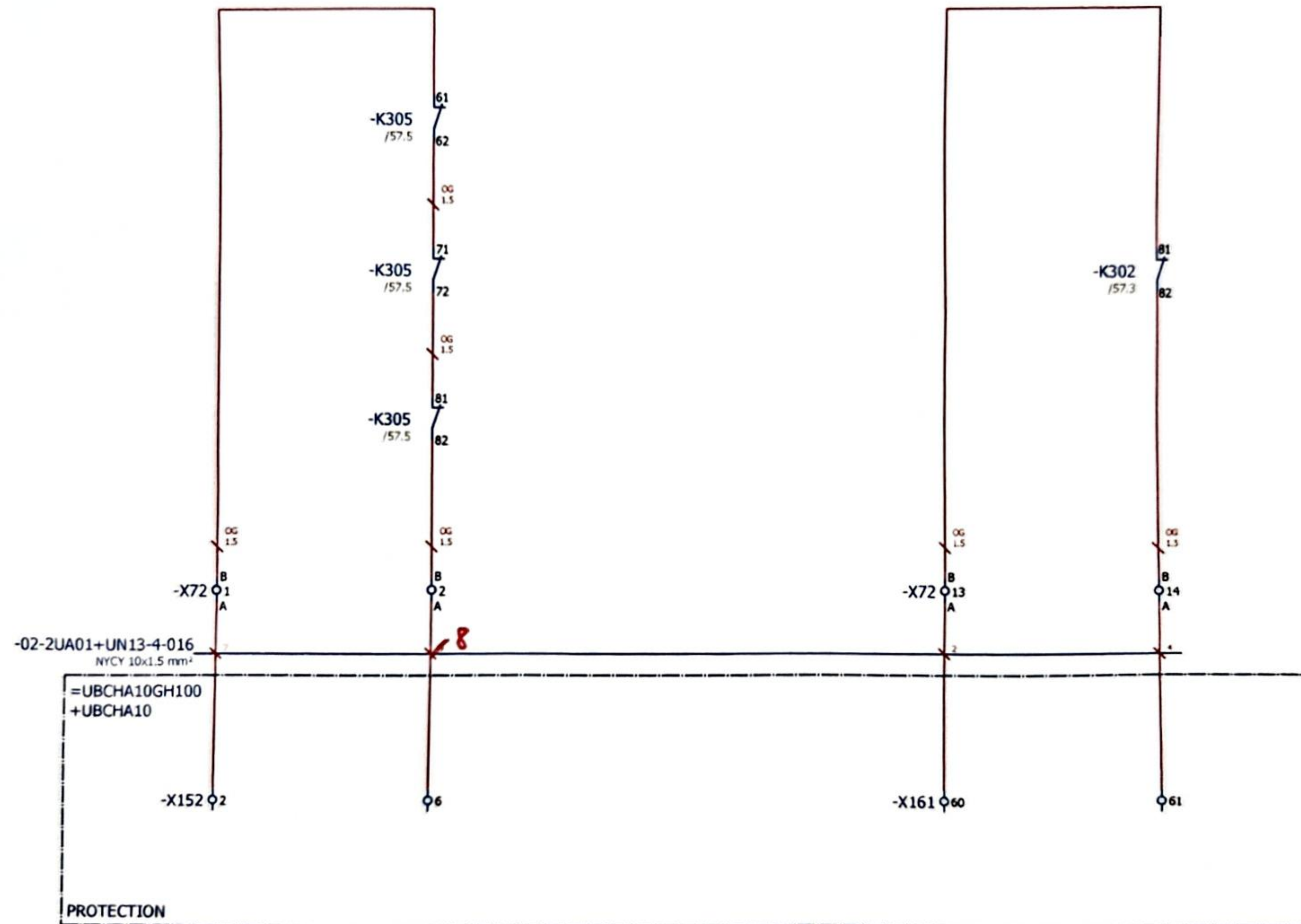
TRANSFORMER CB
CLOSING



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II			Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100 + UBCJA10
Approved by Designer	Grga Kolić	27.7.2026.							
Checked by	Antonio Aleksić	27.7.2026.							
Drawn by	Tamara Trogrlić	27.7.2026.							
			PROJECT NAME	EMPLOYER	DESIGNER	CB CONTROL	Designer No.:	SALES No.:	PAGE

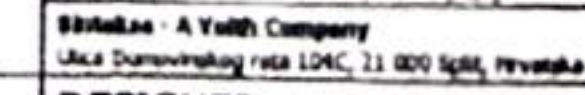
ECB OFF COMMAND
- ESD

ECB OFF COMMAND ESD
FEEDBACK



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A Velebit Company Ulica Društvenog rata 104C, 21 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams EXCITATION CB TRIP COMMAND	DRAWING No.:	REV No.:	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.					PERU-01-19-170-31-002	02	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.					Designer No.:	SALES No.:	PAGE
Drawn by	Tamara Trogrlić	27.7.2026.							55
			PROJECT NAME	EMPLOYER	DESIGNER				PAGE

Format ISO/DIN A3

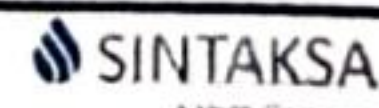


Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		 SINTAKSA A Youth Company Birtak - A Youth Company Jaka Domanjšičeva cesta 104C, 31 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.:	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.						02	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.						SALES No.:	PAGE 57
Drawn by	Tamara Trogrlić	27.7.2026.						DESIGNER	PAGE 57 / 158
PROJECT NAME			EMPLOYER			EMERGENCY SHUTDOWN	Designer No.:		

[illegible]

Format ISO/DIN A3

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II
Approved by Designer	Grga Kolić	27.7.2026.	
Checked by	Antonio Aleksić	27.7.2026.	
Drawn by	Tamara Trondić	27.7.2026.	



DRAWING No.:
PERU-01-19-170-31-002

REV No.:	02
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$$+ \text{UBCIA10}$$

SYNCHRONIZATION
MANUAL

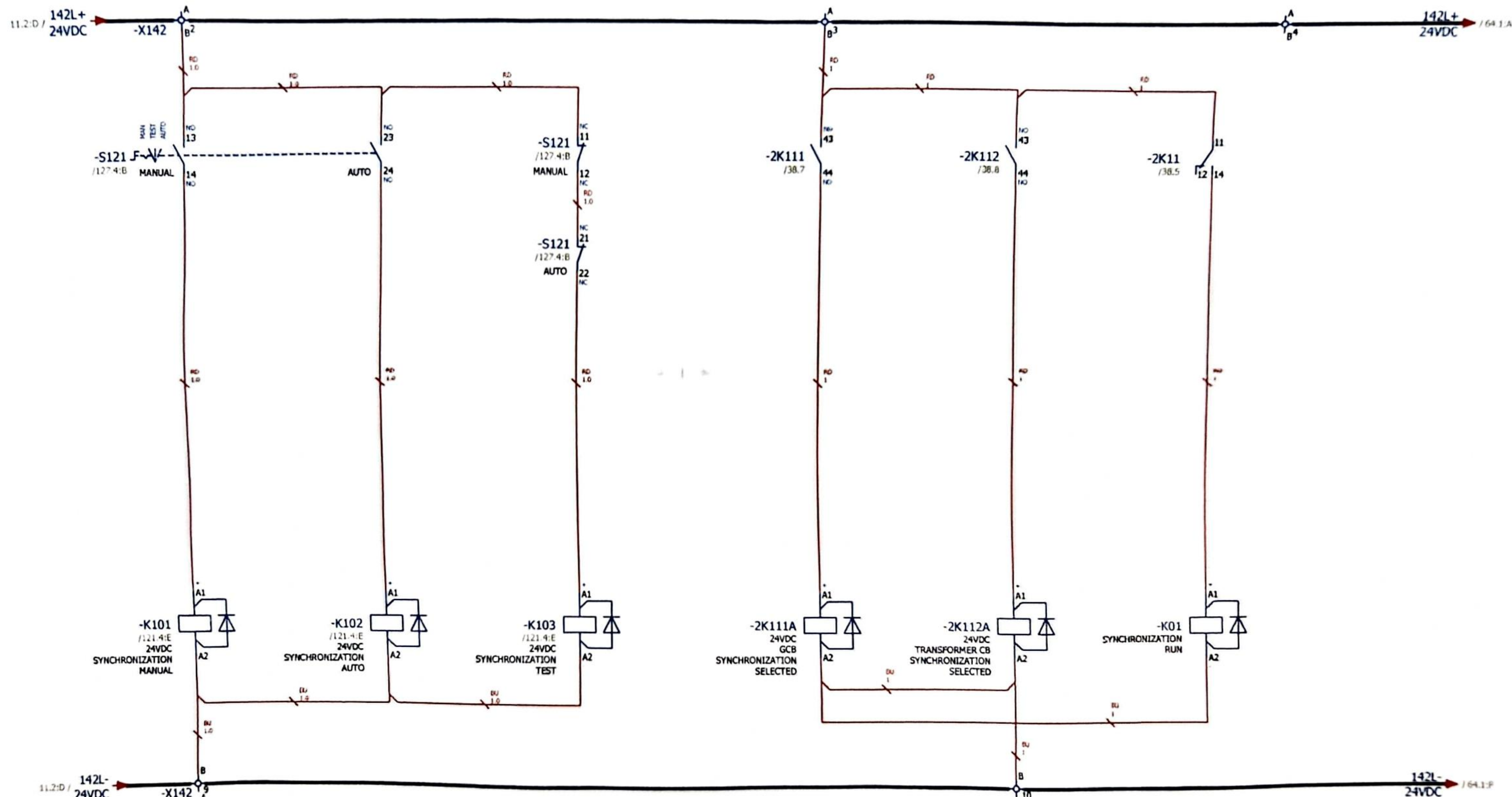
SYNCHRONIZATION
AUTO

SYNCHRONIZATION
TEST

GCB SYNCHRONIZATION
SELECTED

TRANSFORMER CB
SYNCHRONIZATION
SELECTED

SYNCHRONIZATION
RUN



14 - 11 /16.4:D
12 - 21 /64.2:B
24 - 31
32 - 41
44 - 42

14 - 11 /16.6:D
12 - 21 /64.3:B
24 - 31
32 - 41
44 - 42

14 - 11 /16.5:D
12 - 21 /64.4:B
24 - 31 /64.2:D
32 - 41 /64.4:D
44 - 42

13 - 14 /60.5:C
23 - 24 /60.5:C
33 - 34 /61.3:C
43 - 44 /61.4:C

13 - 14 /60.6:C
23 - 24 /60.6:C
33 - 34 /61.4:C
43 - 44 /61.4:C

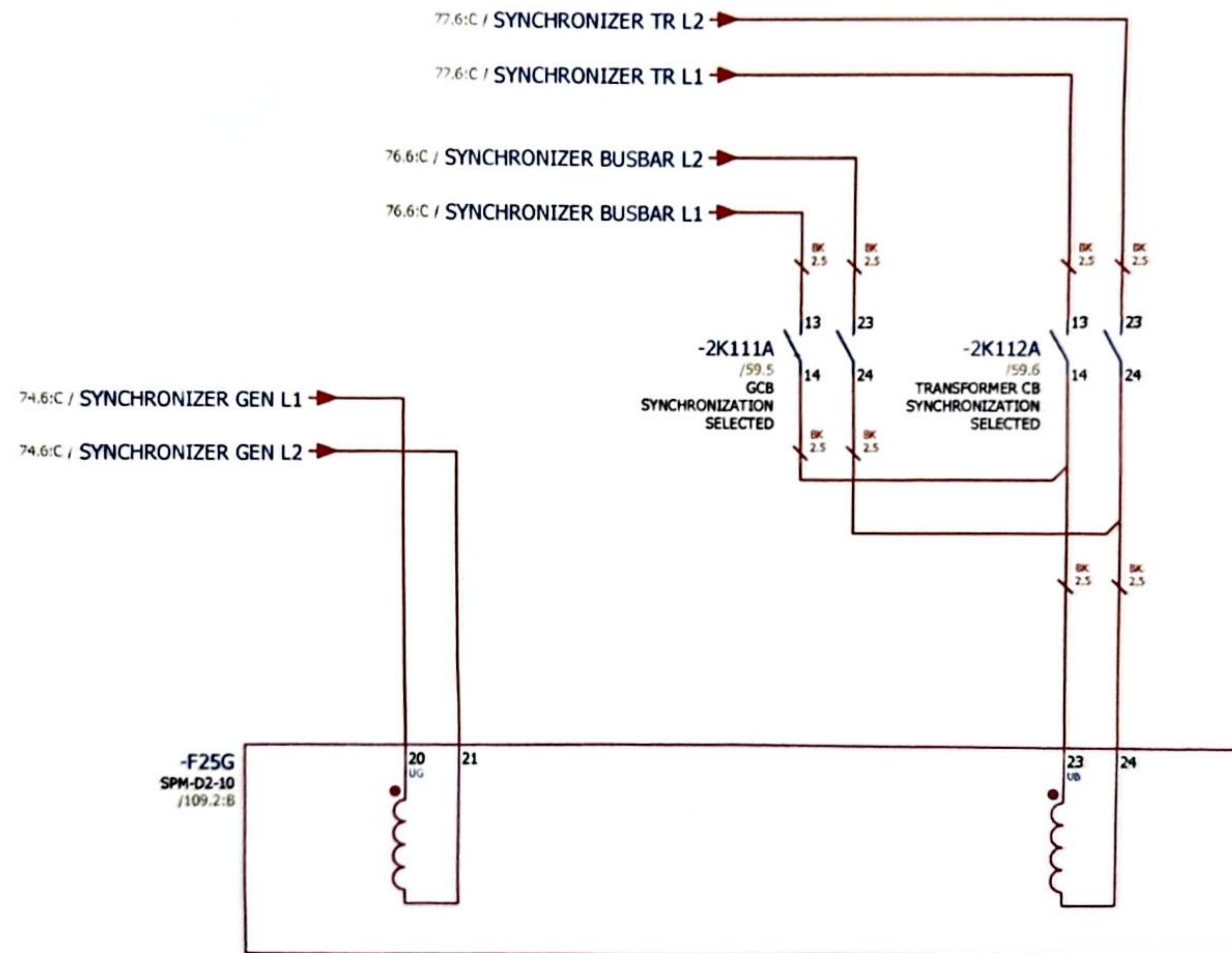
13 - 14 /61.4:D
23 - 24 /61.4:D
33 - 34 /61.5:D
43 - 44 /61.5:D
53 - 54
61 - 62
73 - 74
83 - 84

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		 A Youth Company Sintaksa - A Youth Company Ulica Demotrovačka 104C, 21 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams	DRAWING No.:	REV No.:	= UBCJA10GH100	
Approved by Designer	Grga Kolić	27.7.2026.					PERU-01-19-170-31-002	02	+ UBCJA10	
Checked by	Antonio Aleksić	27.7.2026.					SYNCHRONIZATION CONTROL	Designer No.:	SALES No.:	PAGE 59
Drawn by	Tamara Trogrlić	27.7.2026.								PAGE 59 / 158

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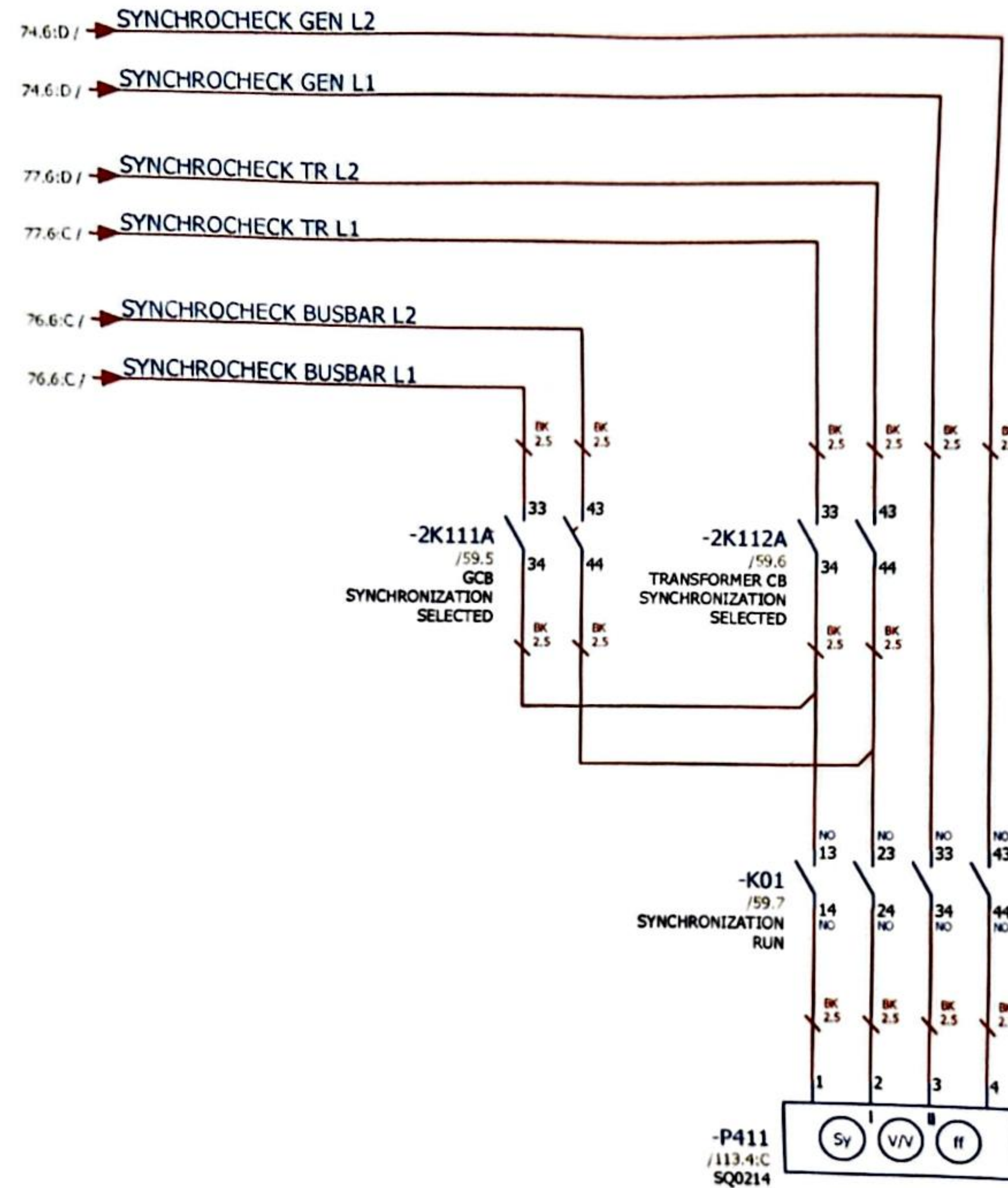
Format ISO/DIN A3

SYNCHRONIZER VOLTAGE MEASUREMENT



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II	EMPLOYER	DESIGNER	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.				SYNCHRONIZATION CONTROL	Designer No.:	SALES No.:	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME						PAGE 60
Drawn by	Tamara Trogrlić	27.7.2026.							PAGE 60 / 158

SYNCHROCHECK VOLTAGE MEASUREMENT



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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II			Unit KA2 control cubicle circuit diagrams	DRAWING No.:	REV No.:	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.					PERU-01-19-170-31-002	02	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.					Designer No.:	SALES No.:	PAGE
Drawn by	Tamara Trogrlić	27.7.2026.					PROJECT NAME	EMPLOYER	DESIGNER

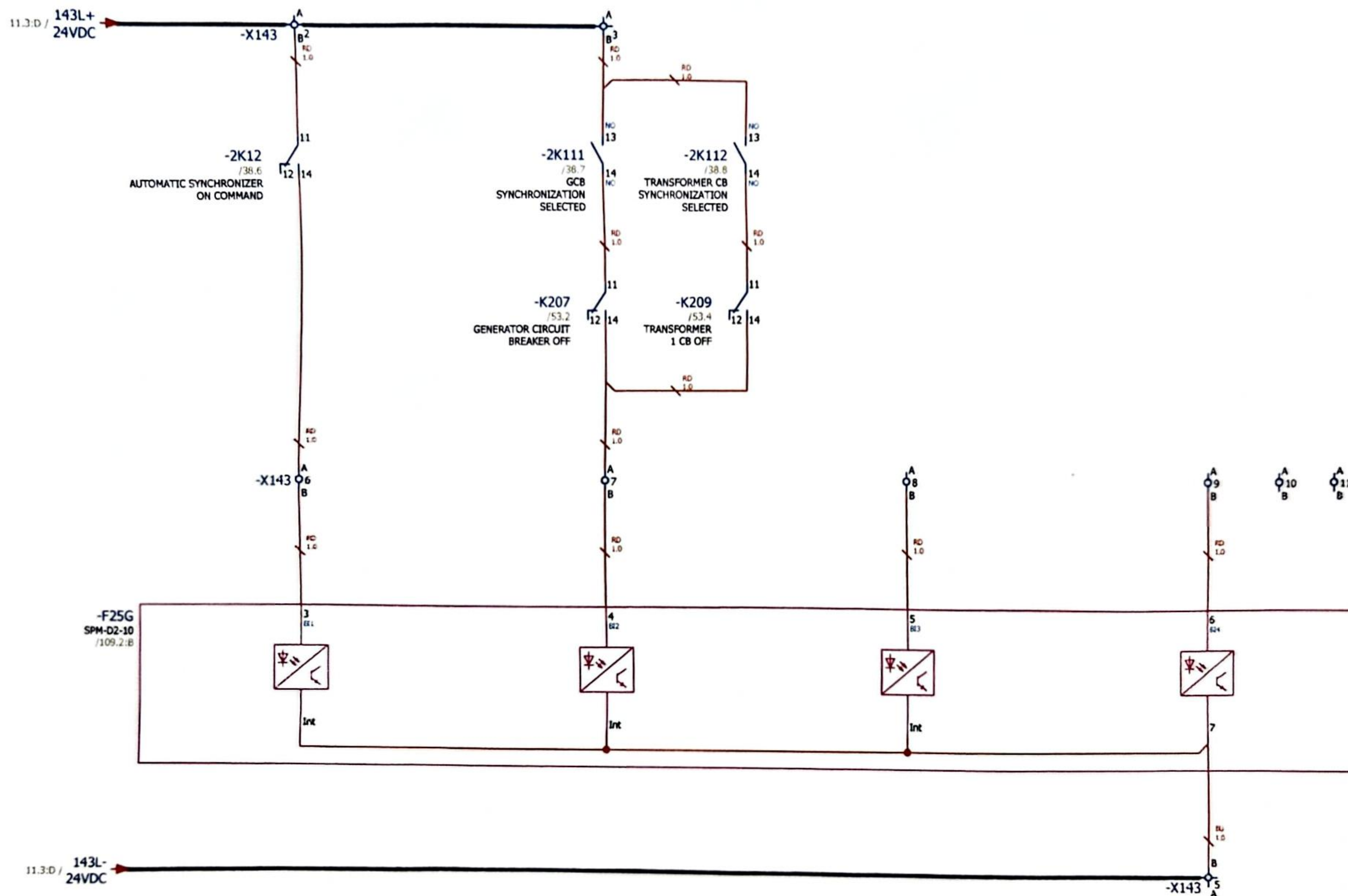
SYNCHRONIZATION

AUTOMATIC SYNCHRONIZER
START

CB OPEN

SPARE

SPARE

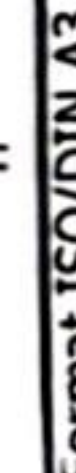


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Format ISO/DIN A3

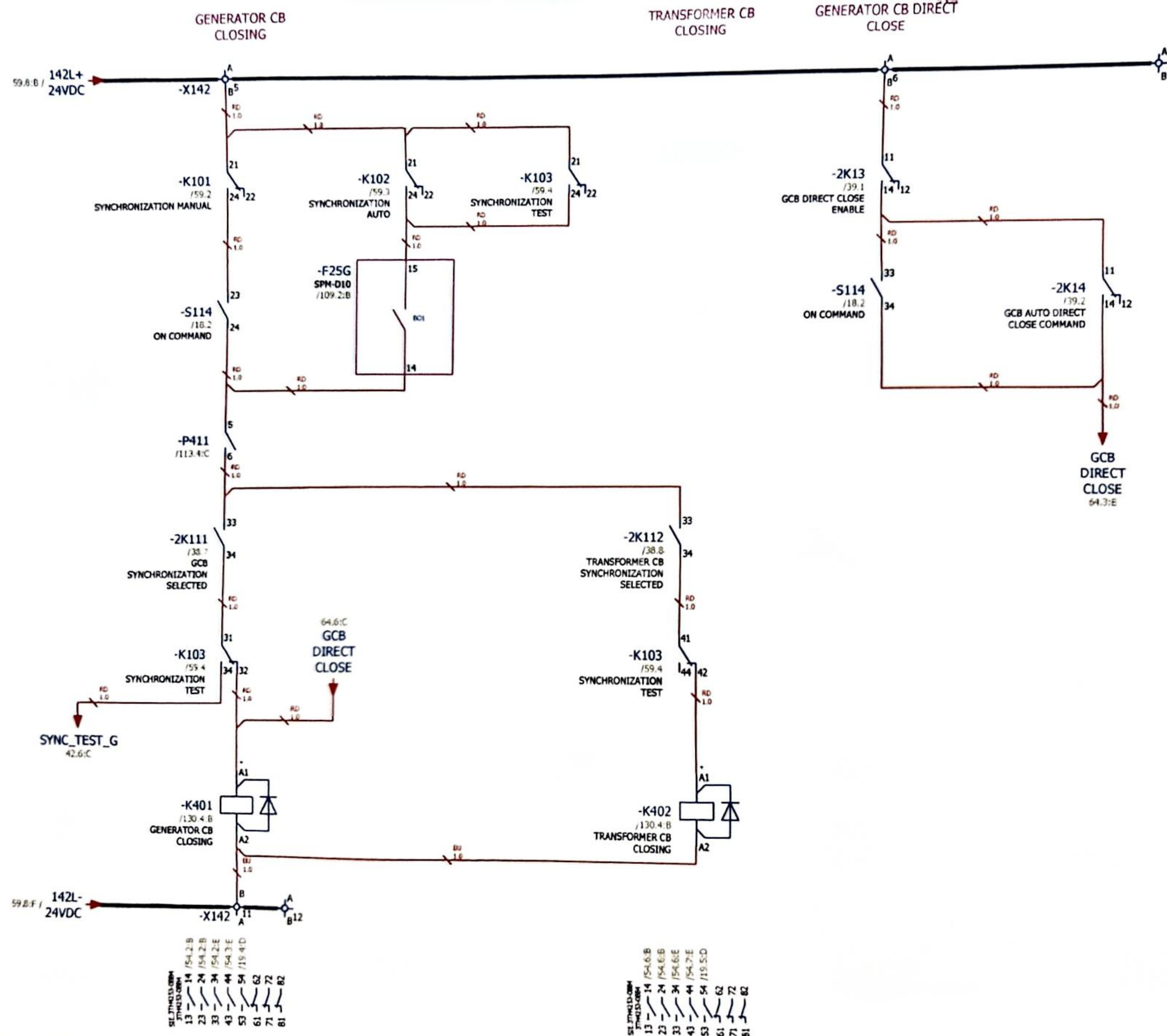
Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A Youth Company Ulica Domovinskega rata 126C, 21 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams SYNCHRONIZATION	DRAWING No.:	REV No.:	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.					PERU-01-19-170-31-002	02	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.					Designer No.:	SALES No.:	PAGE
Drawn by	Tamara Trogrlić	27.7.2026.							62 / 158

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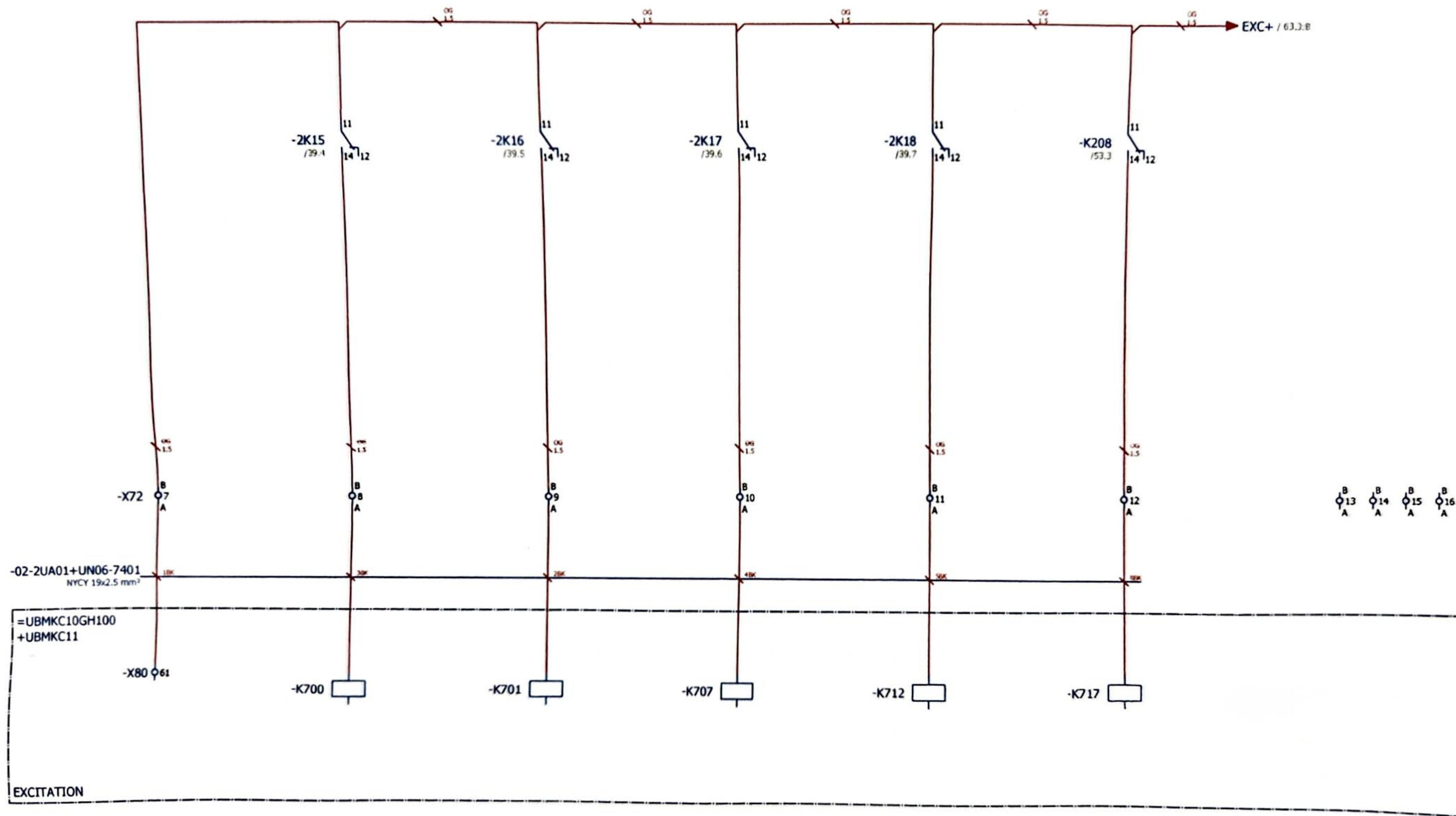
10

16



EXCITATION ON

EXCITATION OFF

SPEED > EXCITATION
START LIMITEXCITATION RESET OF
ALARMSGENERATOR CIRCUIT
BREAKER ON

Approved by Contractor
Approved by Designer
Checked by
Drawn by

Grga Kolić
Antonio Aleksić
Tamara Trogrlić

27.7.2026.
27.7.2026.
27.7.2026.

Reconstruction and Modernisation
of HPP Perucica - Phase II
PROJECT NAME

EMPLOYER



SINTAKSA
A Yutth Company
Sintaksa - A Yutth Company
Ulica Demotinskog rata 104C, 21 000 Split, Hrvatska

DESIGNER

Unit KA2 control cubicle circuit diagrams

SIGNALIZATION - EXCITATION

DRAWING No.:
PERU-01-19-170-31-002
Designer No.:

REV No.:
02
SALES No.:

= UBCJA10GH100
+ UBCJA10

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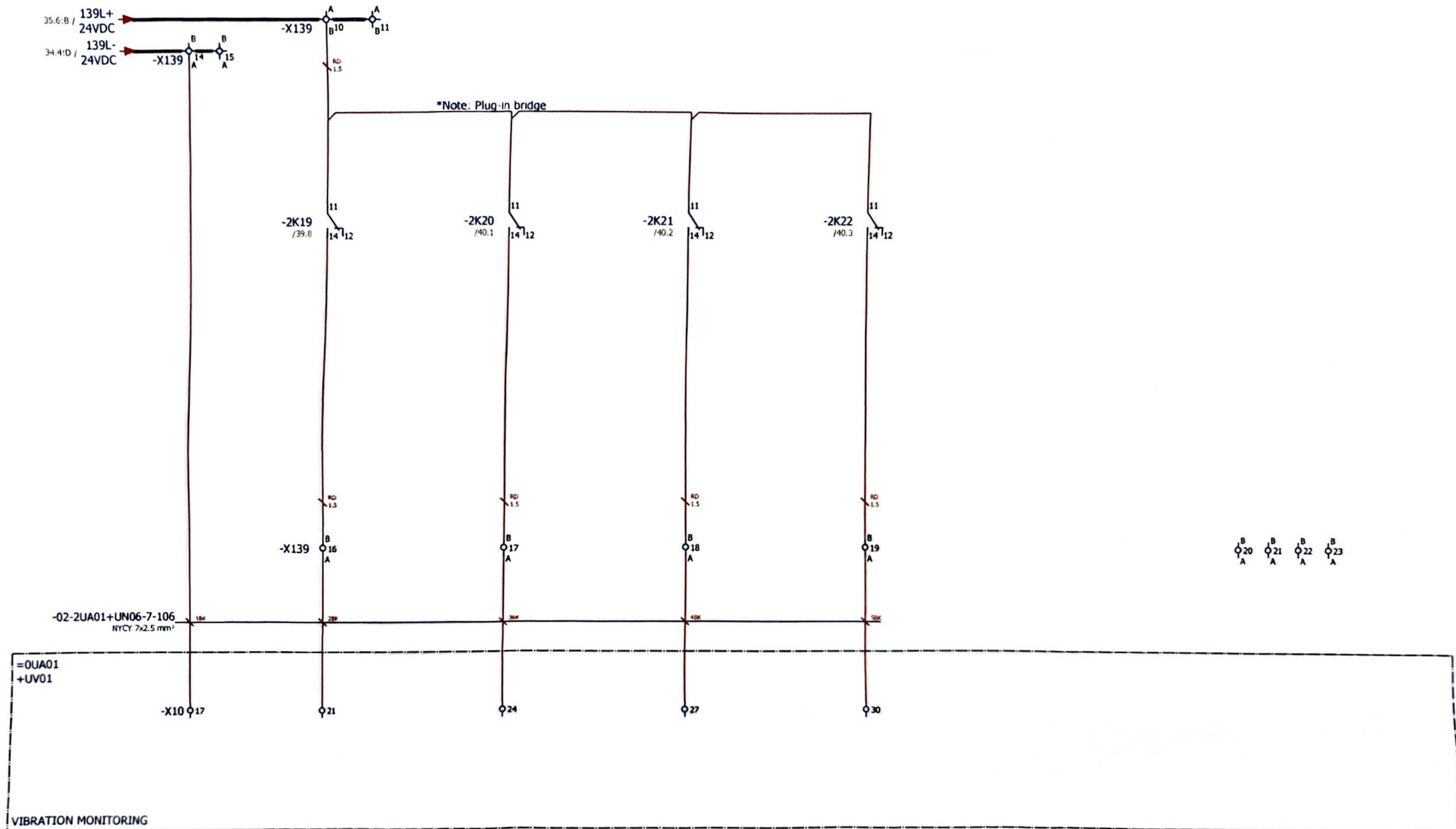
Format ISO/DIN A3

START COMMAND ON

STOP COMMAND ON

ESD COMMAND ON

ACTIVE POWER > 50%



20 21 22 23
A A A A

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		 A Voith Company Sintaksa - A Voith Company Uluka Domovinskog rata 104/C, 21 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams	DRAWING No.:	REV No.:	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.					PERU-01-19-170-31-002	02	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.					Designer No.:	SALES No.:	PAGE 67
Drawn by	Tamara Trogrlić	27.7.2026.					SIGNALIZATION		PAGE 67 / 158

A

B

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SIREN

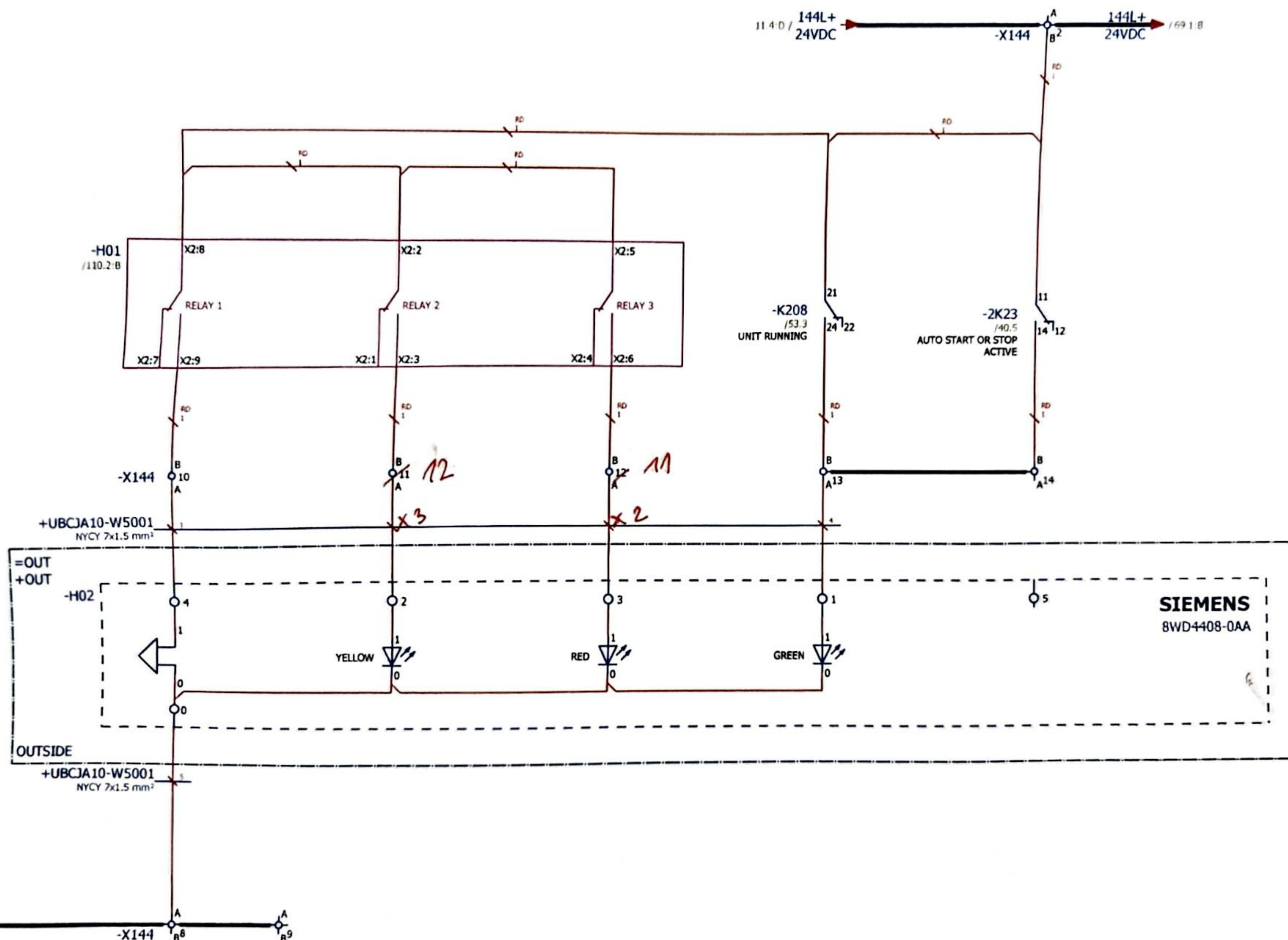
WARNING ACTIVE

ALARM ACTIVE

UNIT RUNNING

ALARM

WARNING



Approved by Contractor

Approved by Designer

Checked by

Drawn by

Grga Kolić

Antonio Aleksić

Tamara Trogrlić

27.7.2026.

27.7.2026.

27.7.2026.

Reconstruction and Modernisation
of HPP Perucica - Phase II

PROJECT NAME

epcg

SINTAKSA

A Voth Company

Sintaksa - A Voth Company

Ulica Dalmatinske 104C, 21 000 Split, Hrvatska

Unit KA2 control cubicle circuit diagrams

SIGNAL COLUMN

DRAWING No.:

PERU-01-19-170-31-002

Designer No.:

REV No.:

02

SALES No.:

= UBCJA10GH100

+ UBCJA10

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SPARE



VIBRATION
MONITORING TRIP

EXCITATION TRIP

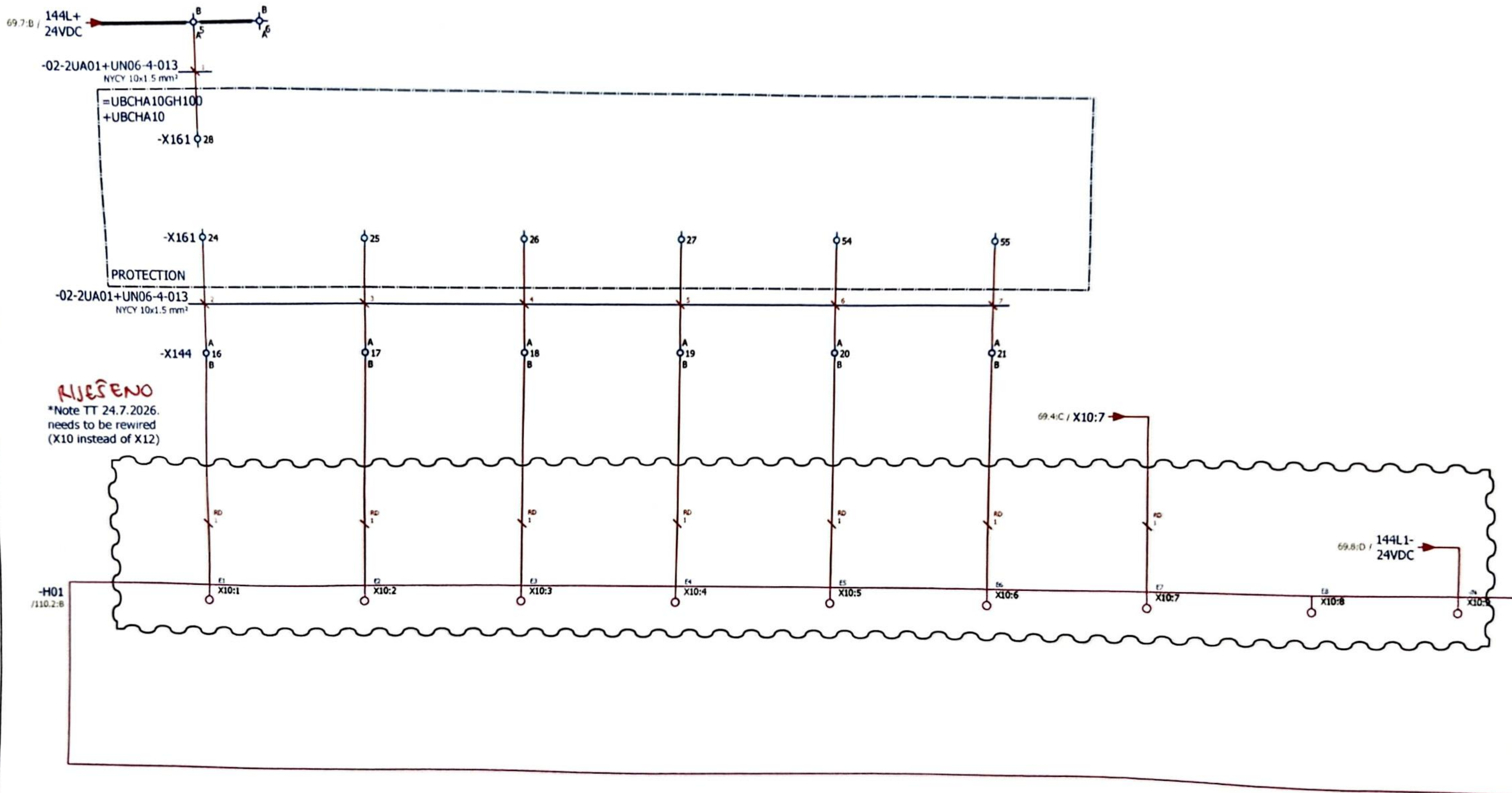
OVERSPEED TRIP

GCB PRIMARY
PROTECTION TRIP

ESD RELAY GROUP TRIP

PROTECTION SYSTEM
FAULT

24 VDC MCB TRIP
GROUP OK



Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II
Approved by Designer	Grga Kolić	27.7.2026.	
Checked by	Antonio Aleksić	27.7.2026.	
Drawn by	Tamara Trogrlić	27.7.2026.	

PROJECT NAME



EMPLOYER



Sintaksa - A Voith Company

Ulica Demotroškova 104C, 21 000 Split, Hrvatska

DESIGNER

Unit KA2 control cubicle circuit diagrams

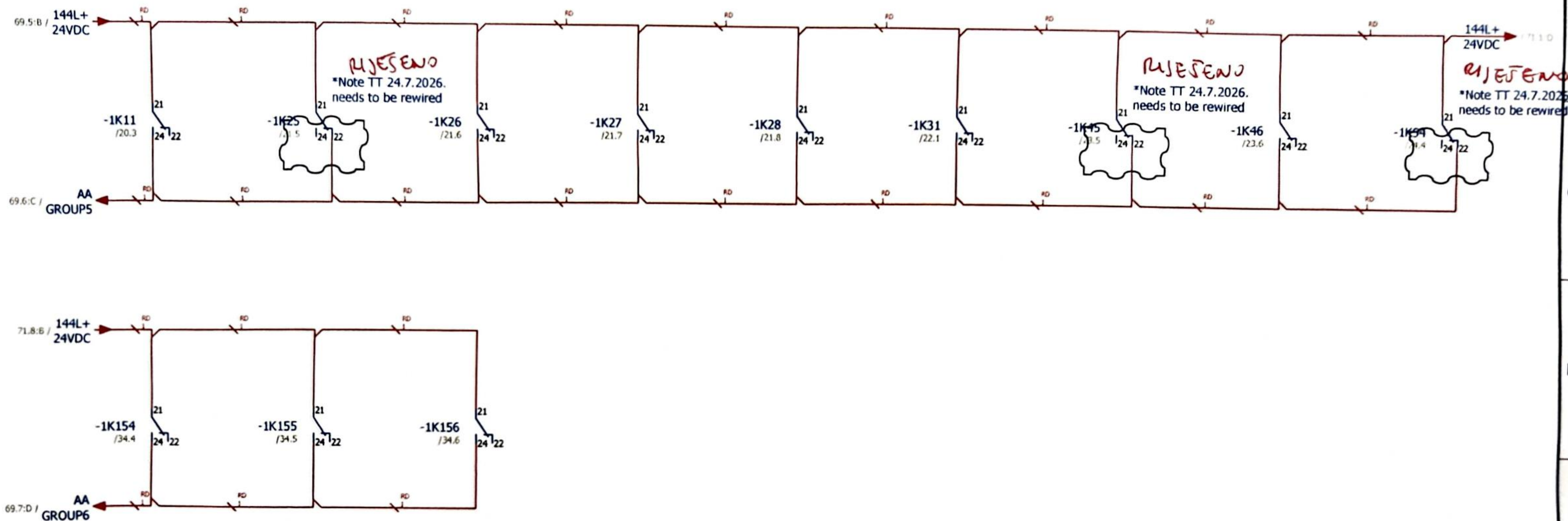
ALARM ANNUNCIATOR - TRIP

DRAWING No.:
PERU-01-19-170-31-002
Designer No.:

REV No.:
02
SALES No.:

= UBCJA10GH100
+ UBCJA10
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ALARM ANNUNCIATOR
ALARM GROUPS

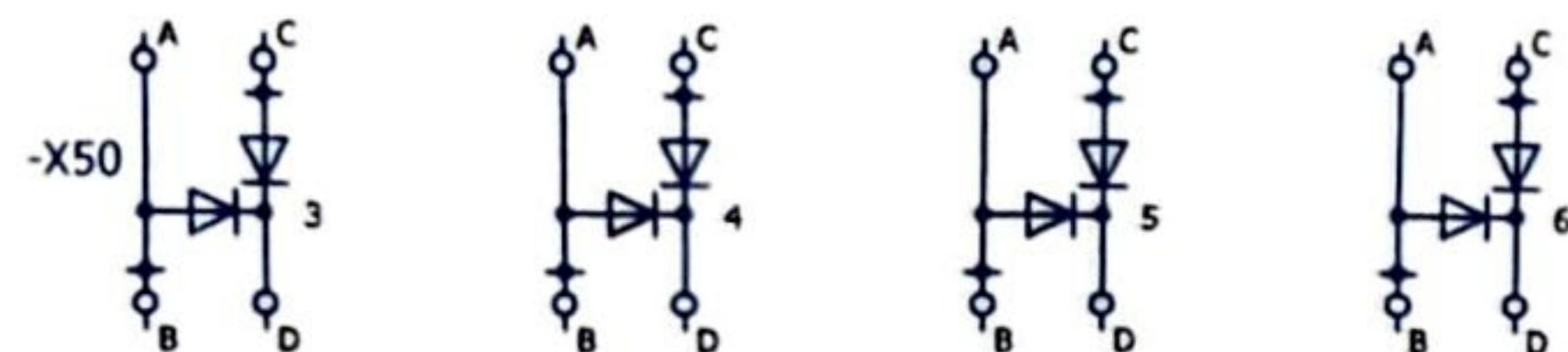


Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A Vieth Company	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100 + UBCJA10
Approved by Designer	Grga Kolić	27.7.2026.			Sintaksa - A Vieth Company Jaka Damaštrakog rata 104C, 21 000 Split, Hrvatska	ALARM ANNUNCIATOR - TRIP	Designer No.:	SALES No.:	PAGE 71 / 158
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER				
Drawn by	Tamara Trogrlić	27.7.2026.							

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-X110 ϕ_1^A ϕ_2^A ϕ_3^A ϕ_4^A ϕ_5^A ϕ_6^A ϕ_7^A ϕ_8^A ϕ_9^A ϕ_{10}^A ϕ_{11}^A ϕ_{12}^A ϕ_{13}^A ϕ_{14}^A ϕ_{15}^A ϕ_{16}^A ϕ_{17}^A ϕ_{18}^A ϕ_{19}^A ϕ_{20}^A

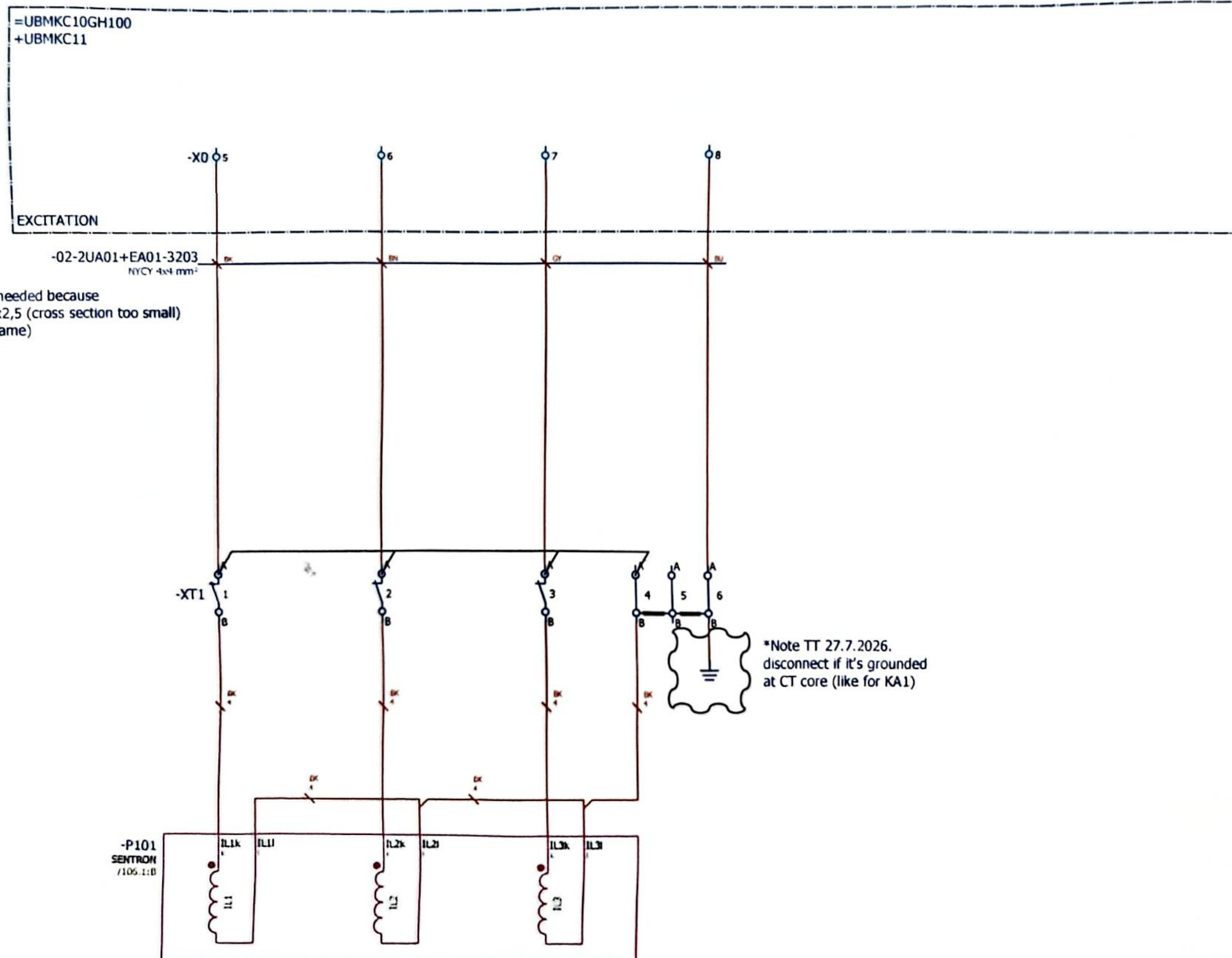


-X151 ϕ_5^A ϕ_6^A ϕ_7^A ϕ_8^A

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A Vukob Company Sintaksa - A Vukob Company Ulica Dalmatinskeg rata 104C, 21 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100 + UBCJA10
Approved by Designer	Grga Kolić	27.7.2026.							
Checked by	Antonio Aleksić	27.7.2026.							
Drawn by	Tamara Trogrlić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER	SPARE CONTACTS	Designer No.:	SALES No.:	PAGE 72 / 158
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GENERATOR
PHASE 1 PHASE 2 PHASE 3
CURRENT

MULTIMETER



*Note TT 24.7.2026.
new cable NYCY 4x4 is needed because
the old cable is NYCY 7x2,5 (cross section too small)
(cable name stays the same)

*Note TT 27.7.2026.
disconnect if it's grounded
at CT core (like for KA1)

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Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A Youth Company	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBC1A10GH100 + UBC1A10
Approved by Designer	Grga Kolić	27.7.2026.					Designer No.:	SALES No.:	PAGE 73
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER	CT CIRCUITS			PAGE 73 / 158
Drawn by	Tamara Trogrlić	27.7.2026.							
1	2	3	4	5	6	7	8		

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MULTIMETER

SPEED MEASUREMENT
GENERATOR RESIDUAL VOLTAGE=NE10
+NE10AUXILIARY DISTRIBUTION -X2
GENERATOR B-02-2UA01+NE10-3-601
NYCY 5x2.5 mm²

8x2.5

-XT2

ČEKANO
POBUDU

EXCITATION L1 75.2;C

EXCITATION L2 75.2;C

EXCITATION L3 75.2;C

EXCITATION N 75.2;C

SYNCHRONIZER GEN L1 60.3;C

SYNCHRONIZER GEN L2 60.3;D

SYNCHROCHECK GEN L1 61.2;B

SYNCHROCHECK GEN L2 61.2;B

-P101
SENTRON
/106.1;B

V1

V1.1

V2

V2.2

V3

V3.3

VN

*Note TT 27.7.2026.
needs to be rewired
(connect cable instead
of wires)UYCY 4x0.75 mm²

-XF201

UYCY 4x0.75 mm²

-U201

Volth
VRU1
/134.4;C

Approved by Contractor

Approved by Designer

Checked by

Drawn by

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Grga Kolić
Antonio Aleksić
Tamara Trogrlić27.7.2026.
27.7.2026.
27.7.2026.Reconstruction and Modernisation
of HPP Perucica - Phase II

PROJECT NAME

EMPLOYER

epcg

SINTAKSA
A Voth CompanySintaksa - A Voth Company
Ulica Dalmatinskega rata 104C, 21 000 Split, Hrvatska

DESIGNER

Unit KA2 control cubicle circuit diagrams

VT CIRCUITS

DRAWING No.:

PERU-01-19-170-31-002

Designer No.:

REV No.:

02

SALES No.:

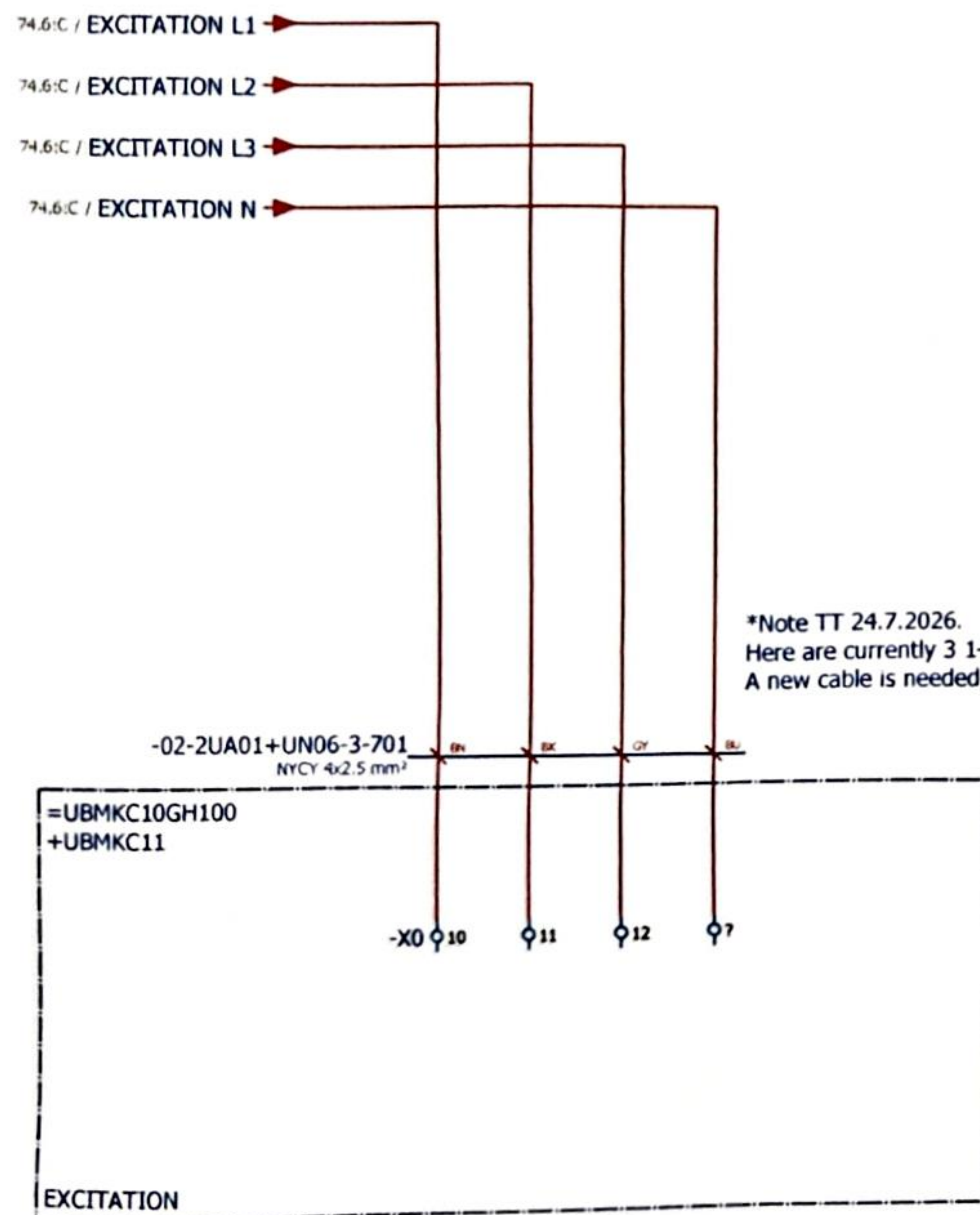
= UBCJA10GH100

+ UBCJA10

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EXCITATION VOLTAGE



*Note TT 24.7.2026.
Here are currently 3 1-core cables (one per phase, neutral not connected)
A new cable is needed, e.g. NYCY 4x2,5 or 5x2,5

Approved by Contractor			Reconstruction and Modernisation of HPP Perucica - Phase II		 SINTAKSA A Youth Company Sintaksa - A Youth Company Ulica Demotivirskog rata 104C, 21 000 Split, Hrvatska	Unit KA2 control cubicle circuit diagrams VT CIRCUITS	DRAWING No.:	REV No.:	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.					PERU-01-19-170-31-002	02	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.					Designer No.:	SALES No.:	PAGE 75
Drawn by	Tamara Trogrić	27.7.2026.					PROJECT NAME	EMPLOYER	DESIGNER

NYCY-J 8x25
(ima 9AYE)

0,4 kV BUSBAR VOLTAGE

= NE09
+ NE09

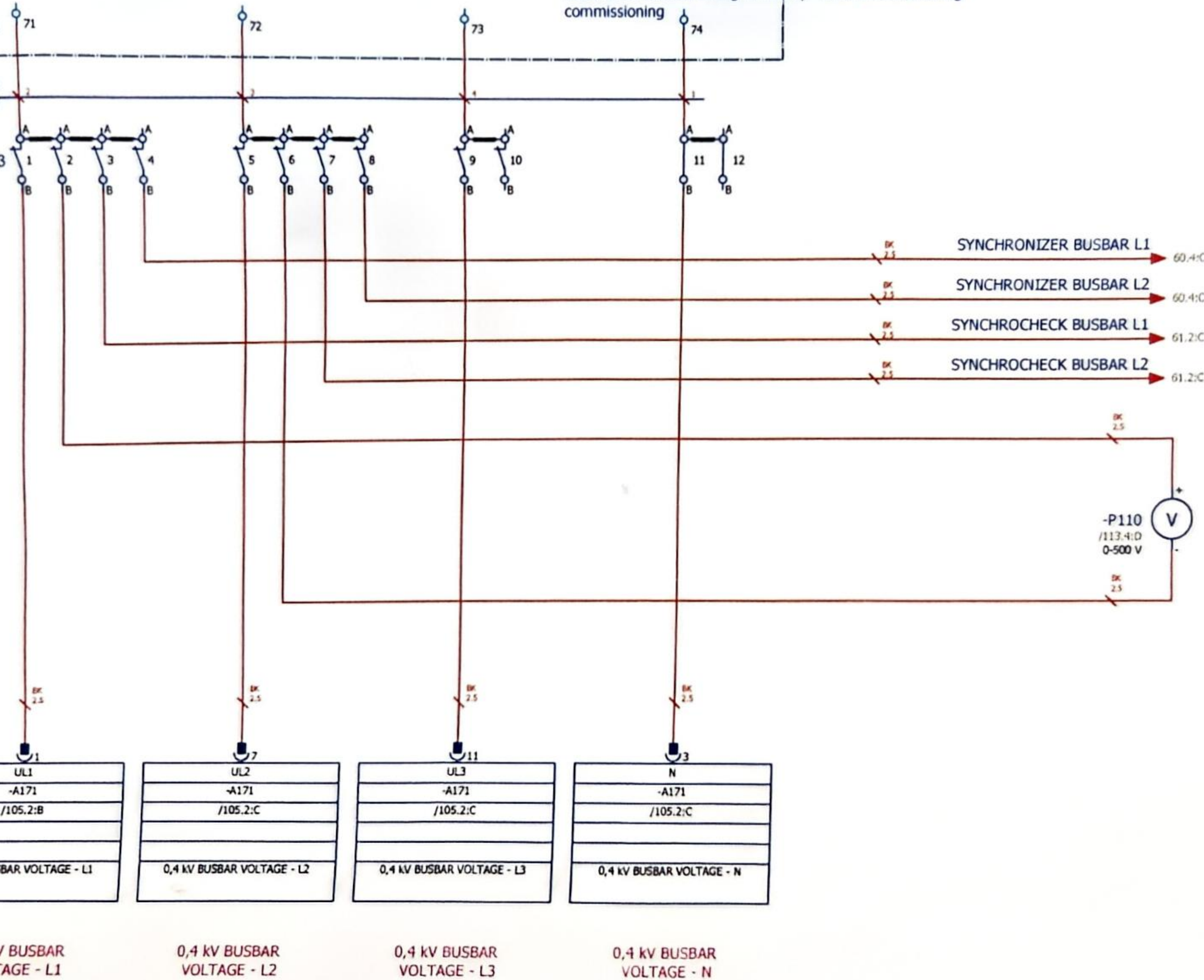
AUXILIARY DISTRIBUTION - X2
COUPLER

*Note TT 24.7.2026.
Cable core 1 is connected to X2:75, but should probably
be connected to X2:74 (according to documentation, plus
there is no internal wiring in X2:75). To be checked during
commissioning

-02-2UA01+NE09-3-607
NYCY-8x25 mm²

8x25

-XT3



0,4 kV BUSBAR
VOLTAGE - L1

0,4 kV BUSBAR
VOLTAGE - L2

0,4 kV BUSBAR
VOLTAGE - L3

0,4 kV BUSBAR
VOLTAGE - N

Approved by Contractor	Grga Kolić	27.7.2026.	Reconstruction and Modernisation of HPP Perucica - Phase II	epcg	SINTAKSA A Youth Company	Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100 + UBCJA10
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER	VT CIRCUITS	Designer No.:	SALES No.:	PAGE 76 / 158
Drawn by	Tamara Trogrlić	27.7.2026.							

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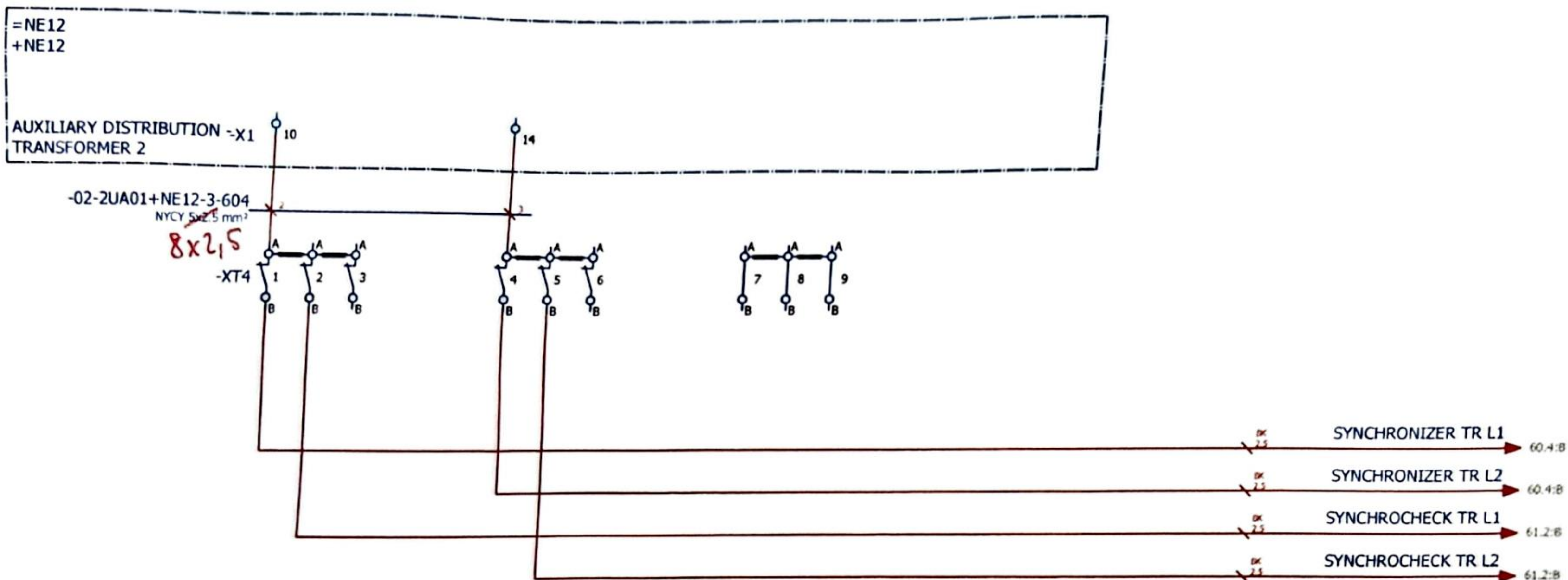
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TRANSFORMER VOLTAGE



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Approved by Designer	Grga Kolić	27.7.2026.				VT CIRCUITS	Designer No.:	SALES No.:	PAGE
Checked by	Antonio Aleksić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER				77
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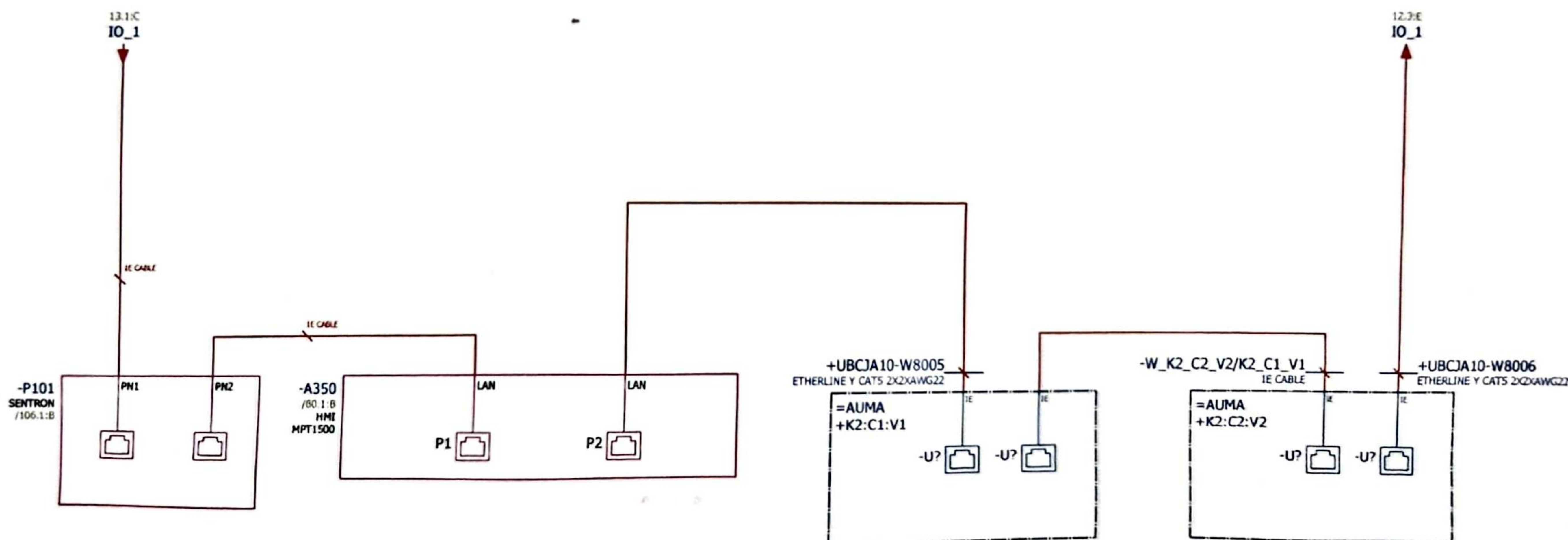
D

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Approved by Contractor					Unit KA2 control cubicle circuit diagrams	DRAWING No.: PERU-01-19-170-31-002	REV No.: 02	= UBCJA10GH100
Approved by Designer	Grga Kolić	27.7.2026.	Reconstruction and Modernisation of HPP Perucica - Phase II		SINTAKSA A youth company	Designer No.:	SALES No.:	+ UBCJA10
Checked by	Antonio Aleksić	27.7.2026.						PAGE 79
Drawn by	Tamara Trogrlić	27.7.2026.	PROJECT NAME	EMPLOYER	DESIGNER			PAGE 79 / 158
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